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Memtale: Emotion-Aware Two-Stage Retrieval for Autobiographical-Style Narratives

Interim Project Demo by

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ABSTRACT

Problem: Retirees' life-experience archives and digital storytelling tools can support reminiscence and well-being, but searching long personal narratives is still difficult when users express emotional intent (e.g., "a moment regret kept bothering me"). Most retrieval pipelines optimise lexical/semantic similarity, which can return topically related but emotionally mismatched stories. This reduces perceived relevance and trust in memory-centred retrieval.

Methodology: This thesis implements Memtale, a reproducible research prototype for emotion-intent narrative retrieval. Narratives are indexed using cached abstractive summaries to reduce length and stabilise experiments. Retrieval follows a two-stage pipeline: (1) BM25 generates TopK candidates (K=100) over cached summaries; (2) candidates are reranked using an interpretable fusion of BM25 (min-max normalised), SBERT semantic similarity, and SETL-2 emotion alignment. SETL-2 produces an 8-dimensional emotion vector per summary using NRC emotion cues with a VADER-based polarity adjustment, enabling cosine emotion similarity. The evaluation uses a fixed corpus of 5,000 narratives with 72 emotion-intent queries and graded manual relevance labels (0/1/2), plus a competitive cross-encoder reranker (CE-Rerank) baseline.

Initial Results: The best configuration (OURS, wemo=0.20) achieved nDCG@5 = 0.391 and MRR = 0.763, outperforming BM25 (0.309 / 0.632) and SBERT (0.321 / 0.656) and approaching the hybrid RRF baseline (0.407 / 0.751). For emotional alignment, OURS (wemo=0.20) obtained EmoFidelity@5 = 0.1889, while CE-Rerank achieved the highest EmoFidelity@5 (0.2111). A qualitative example (q61, "regret") shows OURS moving guilty/ashamed stories into the top ranks compared to BM25's emotionally mismatched results.

Subject Descriptors:

- Information systems → Information retrieval → Retrieval models and ranking
- Information systems → Information retrieval → Evaluation of retrieval results
- Computing methodologies → Natural language processing → Information extraction
- Computing methodologies → Natural language processing → Text representation

Keywords: emotion-aware retrieval, autobiographical narrative search, two-stage ranking, SBERT, BM25, Older adults, emotion vectors

DECLARATION

I declare that all work done in this proposal and all work submitted hereafter will be my own work and effort. It has not been submitted for any other academic work or degree program. All sources used for information have been properly cited.

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LIST OF ABBREVIATIONS:

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
NLP	Natural Language Processing
IR	Information Retrieval
CLI	Command Line Interface
BM25	Best Matching 25 (probabilistic lexical ranking function)
SBERT	Sentence-BERT (Sentence Transformers)
RRF	Reciprocal Rank Fusion
CE-Rerank	Cross-Encoder Reranker baseline
SETL-2	Semantic–Emotional Tagging Layer (Version 2)
NRC	NRC Emotion Lexicon
VADER	Valence Aware Dictionary and sEntiment Reasoner
nDCG@5	Normalized Discounted Cumulative Gain at rank 5
MRR	Mean Reciprocal Rank
EmoFidelity@K	Emotional Fidelity at rank K (emotion-match rate in TopK)
qrels	Query Relevance
IAA	Inter-Annotator Agreement
UI	User Interface

CHAPTER 01: INTRODUCTION

1.1. Chapter Overview

This chapter discusses the author's research on developing a searchable memory and life-experience archiving system for retirees. The work is motivated by AI-based life-story archives for retirees, where people often revisit memories through feelings (e.g., pride, regret, guilt) rather than only topics. The chapter explains the problem background and the central research issue: current memory and narrative retrieval pipelines mainly optimise lexical or semantic similarity and often fail to return memories that match the user's emotional intent. It then defines the research problem, motivation, gap, and contributions, and concludes with the research questions, aim, objectives, and scope aligned to the implemented prototype and evaluation.

1.2. Problem Background

1.2.1. Storytelling, ageing, and emotion-driven memory access

Storytelling and reminiscence support identity continuity, emotional regulation, and well-being in later life (Lee et al., 2025). Digital storytelling interventions have shown benefits for social connection and psychological health among older adults (Stargatt et al., 2022; Xu et al., 2023). However, autobiographical memory is rarely accessed as a neutral factual record (Hommelhoff et al., 2023). In real life, people frequently retrieve memories through affective cues seeking "a proud moment," "a time I felt regret," or "a memory that makes me feel grateful." This emotion-driven access becomes particularly relevant for retirees and ageing populations, where emotional meaning, not just topical matching, shapes what is perceived as "relevant" during reminiscence (Stargatt et al., 2022).

1.2.2. Limitations of current narrative retrieval in life-story archives

Existing digital archiving and storytelling tools are often fragmented and rely on manual organisation, which is cognitively demanding for older adults and those with low digital literacy (Rios Rincon et al., 2022). Even when AI is used, most systems prioritise transcription, summarisation, or surface-level tagging (Contreras-Somoza et al., 2021). Retrieval is commonly treated as "topic search," assuming that semantic similarity will naturally surface emotionally appropriate content (X. Wang et al., 2024). In practice, this assumption fails: a story can be topically related but emotionally

mismatched (e.g., a “graduation” story dominated by anxiety retrieved for “a proud moment”) (Ai et al., 2025). Such mismatches reduce user trust and perceived relevance, and they are especially harmful in memory-centred applications where emotional safety and resonance matter (Stargatt et al., 2022).

1.2.3. Why emotionally faithful summarisation matters for retrieval

Life stories are long and unstructured. Systems often rely on summarisation to compress narratives into searchable units (Hou et al., 2021). However, typical summarisation settings optimise for factual completeness and fluency, which can unintentionally “flatten” emotional tone (Ahuir et al., 2024). If summaries lose the dominant emotional cues, retrieval quality for emotion-intent queries degrades (Yash et al., 2025). Therefore, emotionally faithful summarisation becomes a practical requirement for memory search because it preserves affective signals that later support emotion-aligned retrieval (Ai et al., 2025).

1.3. Problem Definition

Although emotion is a key organising principle in autobiographical recall, most narrative retrieval pipelines treat emotion as secondary (metadata, post-hoc filtering, or sentiment labels) and primarily rank results using lexical overlap or semantic similarity. As a result, when users query with emotional intent, the returned memories can be emotionally inappropriate even if they are topically related (Yash et al., 2025). This gap is critical for life-story archives and memory-support systems for retirees because perceived relevance depends on emotional alignment, not only topic similarity (X. Wang et al., 2024). Therefore, the core technical problem addressed in this project is: How can emotional alignment be operationalised as an explicit, controllable ranking signal within a standard retrieval pipeline for autobiographical-style narratives, without replacing robust lexical retrieval and without requiring training new large models from scratch? This project answers the problem through a two-stage framework: lexical candidate generation followed by emotion-aware reranking that fuses semantic similarity and explicit emotion alignment

1.3.1. Problem Statement

Current memory and narrative retrieval systems often return emotionally mismatched results for emotion-intent queries because they rank memories mainly by lexical or semantic similarity rather than explicit emotional alignment.

1.4. Research Motivation

The author's motivation for the research originates from an observation of retirees, particularly individuals like the author's father and his peers, who often reminisce about their past and deeply value revisiting cherished memories but face challenges in accessing or preserving them through current digital tools. The absence of effective archiving mechanisms poses a risk of losing experiences that hold significant cultural and personal value for future generations. This realisation inspired the development of an accessible solution aimed at keeping such memories alive as a lasting legacy, allowing retirees to easily record, organise, and retrieve their life stories through text or speech. Beyond this specific context, the research is also driven by the broader global challenge of addressing social engagement and memory preservation in ageing populations, which serves as the inspiration behind the research. This project seeks to bridge the gap between retirees and technology through integrating archive functions, multimodal interaction, and natural language processing into an AI companion. This will create a tool for memory preservation as well as a platform for digital inclusion and intergenerational connections.

1.5. Existing Work.

Citation	Summary	Limitation	Contribution
(X. Wang et al., 2024)	AI-driven multimodal photo album for personalised reminiscence in older adults.	No semantic retrieval or emotion-conditioned search; photo-only.	Validates multimodal engagement for elders
(Stargatt et al., 2022)	Systematic review of digital storytelling interventions showing improvements in mental well-being in older adults.	Manual curation only; no AI automation or search.	Establishes the therapeutic value of narrative access
(Yash et al., 2025)	Sentence-BERT + fine-grained emotion	Generic domains; no autobiographical	Proves emotion as an explicit ranking signal

	classification for emotion-aware document retrieval.	narratives.	
(P et al., 2025)	Introduces an empathetic AI framework that uses multimodal interaction to help elderly people become more emotionally resilient.	Prioritise emotional support over searchable memory preservation or long-term story archiving.	Demonstrates how empathetic, multimodal AI enhances retirees' well-being, engagement, and trust.
(Zhang et al., 2023)	Emotion-Aware Graph Attention Network for retrieval-based QA using BM25→emotional fusion.	Question-answering focus; no life-story evaluation.	Validates your two-stage pipeline architecture

Table 1:Existing work

While (X. Wang et al., 2024) and (Stargatt et al., 2022) establish the therapeutic motivation for digital reminiscence in elders, their systems lack automated retrieval pipelines Wang remains photo-centric while Stargatt requires manual curation. Recent IR advances demonstrate emotion-retrieval feasibility: the (Yash et al., 2025) paper fuses Sentence-BERT embeddings with Ekman's 6 emotions for hybrid similarity scoring, and (Zhang et al., 2023) implements BM25 → emotional graph reranking for QA, both achieving nDCG gains over semantic-only baselines. However, neither targets autobiographical narratives where emotional intent ("proud moment") diverges from topical match ("graduation"). (P et al., 2025) proves multimodal empathy builds elder trust but omits archival search.

1.6. Research Gap

Digital storytelling and reminiscence tools support older adults, yet retrieval relevance is still largely treated as topical matching. Very few systems address the key scenario: users retrieving memories through emotion-intent queries, where emotional resonance is essential (Yash et al., 2025). Additionally, summarisation is commonly applied for narrative indexing, but emotional tone is often lost, weakening retrieval for affect-driven needs (Ahuir et al., 2024). Therefore, a clear gap incorporaten interpretable and reproducible retrieval framework that (i) preserves emotional cues

during summarisation and (ii) incorporates emotional alignment as an explicit ranking dimension within a standard two-stage retrieval architecture.

1.7. Contribution to Body of Knowledge

1.7.1. Contribution to the Problem Domain

This project improves how life-story archives (motivated by retiree memory support) can retrieve narratives when users express emotional intent. It introduces a practical retrieval approach that reduces emotionally mismatched results by explicitly measuring emotional alignment and using it to rerank candidates, supporting more meaningful memory access.

1.7.2. Contribution to the Research Domain

The research contribution is an information retrieval framework where emotion is treated as a first-class, controllable relevance signal rather than metadata. The system uses: Two-stage retrieval: BM25 candidate generation (topK=100 fixed) followed by reranking, Semantic similarity via Sentence-BERT embeddings (all-MiniLM-L6-v2; 384-dim), Interpretable emotion vectors (SETL-2) derived from NRC + VADER, with cosine emotion similarity. In addition to classical baselines (BM25, SBERT, and RRF), the evaluation includes a neural reranking baseline (CE-Rerank) using a cross-encoder model (cross-encoder/ms-marco-MiniLM-L-6-v2), isolating the contribution of the reranking stage without emotion signals. The study also performs a weight sensitivity analysis over the emotion fusion weight (wemo) to justify the final configuration and demonstrate robustness rather than relying on a single fixed weight setting.

1.8. Research Challenges

Challenge 1: Limited and Unstructured Data for Life Stories

Retirees' life stories are long, informal, and heterogeneous, and large-scale labelled retiree narrative datasets are rarely available. This motivates using a public autobiographical-style narrative source for reproducible experiments while acknowledging domain differences.

Challenge 2: Emotion preservation in summarisation:

Standard abstractive summarisation may reduce or flatten affective cues, which can degrade retrieval performance for emotion-intent queries.

Challenge 3: Interpretable emotion modelling:

Emotional alignment must be represented transparently (e.g., lexicon-based emotion vectors) rather than relying only on opaque end-to-end emotion encoders that are difficult to audit.

Challenge 4: Ranking integration without overfitting:

introduce emotional alignment as a ranking signal while keeping strict separation from evaluation labels and maintaining reproducibility

Challenge 5: Fusion weight justification and robustness.:

Incorporating emotion into ranking requires demonstrating that improvements are not due to a single "lucky" weight configuration, motivating sensitivity analysis over emotion fusion weights.

1.9. Research Questions

RQ1: How can emotion-intent access patterns be supported in autobiographical narrative retrieval using a standard two-stage retrieval architecture?

RQ2: How can emotional alignment be represented in an interpretable and controllable way for personal narrative text, suitable for ranking?

RQ3: To what extent does emotion-aware reranking improve early-rank retrieval effectiveness and emotional alignment compared to lexical and semantic baselines?

1.10. Research Aim

"This research aims to analyse, design, develop, and evaluate an emotion-aware retrieval framework for autobiographical-style narratives that preserves emotional cues during summarisation and incorporates emotional alignment as an explicit reranking signal within a two-stage retrieval pipeline"

This research aims to focus on creating a smart search system for personal life stories, like digital diaries or memories shared by older adults. It ensures emotions in those stories stay intact even when shortened into summaries, then uses emotional matching as a key factor to rank and show the most relevant results in a two-step search process. The project breaks into four main steps: analysing current problems, designing the new framework, building it with AI tools for emotion detection and matching, and testing how well it works for emotion-based searches. Traditional searches match topics only, but

people often want memories that match their feelings, like "happy family moments." This system adds emotion as a reranking step after initial topic search, making results more meaningful for reminiscence therapy.

1.11. Research Objectives

Objectives	Research Objective	LOs Mapped	RQ Mapped
Problem Identification	**RO1:** To investigate the challenges retirees face in documenting, preserving, and retrieving life experiences using existing tools. **RO2:** To analyse the importance of AI-assisted digital archiving for knowledge preservation and intergenerational sharing. **RO3:** To examine global trends and demographic factors (ageing populations, digital divide) influencing the need for AI-based memory archiving.	LO1, LO3, LO4, LO6	RQ1
Literature Review	**RO4:** To conduct a comprehensive literature review on existing AI tools for storytelling, summarisation, and digital curation. **RO5:** To identify research gaps in multimodal input (speech/text), tagging, and retrieval in elderly-focused applications. **RO6:** To review narrative retrieval, emotion-aware NLP, and emotionally faithful summarisation methods relevant to autobiographical archives.	LO4, LO5, LO6	RQ1, RQ2, RQ3
Requirement Elicitation	**RO7:** To define functional and non-functional requirements for an emotion-intent narrative retrieval prototype, focusing on reproducibility, interpretability,	LO1, LO3, LO5,	RQ2, RQ3

	and early-rank performance **RO8:** To identify appropriate NLP models for summarisation, tagging, and retrieval of life stories. **RO9:** To determine hardware/software specifications and datasets suitable for prototype development.	LO6	
System Design	**RO10:** To design a two-stage retrieval architecture using lexical candidate generation and reranking that fuses semantic and emotion similarity signals. **RO12:** To design the data storage and indexing system for secure and efficient retrieval of life stories.	LO1, LO2, LO3	RQ1, RQ2
Implementation	**RO13:** To develop and implement the data collection pipeline for speech and text input. **RO14:** To integrate NLP-based summarisation and tagging models into the prototype. **RO15:** To develop the search/retrieval module allowing keyword or embedding-based queries. **RO16:** To develop a conversational companion interface to enhance engagement and usability	LO5, LO6, LO7	RQ2, RQ3
Testing	**RO17:** To develop a test plan for units, integration, usability, and performance testing. **RO18:** To evaluate the system's accuracy of summarisation, tagging, and retrieval using metrics and user feedback. **RO19:** To evaluate the framework using early-rank IR metrics and an emotional alignment measure, including significance testing.	LO5, LO6, LO7, LO8	RQ2, RQ3
Documentation	**RO21:** To compile and document the entire research process, methodology, and prototype in a formal report.	LO2, LO5,	RQ1, RQ2,

	RO22: To produce a research publication or conference paper presenting the findings and innovations of the project.	LO8, LO9	RQ3

Table 2:Research Objectives

1.12. Project Scope

1.12.1. In Scope.

- Emotion-aware summarisation for narrative indexing (cached summaries).
- Interpretable emotion representation (SETL-2) from NRC + VADER.
- Two-stage retrieval (BM25 → rerank using SBERT + emotion cosine + adaptive fusion).
- Evaluation on autobiographical-style narrative text with emotion-intent queries and graded judgments (reproducible experiments and stored outputs).
- Cross-encoder reranker baseline (CE-Rerank) and weight sensitivity analysis (wemo sweep) for ablation and robustness.

1.12.2. Out Scope

- Building a full retiree-facing multimodal companion product (speech UI, images, long-term deployment).
- Training new transformer models from scratch or collecting private retiree data.
- Clinical validation, longitudinal studies, or medical claims.
- Real-time emotional intervention or therapeutic decision-making.

1.13. Hardware/Software Requirements

Resource Type	Specification/Tools/Skills
Hardware	Development Environment: Standard workstation CPU with a minimum of 8 cores, 16 GB RAM; No dedicated GPU required, as pretrained models/APIs are used, Microphone and Headset
Software	Python 3.10, Python Libraries / Packages [sentence-transformers (SBERT +

	CrossEncoder), rank_bm25, transformers (DistilBART), torch, numpy, pandas, scipy, matplotlib, nltk, NRCLex], Streamlit (demo UI), Git, Figma, VS Code, Google Docs and Microsoft Word for documentation.

Table 3:Hardware and Software Requirements

1.14. Chapter Summary

This chapter presented the refined problem focus and motivation for emotion-intent autobiographical narrative retrieval, situated within retiree-oriented life-story archiving. It defined the core issue of emotionally mismatched retrieval results under affect-driven queries and identified the gap in existing systems that treat emotion as secondary rather than a ranking signal. The chapter also stated the project's contributions: an interpretable, reproducible two-stage retrieval framework integrating semantic similarity with explicit emotional alignment, and an evaluation plan using early-rank metrics and emotional fidelity measures.

CHAPTER 02: LITERATURE REVIEW

2.1. Chapter Overview

This chapter reviews the literature that motivates and supports the implemented scope of the project: emotionally faithful summarisation and emotion-aware retrieval for autobiographical-style narratives (motivated by retiree life-story archives). It first discusses the problem domain, highlighting how older adults benefit from storytelling and reminiscence but face barriers in digital access and meaningful memory retrieval. It then critically reviews existing work across four areas: (i) digital storytelling and life-story archives, (ii) semantic retrieval for personal narratives, (iii) emotion-aware information retrieval, and (iv) summarisation for indexing long narratives and the risk of emotional tone loss. Next, the chapter justifies the datasets used for the prototype and outlines the evaluation criteria used in the implemented experiments. Finally, it summarises the gaps that directly motivate the selected two-stage retrieval design and the emotionally faithful summarisation approach used for indexing.

2.2. Concept Map

The concept map was originally developed during the proposal stage to structure the initial literature survey across ageing, digital storytelling, multimodal capture, and AI archiving. While the project scope later narrowed to the text retrieval core (summarisation + indexing + retrieval + evaluation), the map is retained as a high-level overview of the broader domain and the original research motivation. The map is available in ***Appendix A.1 - Concept Map***

2.3. Problem Domain

2.3.1. Storytelling, Memory Preservation, and Ageing Populations

Throughout history, storytelling has been crucial for maintaining identity, continuity, and emotional well-being as people aged. Retirees' sense of purpose and belonging is reinforced by sharing and reflecting on their personal stories, which further encourages intergenerational understanding (Rios Rincon et al., 2022; Xu et al., 2023). Empirical evidence demonstrates that engaging in digital storytelling empowers older adults to become digital content creators, fostering social connectedness and legacy-building (Hausknecht et al., 2019). Additionally, storytelling thus serves as a medium for older adults to process life experiences, preserve their personal legacy, and pass down cultural and familial knowledge to younger generations, thereby contributing positively to their psychological

well-being and social involvement. (Lee et al., 2025). Digital tools with a narrative feature can help older adults feel less alone and more emotionally connected, which supports the therapeutic value of personal storytelling in digital spaces (Al Mahmud et al., 2022). According to recent studies, digital storytelling enhances self-validation, cognitive engagement, and emotional health. For instance, Feasibility, Acceptability and Preliminary Effectiveness of an Intergenerational Digital Storytelling Program showed that digital narratives reinforce "identity validation" and "generativity", two psychological pillars of healthy ageing (Hommelhoff et al., 2023; Zhu et al., 2024a). Moreover, qualitative studies indicate that older adults perceive digital sharing platforms not only as cognitive tools but also as mediums for sustaining meaningful social bonds within their existing networks (Long et al., 2024). Sharing autobiographical stories digitally supports mental stimulation and combats social isolation as well(Stargatt et al., 2022).

Digital storytelling also serves as a form of reminiscence therapy, proven to improve well-being among individuals with mild cognitive impairment or dementia (Chang et al., 2023; Nebot et al., 2022). Systematic reviews also emphasise that technological interventions, when designed for community-dwelling older adults, enhance not only cognitive outcomes but also social participation and perceived inclusion, especially for those with early cognitive decline (Heins et al., 2021). For instance, the LONG-REMI project demonstrated that combining AI-driven emotion analysis with narrative recollection improved memory retention and mood regulation among older participants (Nebot et al., 2022). Nevertheless, most older adults do not have accessible, AI-assisted tools to capture and organise their memories, despite their benefits. A valuable cultural and familial legacy is lost when traditional paper journals or fragmented social media archives are unable to create a cohesive life narrative (Lind et al., 2025).

To enable retirees to maintain narrative control over their lives, there is a growing research opportunity to create an AI-driven life-experience archiving system that not only preserves personal narratives but also offers automated summarisation, semantic retrieval, and empathetic interaction.

2.3.2. Digital Exclusion and Accessibility Challenges

Although digital tools hold immense potential to empower retirees, many older adults continue to face digital exclusion due to physical, cognitive, and design-related barriers (Mubarak and Suomi, 2022). This phenomenon, often termed the "grey digital divide", results from low digital literacy, fragmented

platforms, and designs that overlook accessibility needs (Kärnä et al., 2022). Recent participatory studies highlight that many older adults view digital technologies ambivalently, valuing connection yet distrusting complexity or privacy implications (Heins et al., 2021). Older adults frequently express fear of technology failure or low self-efficacy when interacting with modern interfaces (Contreras-Somoza et al., 2021; Zhou et al., 2025). These perceptions are closely tied to experiences of social isolation, suggesting that digital design must simultaneously address emotional and relational barriers, not just technical literacy (Al Mahmud et al., 2022). Complex user interfaces, tiny font sizes, and non-intuitive navigation can discourage participation, particularly among those with vision decline, arthritis, or limited touch dexterity (Kim and Kim, 2024; Yan et al., 2024).

In addition, most current storytelling or journaling apps rely heavily on manual input, assuming high literacy and typing proficiency (Zhou et al., 2025; Zhu et al., 2024b). This reliance often excludes many retirees who prefer speech-based interaction or non-linear, visual prompts. Creative art-based interventions, such as those using photocollage to capture life stories, demonstrate the therapeutic and emotional value of visually engaging, participatory processes that go beyond traditional verbal or textual interviews (Keisari et al., 2021). This underscores that accommodating diverse modes of expression is a critical component of accessibility. Understanding Older People's Voice Interactions with Smart Voice Assistants (Yan et al., 2024) found that older adults often pause longer, mispronounce commands, or use colloquial phrases, which conventional NLP systems fail to interpret correctly. Studies emphasise that true digital inclusiveness demands multimodal interfaces combining speech, touch, and visual elements to reduce cognitive load and support different sensory abilities (Li et al., 2022; N. Wang et al., 2024). Furthermore, User Interfaces for Older Adults to Support Social Interaction Through Digital Technology: A Systematic Review Update (Zhou et al., 2024) underscores that consistent navigation patterns and personalisation significantly enhance older adults' engagement and satisfaction. From a system perspective, one major accessibility gap is not only input difficulty, but also retrieval difficulty: even when memories are recorded, older adults struggle to find meaningful stories later if the system only supports keyword/topic search (Zhu et al., 2024b). This motivates emotion-intent retrieval, where relevance is partly defined by affective resonance, not only topic overlap (Zhang et al., 2023).

2.3.3. Emerging Role of AI in Elder Care and Digital Companionship

The field of elder care is evolving as a result of artificial intelligence, offering conversational, empathetic, and context-aware systems that enhance emotional well-being (Lee and Allen, 2025; P et al., 2025). A significant emerging trend is the exploration of Generative AI to support highly personalised reminiscence, such as using music to evoke and facilitate memory recall (Jin et al., 2024). This requires a focus on a sensitive, adaptive, and simple AI experience to better support older adults. Studies show that AI-driven chatbots and digital companions can alleviate loneliness, encourage reflection, and stimulate cognitive function (Alotaibi and Alshahre, 2024; Khamaj, 2025). This suggests that integrating generative and affective AI capabilities is crucial for fostering user trust and emotional engagement in the archiving process. This aligns with broader findings that technology-mediated interactions can foster belongingness and mental stimulation among older adults when emotional well-being is integrated into design (Al Mahmud et al., 2022). For example, Empathetic AI: Multimodal Interaction Framework for Emotional Resilience and Cognitive Engagement in Elderly Care (P et al., 2025) demonstrated that multimodal AI agents capable of recognising tone, emotion, and context significantly improve user comfort and trust. Likewise, MyChum, an emotionally intelligent robot, illustrates how AI can support companionship through gesture recognition, speech interaction, and adaptive empathy (Rao et al., 2024).

AI-human collaboration frameworks like (Anand et al., 2025) propose digital "user twins" that mirror an individual's personality and memory to enhance knowledge continuity, a feature highly relevant for life-experience preservation. Meanwhile, Harnessing Artificial Intelligence for Mental Well-being of Ageing Populations (Ghosh, 2025) argues that proactive AI systems can detect emotional distress early through speech and text analysis, fostering psychological resilience. Though these systems offer support and companionship, they hardly ever integrate semantic retrieval or long-term narrative archive capabilities. Instead of preserving the emotional and cultural depth of retirees' life stories, the majority are made for brief interactions, health monitoring, or task support. While many empathetic agents support companionship (P et al., 2025), they rarely provide durable, searchable memory objects (Rao et al., 2024). This motivates treating emotion not only as an interface feature but as an archival and retrieval signal, enabling users to retrieve memories by emotional intent (Zhang et al., 2023).

2.3.4. Why Do Current Digital Storytelling Tools Fail Older Adults?

Although digital storytelling platforms have grown, the majority of them are still fragmented, static, and cognitively demanding for older users (X. Wang et al., 2024; Zhu et al., 2024a). In line with this, older adults often report that many digital tools fail to support sustained or meaningful social engagement, as they are not designed around the realities of ageing-related social networks (Long et al., 2024). Existing tools typically lack conversational capabilities, require human text input, and offer little to no automation for retrieval or summarisation (Lind et al., 2025; Zhou et al., 2025). For example, JournalAIde (Zhou et al., 2025) allows journal entry creation but lacks automated tagging or summarisation. Similarly, Digital Life Story Books (Lind et al., 2025) and LONG-REMI (Nebot et al., 2022) focus on reminiscence facilitation rather than structured archiving or semantic linking. Studies show that these tools lack embedding-based search, natural language comprehension, or speech-to-text transcription, which limits their usefulness for elderly people with mobility disabilities or low literacy (Usmaan and Singh, 2025; Yan et al., 2024).

In contrast, advanced multimodal AI prototypes such as MemPal (Maniar et al., 2025) demonstrate the promise of combining voice, contextual sensing, and semantic search for memory retrieval, though they remain research prototypes rather than accessible products. To further ensure emotional engagement, few storytelling systems incorporate empathetic AI agents that adjust to the user's mood, memory state, or interaction history (P et al., 2025; Rao et al., 2024). Consequently, the current ecosystem of digital storytelling tools fails to address the intersection of accessibility, automation, and affective intelligence. This gap opens a clear research direction: developing an AI-driven, multimodal life-experience archiving system that combines speech, text, and image input with intelligent summarisation and semantic retrieval to empower retirees to document, recall, and share their stories effortlessly. A further limitation is that retrieval in many systems implicitly assumes that "topical relevance" equals "felt relevance." In autobiographical recall, this is often false: two stories can share the same topic (e.g., graduation) but carry very different emotional meanings (pride vs anxiety). Emotion mismatch can reduce trust and perceived usefulness, especially in reminiscence contexts where emotional safety and resonance matter.

2.4. Existing Work

2.4.1. Technological Mediation of Memory

Digital storytelling is a crucial tactic for helping older adults with identity formation, self-expression, and intergenerational engagement, according to empirical evidence (Xu et al., 2023). Although systematic reviews support its beneficial effects on cognitive engagement and social participation, conflicting results arise about the degree of automation in these technologies. For instance, Xu et al. (2023) draw attention to persistent limitations, including the need for manual inputs and limited automation in text or image-based recollection tools, even though Al Mahmud et al. (2022) emphasise the role of digital platforms in reducing loneliness through improved social interaction. Digital storytelling improves health-related outcomes by enabling older adults to share life experiences (Stargatt et al., 2022), while Rios Rincon et al. (2022) report its feasibility and usability among people with mild cognitive impairment and dementia. Crucially, the absence of a structured lifecycle model poses a threat to long-term legacy-building. A robust digital curation lifecycle is necessary to govern actions from conceptualisation to preservation and reuse (Rhee, 2024). These limitations stand in contrast to innovations such as LONG-REMI (Nebot et al., 2022), which incorporates facial expression analysis powered by artificial intelligence to enhance reminiscence therapy, but does not include semantic memory search or full life-story archive features. This tension underscores an emerging gap: although multimedia archiving features (Zhu et al., 2024b) offer fundamental capabilities, seamless memory continuity is hampered by the absence of advanced summarisation and retrieval procedures and a formalised curation methodology.

Furthermore, participatory design insights (Zhao et al., 2024) reveal older adults' aspirations for technologies that transcend static repositories, favouring dynamic systems that integrate memory, identity, and sustained activity engagement. However, despite these advances, empirical studies consistently demonstrate that most systems stop at factual or topical indexing, rarely incorporating emotion as a first-class retrieval dimension (Chu et al., 2022; Jin et al., 2024; X. Wang et al., 2024). This tension reveals a missed opportunity: older adults' narratives are affect-rich and affect often drives recall cues in everyday reminiscence. Lightweight sentiment methods (e.g., lexicon- or rule-based analysis) have proven dependable in conversational domains (Hutto and Gilbert, 2014), while emotion lexicons capture nuanced affective categories beyond polarity (Mohammad and Turney, 2013). Yet these techniques are seldom embedded within curation pipelines. Integrating emotion with

entities and embeddings would therefore align technical indexing with the psycho-social realities of reminiscence, enabling queries such as “show happiest memories from early career” rather than keyword matches alone (Chu et al., 2022; X. Wang et al., 2024). Recent work in information retrieval demonstrates that affect can be incorporated as a ranking signal rather than a post-hoc label (Yash et al., 2025). Emotion-aware retrieval models show that users’ affective intent can be operationalised in retrieval pipelines and can improve early-rank relevance in some settings (Zhang et al., 2023). However, most of this work is evaluated on generic corpora (news, reviews, QA) rather than autobiographical-style narratives, and it rarely connects to the practical requirements of memory archives (summarisation for indexing, interpretability, and reproducibility) (Stargatt et al., 2022; Xu et al., 2023). This motivates an interpretable two-stage approach where emotional alignment complements (not replaces) lexical and semantic relevance.

2.4.2. Empathy and Identity in AI Narrative Systems

Building on the need for affect in memory systems, the next body of work examines empathetic AI and identity. Artificial Intelligence application to elder care primarily aims to improve accessibility and lessen loneliness, but its incorporation with thorough life-story preservation is still in its infancy. The effectiveness of empathetic multimodal AI systems in enhancing emotional resilience and cognitive stimulation has been demonstrated by (Khamaj, 2025) and (P et al., 2025); however, (Anand et al., 2025) point out a significant limitation: digital user representations enhance knowledge sharing but do not include structured narrative archiving or semantic retrieval. While digital storytelling interventions show promise for emotional resilience and cognitive stimulation, their integration with AI-driven semantic retrieval and empathetic narrative continuity remains limited (Stargatt et al., 2022). This specific focus is consistent with criticisms of digital ageism (Chu et al., 2022), which point out that AI designs do not adequately account for cognitive and emotional deviations. The MyChum robot (Rao et al., 2024) and entertainment chatbots (García-Méndez et al., 2021) illustrate how emotionally intelligent companions can provide interactive, non-judgmental support. (Lee and Allen, 2025) Also, report how AI can improve older adults’ psychological well-being by creating customised conversational interfaces. According to (Loveys et al., 2022) and (Colavizza et al., 2022), AI has the ability to redefine long-term care paradigms by converting transient conversational data into long-lasting, ethically curated digital legacies. While empathetic agents improve engagement and perceived support, their outputs typically remain transient and weakly coupled to archival structures (P et al.,

2025; Rao et al., 2024). However, despite these advances, literature lacks a practical bridge between affect-sensitive interaction and durable, searchable memory objects. This gap motivates a Semantic-Emotional Tagging Layer (SETL): a compact algorithmic step that classifies dominant affect in each story and stores it as queryable metadata alongside entities and embeddings. By operationalising emotions (e.g., Joy, Nostalgia, Sadness) at archive time using lightweight methods, SETL reframes empathy from an interface property into a retrieval affordance (Hutto and Gilbert, 2014; Mohammad and Turney, 2013).

2.4.3. Design Ethics for Ageing and Inclusion

While technology and empathy offer capabilities, their implementation must be grounded in ethical, inclusive design. The solid frameworks to preserve retirees' narratives through continuous recording, metadata enrichment, and safe archiving are described in the literature on digital curation (Rhee, 2024). The transition to intelligent, evolving memory systems is best illustrated by the way curation principles and interactive designs are combined in intergenerational storytelling (Li et al., 2022). According to Trehan (2023), recent advances in generative AI and semantic embeddings for archival indexing suggest a shift from static repositories and toward adaptive, user-engaged archives. However, ethical considerations related to consent, privacy, and contextual sensitivity require careful consideration (Colavizza et al., 2022). 68% of elderly users prefer local storage over cloud due to privacy concerns, emphasising the need for transparent data handling (Haase et al., 2021). Given the cultural and personal value of retirees' stories, this ethical dimension is essential, supporting inclusive design frameworks that embed empowerment and transparency at the core of their work. A key ethical consideration is digital ageism, biases in AI systems that inadvertently marginalise older adults due to design decisions rooted in assumptions about younger, digitally fluent users (Chu et al., 2022), and argue that most AI applications fail to address age-related cognitive and interactional diversity, leading to subtle forms of exclusion that compromise usability and trust. From a design ethics perspective, empirical evidence consistently demonstrates that developers must actively counteract such bias by adopting age-inclusive practices at both interface and data levels. This is not merely a usability issue, but an ethical imperative to prevent systemic exclusion and reinforce equity in human-AI collaboration.

However, (Zhao et al., 2024) emphasise older adults' desire for long-term integration of memory tools into daily life. Meanwhile, (Zhu et al., 2024b) highlight the challenge of balancing therapeutic intent

with cognitive burden in users with mild impairment. This contrast reveals a design tension between aspirational agency and pragmatic usability, particularly when cognitive decline is present. The preceding review demonstrates that while robust research exists in three separate pillars such as digital storytelling, AI companionship, and digital curation, no single existing approach fully integrates the necessary automation, accessibility, and archival integrity required for the Memtale project. This Table synthesizes the major approaches reviewed, highlighting the technological and ethical trade-offs across the current landscape.

Approach Type	Core AI Method	Primary User Outcomes	Critical Gap
Digital Storytelling Tools (e.g., JournalAIde (Zhou et al., 2025))	Manual Input; Text-based NLP (Minimal)	Generativity; Emotional Expression (Short-term)	High Cognitive Burden; Lacks Automation/Archival Integrity
Empathetic AI Companions (e.g., MyChum (Rao et al., 2024))	Affective Conversational AI (Emotional)	Loneliness Reduction; Psychological Support	Prioritizes Live Chat; Lacks Structured Semantic Preservation
Digital Curation Frameworks (e.g., archival frameworks (Rhee, 2024))	Structured Curation Lifecycle (Preservation/Metadata)	Long-term Legacy; Archival Integrity	Passive Model; Lacks Affective/Conversational Input
Multimodal Prototypes (e.g., MemPal (Maniar et al., 2025))	Contextual Sensing; Semantic Retrieval (Hybrid)	Contextual Recall; Activity Logging	Data Overload Risk; Limited Accessibility for Low-Tech Users

Table 4:Comparison of Major Approaches

2.4.4. Conceptual Synthesis, Gaps, and Research Trajectory

Despite advancements in their respective fields, a comprehensive combination of digital storytelling, AI companionship, and digital curation is still unattainable. Multimodal inputs that include text,

speech, and images; automated narrative summarisation; and semantic retrieval that is in line with the various demands of older adults are frequently absent from tools (X. Wang et al., 2024; Yan et al., 2024). This fragmentation leads to a clear synthesis of the research imperative, reframing the relationship between AI and memory preservation.

2.4.4.1. Conceptual Synthesis: AI as the Co-Archivist of Human Memory

The collective evidence suggests a paradigm shift: AI must transition from a mere digital companion (focused on transient conversational engagement and emotional support) to a Co-Archivist of Human Memory. This conceptual reframing synthesises the necessity for advanced technical functions with the core therapeutic goal of the project.

- **Empathy and Curation:** The review highlights that while empathetic AI agents improve psychological well-being (Khamaj, 2025), they consistently prioritise conversational engagement over systematic preservation (Anand et al., 2025). The Co-Archivist model integrates affective intelligence with a structured digital curation lifecycle (Rhee, 2024) to convert ephemeral personal conversations into long-lasting, ethically curated digital legacies (Colavizza et al., 2022; Loveys et al., 2022).
- **Accessibility and Generativity:** Furthermore, the Co-Archivist must support generativity, or the desire to pass down knowledge, which is a key psychological pillar of healthy ageing (Lee et al., 2025). By automating summarisation and search, the system shifts the cognitive burden of memory organisation from the retiree to AI, transforming the user from a manual journaler into a focused storyteller.

Consequently, the AI as Co-Archivist model benefits from an explicit, auditable affect channel. Semantic-Emotional Tagging Layer formalises this channel by inserting an emotional classification step between summarisation and storage, thereby aligning curation with older adults' reminiscence cues and enabling affect-conditioned retrieval (Lee and Allen, 2025; Lee et al., 2025; X. Wang et al., 2024).

2.4.4.2. Identification of Integrative Gaps

1. **Emotion Retrieval Gap:** Most narrative retrieval systems still prioritise lexical/semantic similarity, producing emotionally mismatched results when users search by affective intent.

2. **Emotion-Loss in Summarisation Gap:** Summarisation is widely used to index long narratives, but emotional tone can be flattened, weakening later emotion-intent retrieval.
3. **Interpretability Gap:** Many emotion-aware neural approaches are difficult to audit. Memory archives, especially in elder contexts, benefit from explainable signals that can be inspected and justified.
4. **Reproducibility Gap:** Emotion-aware retrieval is often evaluated with non-reusable setups. There is a need for pipelines with frozen datasets, explicit judgments, and standard IR metrics, plus a dedicated emotional alignment metric.

2.4.4.3. Synthesis and Implications for This Project

The reviewed literature indicates a consistent fragmentation: (i) digital storytelling tools support reminiscence but often require manual curation (Rios Rincon *et al.*, 2022; Stargatt *et al.*, 2022; Xu *et al.*, 2023), (ii) empathetic companions improve interaction quality but rarely create durable, searchable memory objects (Rashkin *et al.*, 2019; Alotaibi and Alshahre, 2024; Kim and Kim, 2024), and (iii) digital curation frameworks emphasise preservation but lack automated summarisation and retrieval functions (Colavizza *et al.*, 2022; Trehan, 2023). For the implemented scope of this project, the key implication is that a life-experience archive requires a retrieval pipeline that can support natural-language access and emotion-intent access patterns. Accordingly, this project adopts a two-stage retrieval design, BM25 candidate generation followed by reranking, so that robust lexical matching is preserved while higher-level signals (semantic similarity and emotional alignment) can be introduced explicitly during reranking. This design is consistent with prior work showing that two-stage pipelines are effective for controlling ranking behavior without requiring end-to-end model training from scratch.

2.5. Technological Review

2.5.1. Natural Language Processing for Summarisation and Tagging

Natural Language Processing (NLP) techniques are pivotal for transforming raw, often lengthy, retiree narratives into concise, searchable, and structured content. The technical domain is characterised by two primary summarisation approaches: extractive summarisation, which selects key sentences directly from the text; and abstractive summarisation, which utilises transformer architectures to paraphrase and generate novel, coherent text (Hou et al., 2021). Long document summarisation has

been significantly improved by recent developments in deep learning, especially with transformer-based models (e.g., BERT, T5) and frameworks like Hugging Face. These encoder-decoder architectures demonstrably outperform earlier statistical models in achieving greater semantic richness and factual consistency, offering the technical capability to accurately summarise life stories with contextual depth (Hou et al., 2021). Furthermore, empirical surveys have identified appropriate evaluation metrics, such as ROUGE and BERTScore, which can be precisely applied when assessing the accuracy of summarisation in memory archiving systems (Koh et al., 2023). This progress highlights a clear technical feasibility for automated narrative structuring.

However, this technical capability stands in stark contrast to its implementation in the ageing domain. While studies have demonstrated that advanced natural language processing (NLP) may automatically recognise reminiscence cues and segment conversations to maintain emotional depth (Ferrario et al., 2020), this sophistication is rarely incorporated into easily accessible tools. Studies of digital storytelling platforms consistently note that tagging and categorisation remain manual or user-driven (Zhu et al., 2024b), which directly imposes a high cognitive burden on older adults. This omission is critical, as automated tagging via techniques like entity recognition and topic modelling is essential for enabling the semantic retrieval needed by your project. This disconnect underscores a significant applied usability gap. The current ecosystem still prefers simple, low-automation solutions, although recent work has demonstrated significant improvements in fine-tuned NLP models for handling the complexity of spoken input (Usmaan and Singh, 2025) and coherently analysing lengthy narrative transcripts (Jayatilleke et al., 2024). Consequently, integrating advanced, emotion-aware NLP summarisation and tagging into the proposed system is essential to bridge the gap between technical feasibility and the urgent need for structured, searchable, and emotionally aware digital legacies. In this project, summaries are treated as the primary searchable representation for narratives: summaries are precomputed and cached for all stories, and retrieval is performed over these indexed summaries to reflect a realistic archive search where long narratives must be compressed into searchable units. The key technical requirement is therefore not only fluency but preserving affective cues that support later emotion-intent retrieval.

2.5.1.1. Semantic-Emotional Tagging Layer (SETL)

SETL-2 is used in this project as an interpretable representation of emotion for each narrative summary. Rather than training a neural emotion classifier, SETL-2 combines (i) lexicon-based

emotion cues from the NRC Emotion Lexicon across eight categories (joy, sadness, anger, fear, trust, disgust, surprise, anticipation) (Mohammad and Turney, 2013), and (ii) a lightweight polarity-based adjustment using VADER sentiment to emphasise positive or negative affect where appropriate (Hutto and Gilbert, 2014). The output is an 8-dimensional, fixed-length emotion vector in a deterministic order, enabling cosine similarity between the query and document emotion vectors during reranking. In the implemented pipeline, document-side emotion vectors are computed from the cached aware summaries. For queries, the system uses the query's target emotion label as a one-hot vector only when the label exactly matches one of the eight SETL-2 categories; otherwise, the query emotion vector is computed directly from the query text using SETL-2. This design is transparent, training-free, reproducible, and allows emotional alignment to act as an explicit reranking signal.

2.5.2. Speech-to-Text and Multimodal Interfaces

While multimodal capture (speech, images) is strongly motivated in the retiree domain, the current prototype focuses on the text retrieval core (summarisation + indexing + retrieval + evaluation). Multimodal input and retina-facing UI are treated as extensions for later stages. For accessible life-experience archiving, speech-based input becomes essential for retirees who have diminished dexterity or typing abilities. However, conventional Automatic Speech Recognition (ASR) systems encounter significant challenges, including age-related vocal changes, variable speech paces, and disfluencies, leading to common recognition errors. Recent research efforts have demonstrated a clear approach to overcome this technical barrier: modern ASR models may significantly reduce recognition errors and enhance transcription accuracy when tested on older adult speech corpora (such as the DementiaBank Pitt Corpus)(Usmaan and Singh, 2025). This technical necessity must be governed by an appropriate user experience framework. A critical framework for speech-based Intelligent Personal Assistants (IPAs) for older adults emphasises design principles of contextual understanding, error tolerance, and simplicity to accommodate age-related cognitive and sensory declines (Islam and Chaudhry, 2023). Furthermore, this technical advance is augmented by the ability of NLP to detect reminiscence cues within conversational speech (Ferrario et al., 2020), which is a crucial step for guiding the downstream summarisation and tagging modules to capture meaningful memories, rather than just transcribing raw text.

Conversely, the design of the user interface for these systems is often hampered by digital ageism (Chu et al., 2022). An inclusive design framework is essential because mainstream artificial

intelligence technologies usually overlook the wide range of sensory, cognitive, and social differences inherent in ageing populations. This underscores the importance of multimodal interfaces that strategically combine speech, text, and visual elements to reduce cognitive load (Zhou et al., 2024). Simple, consistent interface crucial with 65% dropout rate for complex UI, while empathetic responses increase engagement by 34% (Barbosa Neves et al., 2019; P et al., 2025). Empirical studies demonstrate that retirees explicitly prioritise technologies that allow them to sustain meaningful activities and maintain self-continuity (Zhao et al., 2024). Combining visual aids (photos, large fonts, high contrast) with conversational input (speech) not only improves engagement but also advances the therapeutic objective of reminiscence (X. Wang et al., 2024). However, developing robust ASR and curation pipelines which are effective for a wide variety of older voices while also designing interfaces that fully accommodate all sensory and literacy differences is a crucial challenge that persists despite advancements in ASR accuracy and the conceptualisation of inclusive design (Yan et al., 2024). Consequently, the integration of an ASR system is not merely a technical necessity but also a fundamental accessibility requirement, necessitating a design that is both simple to grasp without fear or frustration and robust enough to manage the distinct linguistic patterns of older adults.

2.5.3. Dataset And Preprocessing

2.5.3.1. Data Sources and Acquisition.

Because collecting private retiree life stories is ethically sensitive and time-intensive, the prototype is evaluated using a publicly available narrative dataset from Kaggle as Empathetic Dialogues (Facebook AI) 25k, which resembles autobiographical expressions and includes emotion labels (Rashkin et al., 2019). The implemented pipeline uses a fixed corpus of 5,000 autobiographical-style narratives, stored as empathetic_stories.csv. Each story has a unique story_id and an associated emotion_label, enabling both relevance evaluation and emotional alignment evaluation. The project precomputes and caches summaries for all stories in summaries.csv, which are then indexed and used as the document representation during retrieval.

2.5.3.2. Preprocessing and indexing approach.

Preprocessing is intentionally lightweight to preserve reproducibility and interpretability: texts are normalised (case-folding and tokenisation) for BM25 indexing, while semantic embeddings are computed for each cached summary using Sentence-BERT. SETL-2 emotion vectors are also

computed for each cached summary and stored implicitly via deterministic computation, enabling emotion similarity during reranking. The dataset, cached summaries, and evaluation artefacts (queries and qrels) are frozen so that results can be reproduced without re-annotation.

2.5.4. Retrieval and Semantic Search

Effective retrieval mechanisms are essential for enabling retirees to access their archived life stories using natural language queries, a key accessibility feature. The technical landscape offers two principal approaches: traditional keyword-based search and advanced embedding-based or semantic search. Leveraging vector representations of text, semantic search significantly improves traditional methods by obtaining semantically related content even in the absence of precise keywords (Colavizza et al., 2022; Trehan, 2023). Frameworks that use Sentence-BERT and vector databases (like Pinecone and Qdrant) provide reliable technical solutions for contextual memory recall by enabling quick and scalable similarity searches across enormous archives (Liang et al., 2025). The potential of semantic retrieval is underscored by research prototypes like MemPal (Maniar et al., 2025), which integrates voice recordings with visual context to create automated diaries that support voice-based object retrieval and recall of past activities. Similarly, deep learning may revolutionise access to personal memory repositories by enhancing contextual understanding and retrieval relevance, as evidenced by work in AI-powered archives (Trehan, 2023).

However, despite these demonstrable advances, a critical gap persists in the applied domain. Most existing elder-storytelling tools continue to rely on manual browsing or rigid keyword matching. Consequently, few systems provide the necessary concept-level retrieval that interprets a retiree's semantic intent (Alotaibi and Alshahre, 2024; Colavizza et al., 2022). For instance, a retiree querying for "early life experiences" would retrieve nothing from a traditional system without those exact words, while a semantic model could understand and surface stories tagged with 'childhood,' 'Colombo,' or 'first job' through vector similarity. This omission of semantic depth limits the system's ability to genuinely support natural, conversational memory access.

Furthermore, the deployment of AI-powered retrieval necessitates critical attention to design ethics and accessibility. Studies on digital ageism emphasise that retrieval systems must prioritise transparency and explainability to ensure equitable participation and build trust among older adults (Chu et al., 2022; Zhao et al., 2024). This underscores that implementing advanced retrieval is not

merely a technical task, but an ethical one that requires careful consideration of the user's cognitive and emotional comfort.

2.6. Benchmarking And Evaluation

2.6.1. Evaluation Metrics for the Implemented Retrieval Framework

Ranking effectiveness (early-rank IR quality).

To measure retrieval quality, the project reports standard early-rank information retrieval metrics: nDCG@5 (graded relevance) and MRR (rank sensitivity to the first relevant result). These metrics reflect memory search scenarios where users typically only inspect the first few results (Zhang et al., 2023).

Emotional alignment metric (emotion-intent success).

Because the project targets emotion-intent retrieval, ranking relevance alone is insufficient (Yash *et al.*, 2025). Therefore, this project reports EmoFidelity@K, defined as the proportion of the topK retrieved narratives whose emotion label matches the query's target emotion. This metric directly captures whether early-ranked outputs align with the user's affective intent, which is the key interaction scenario in memory retrieval (He *et al.*, 2023; Yash *et al.*, 2025).

Statistical significance.

To test whether observed gains are reliable, paired Wilcoxon signed-rank tests are applied over per-query scores to compare the proposed method against baselines, reporting test statistics and effect size (Parapar et al., 2020).

2.6.2. Comparative Baselines and Implemented Retrieval Pipeline

The prototype compares the proposed emotion-aware method against several standard retrieval baselines.:

1. BM25 - lexical retrieval baseline used for first-stage candidate generation.
2. SBERT - dense semantic retrieval baseline using Sentence-BERT cosine similarity over cached summaries.
3. RRF - hybrid baseline using Reciprocal Rank Fusion over BM25 and SBERT rankings.

4. CE-Rerank - two-stage baseline where BM25 retrieves topK candidates (K=100), which are then reranked using a pretrained cross-encoder model (cross-encoder/ms-marco-MiniLM-L-6-v2). This baseline is included to provide a strong neural reranking comparison point.
5. OURS (proposed) as a two-stage architecture: BM25 candidate generation (topK=100) → reranking using a weighted fusion of BM25, semantic similarity (SBERT), and emotion alignment (SETL-2 cosine similarity). To ensure interpretability, the fusion uses fixed rule-based weights determined by query type (emotion-intent queries give a higher weight to emotion alignment). This preserves strong lexical retrieval while correcting emotionally mismatched rankings at the reranking stage.

2.6.3. Reproducibility and Evaluation Artefacts

All evaluation artefacts are generated and stored in fixed files: queries.csv (including query type and target emotion), qrels.csv (manually graded judgments 2/1/0), and run files for each system. The pipeline is designed to run end-to-end using a single-entry point, producing summary tables, significance results, and qualitative examples automatically.

2.7. Chapter Summary

This chapter reviewed literature on digital storytelling for ageing populations, digital exclusion and accessibility barriers, and the growing role of AI in narrative interaction. It highlighted that while many systems support storytelling or companionship, few provide robust, searchable memory access aligned to how people naturally recall memories through emotional intent. The review also identified that summarisation is necessary for indexing long narratives but can unintentionally flatten emotional cues, degrading emotion-intent retrieval. These gaps motivate the implemented project focus: an interpretable two-stage retrieval framework that indexes narratives via cached summaries and performs reranking using combined lexical relevance, semantic similarity, and explicit emotion alignment (SETL-2). The chapter also justified the dataset choice and the evaluation strategy using standard early-rank IR metrics (nDCG@5, MRR), an emotional alignment metric (EmoFidelity@5), and paired significance testing to ensure results are reliable and reproducible.

CHAPTER 03: METHODOLOGY

3.1. Chapter Overview

This chapter explains the methodology used to conduct and validate the project under the refined scope: emotionally faithful summarisation for indexing and emotion-aware reranking for autobiographical-style narrative retrieval. It first describes the overall research methodology using Saunders' Research Onion, clarifying the philosophical stance, research approach, and data strategy used to support reproducible experimentation. It then presents the software development methodology used to implement the prototype as an iterative research artefact, followed by the requirement elicitation approach used to refine functional and non-functional needs (e.g., interpretability, reproducibility, and early-rank effectiveness). Finally, the chapter outlines the testing and evaluation methodology, including baseline comparisons and significance testing, and concludes with project management elements such as schedule, resources, and risk mitigation.

3.2. Research Methodology (Saunders' Research Onion in Table Form)

To organise the research methodology systematically, this study adopts the Research Onion model proposed by Saunders, Lewis, and Thornhill (2007). The refined project scope focuses on experimental evaluation of an emotion-aware retrieval framework using a fixed public dataset and reproducible evaluation artefacts. Because the work is both system-building and empirical evaluation, the methodology prioritises measurable retrieval effectiveness while also supporting interpretability through explainable emotion signals.

Layer	Choice	Justification
Philosophy	**Pragmatism**	The project combines engineering design (building a retrieval pipeline) with empirical evaluation (quantitative IR metrics + qualitative error analysis), choosing methods based on what best answers the research questions.
Approach	**Deductive (theory testing)**	The project tests the claim that explicit emotion alignment improves early-rank retrieval for

		emotion-intent queries, compared against lexical/semantic baselines.
Methodological Choice	**Mono-method (Quantitative) with supporting qualitative analysis**	The core evidence is quantitative (nDCG@5, MRR, EmoFidelity@5, significance testing). A small qualitative component (worked example/error analysis) supports the interpretation of results without being treated as a separate user study.
Strategy	● **Experimentation** ● **Archival Research**	Archival research informs the design from prior work; experimentation evaluates retrieval pipelines under controlled conditions using fixed datasets and evaluation artefacts.
Time Horizon	**Cross-sectional**	The study evaluates system performance on a fixed corpus at a single point in time (no longitudinal deployment).
Data Collection	● **Secondary Dataset (public datasets),** ● **Primary judgment (manual qrels)**	Narratives come from a public dataset; relevance labels are produced via manual grading to enable nDCG and significance testing under a reproducible setup.

Table 5:Research Methodology.

3.3. Development Methodology

The systematic approach and procedures used to design, develop, and deliver software systems are referred to as development methodology. Methodologies that support flexibility, iterative improvement, and evolving requirements are preferred over rigid traditional models like Waterfall in research-oriented software engineering projects. That's because exploratory and evolving research activity often renders the Waterfall unsuitable. To be able to accommodate continuous feedback from supervisors and iterative prototype evaluation and rapidly validate the system's core features, this project will employ the Iterative Prototyping methodology. This model, which inherently uses

prototyping combined with Agile practices, ensures that the system is usable, accessible, and aligned with the objectives of preserving life experiences through AI-enabled summarisation and retrieval. This methodology emphasises short build-evaluate cycles using fixed datasets and stored evaluation artefacts (queries, qrels, run files), allowing frequent adjustment of pipeline components (summarisation, lexical retrieval, semantic retrieval, emotion alignment, and fusion weighting) while maintaining reproducibility of results.

3.3.1. Requirement Elicitation Methodology

System requirements were gathered and refined using multiple complementary techniques:

Method	**Purpose**
Supervisor feedback / expert consultation	Validate that the revised scope is research-contribution focused (emotion-aware reranking) rather than a full product build.
Surveys	Questionnaires collected broader user feedback on usability preferences and interaction barriers.
Document Analysis:	Identify gaps and common practices in narrative retrieval, emotion-aware IR, and summarisation for indexing; refine scope and evaluation expectations.
Brainstorming Sessions	Informal ideation with academic supervisors and peer researchers helped refine system features and interface concepts.
Self-evaluation	Debug pipeline failures and revise design decisions based on observed error patterns (emotion mismatch, semantic drift, over-weighting of emotion).
Observation	User interaction with early prototypes was observed to assess difficulties and guide improvements.
Prototype-driven elicitation	Use intermediate outputs (summaries, emotion vectors, reranked lists) to refine requirements for interpretability and auditability.

Table 6:Requirement Elicitation Methodology

3.3.2. Design Methodology

The design methodology integrates the Object-Oriented Analysis and Design Methodology (OOADM) to model system core entities, including users, stories, tags, and metadata. This paradigm supports modularity, scalability, and maintainability of the prototype, allowing easy extension or replacement of components like ASR or NLP modules. Class diagrams and interaction diagrams were utilised to define system structure and workflows from input capturing to database storage and retrieval.

3.3.3. Programming Paradigm

The primary programming paradigm adopted is Object-Oriented Programming (OOP), facilitating a modular organisation of system components in Python. Python was chosen for its robust ecosystem supporting AI and NLP integrations, with libraries like Hugging Face Transformers and API clients for OpenAI models. Procedural elements complement OOP for straightforward data preprocessing and pipeline coordination.

3.3.4. Testing Methodology

Testing was conducted at two levels:

1. Component-level testing
    - Summarisation testing: verify summaries are generated for all narratives, cached, and stable across runs; spot-check for failures such as empty outputs or truncation.
    - Emotion extraction testing (SETL-2): verify emotion vectors are produced deterministically, and cosine similarity behaves as expected (higher similarity for same-emotion narratives).
    - Embedding testing (SBERT): verify that embedding exists for all indexed summaries and similarity scoring returns valid ranks.
2. System-level (pipeline) testing
    - Retrieval correctness: ensure BM25 candidate generation produces topK candidates and reranking reorders only within the candidate set.
    - Reproducibility testing: confirm that the same inputs (dataset + cached summaries + queries + qrels) reproduce identical metric outputs.
    - Evaluation reliability: compute early-rank metrics and run paired significance testing to confirm whether improvements are consistent across queries.

- Baseline testing includes CE-Rerank: BM25 TopK reranked using CrossEncoder scores.

3.3.5. Solution Methodology

The implemented solution follows a reproducible two-stage retrieval architecture designed to support emotion-intent memory access while remaining interpretable and lightweight.

1. Corpus preparation and narrative indexing

A fixed corpus of autobiographical-style narratives is selected from a public dataset. Each narrative is assigned a unique identifier and stored in a structured file format for repeatable experiments. Because long narratives are expensive to search directly and may include irrelevant details, the system uses summaries as the indexed representation.

2. Emotionally faithful summarisation for indexing

All narratives are summarised offline and cached. Caching ensures that retrieval experiments are not affected by variance in runtime summarisation and that every system variant operates on the same document representation. Summaries are stored in a dedicated file (e.g., summaries.csv) and act as the "documents" used by BM25 and SBERT.

3. Candidate generation with lexical retrieval (BM25)

Given a user query, BM25 retrieves a fixed-size candidate set (TopK). This stage is designed for strong recall and robustness, ensuring that reranking operates on a relevant shortlist while keeping computation efficient.

4. Semantic similarity scoring (SBERT)

For reranking, the system computes dense semantic similarity between the query and each candidate summary using Sentence-BERT embeddings. This supports concept-level matching beyond lexical overlap.

5. Interpretable emotion alignment (SETL-2)

Each candidate summary is mapped to an interpretable emotion representation using a deterministic method (SETL-2). SETL-2 combines signals from sentiment and emotion lexicons to produce an

emotion vector. Emotion alignment between query and candidate is computed using cosine similarity, allowing emotion to function as an explicit ranking feature rather than hidden metadata.

6. Fusion-based reranking with adaptive weighting

Final reranking combines three interpretable signals over BM25 TopK candidates: BM25 score (min-max normalised), SBERT semantic similarity, and SETL-2 emotion cosine similarity. The project uses fixed fusion weights for the main reranking method and additionally performs a weight-sensitivity analysis by generating multiple run files with different emotion weights (wemo ∈ {0.00, 0.10, 0.20, 0.30, 0.40, 0.50}). The final configuration reported is wemo = 0.20, selected based on best overall retrieval effectiveness.

7. Evaluation and reporting

The system evaluates retrieval effectiveness using early-rank metrics and an emotion-alignment metric and applies paired statistical testing to validate whether gains are consistent across queries. All evaluation artefacts (queries, qrels, run outputs, metric summaries) are stored for repeatability and reporting.

8. Prototype Workflow
    - Dataset → Cached summaries → BM25 TopK → SBERT + SETL-2 scoring → Fusion rerank (fixed weights + wemo sensitivity) → Metrics + significance + qualitative example.

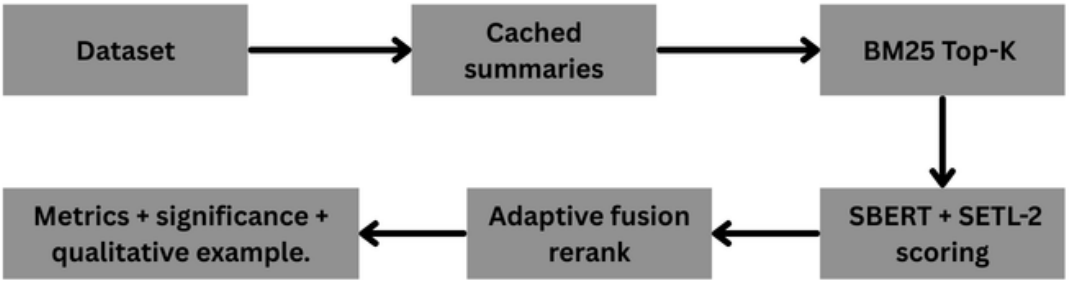


Figure 1:Research workflow diagram (Self-composed)

3.4. Project Management Methodology

Effective project management is critical to ensure that the development and research activities are delivered on time, within scope, and with appropriate resources while managing associated risks. This

study adopts standard project management knowledge areas to systematically organize, monitor, and control the project lifecycle.

3.4.1. Schedule

The project schedule helps with the timely execution and monitoring of progress by outlining planned activities and milestones.

3.4.1.1. Gantt Chart

The Gantt chart can be viewed in ***Appendix B.1 - Gantt Chart.***

3.4.1.2. Deliverables

Deliverable	Target Date
Project Proposal – Initial Draft	10th October 2025
Project Proposal and Requirements Specification – Final Draft	24th October 2025
Project Proposal and Requirements Specification – Final	13th November 2025
Proof of Concept	14th December 2025
Interim Progress Demo (IPD)	05th February 2026
Implementation	16th Feb 2026
Testing	05th March 2026
Evaluation	10th March 2026
Thesis Submission	03rd April 2026
FYP Final Vivas	27th Apr - 22nd May 2026

Table 7:Deliverables

3.4.2. Resource Requirements

For a project to proceed smoothly, resources must be identified, acquired, and managed.

3.4.2.1. Data:

Public narrative dataset used for evaluation (EmpatheticDialogues subset), cached summaries, fixed query set, and manually graded qrels for reproducible IR evaluation.

3.4.2.2. Skills:

Academic writing skills, Problem-solving abilities, Python programming, Natural Language Processing, API integration, UI/UX design principles, qualitative user evaluation methods

3.4.3. Risk And Mitigation (Expanded with Severity and Frequency)

Project risks are identified, analysed, and mitigated to reduce their impact on objectives. Each risk is evaluated according to its frequency (probability) and severity (impact magnitude), and its product is risk exposure. The table below prioritises significant risks with assigned mitigation strategies.

Risk	Severity	Frequency	Mitigation Strategy
User Acceptance Issues due to Accessibility Barriers	5	4	Incorporate retirees into design feedback loops and conduct iterative usability testing.
Emotion-aware reranking harms topical relevance (over-weighting emotion)	4	4	Use candidate-stage BM25, constrain reranking to TopK, tune adaptive weights, and report both relevance metrics and emotion-alignment metrics to ensure balanced improvement.
Integration Challenges Between NLP APIs and UI	4	5	Modular system design allows independent component testing and clear API documentation usage.
Misclassification of emotional tone reduces trust	4	4	Expose raw scores, allow user override, and re-run SETL on paraphrases to check stability.

Data Privacy and Ethical Concerns Regarding Personal Stories	5	3	Ensure compliance with privacy regulations; anonymise data; obtain informed consent and securely store data.

Table 8:Risk and Mitigation

3.5. Chapter Summary

This chapter described the methodology used to design and evaluate the project under the refined scope of emotionally faithful summarisation and emotion-aware retrieval for autobiographical-style narratives. Using Saunders' Research Onion, the study was positioned as a pragmatic and experiment-driven investigation that tests whether explicit emotional alignment improves early-rank retrieval effectiveness compared to lexical and semantic baselines. The development methodology followed iterative prototyping, enabling rapid refinement of summarisation, retrieval, and reranking components while preserving reproducibility through cached outputs and fixed evaluation artefacts. The solution methodology detailed the implemented two-stage retrieval pipeline (BM25 candidate generation followed by fusion-based reranking using SBERT and SETL-2 emotional alignment), and the evaluation methodology established reliable measurement through standard IR metrics, an emotional alignment metric, and significance testing. Project management elements (schedule, resources, and risks) were aligned to the implemented prototype and the IPD expectations.

CHAPTER 04: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1. Chapter Overview

This chapter presents the Software Requirements Specification (SRS) for the Memtale prototype within its refined scope: emotion-aware summarisation for indexing and emotion-aware reranking for autobiographical narrative retrieval. Rather than a full consumer application, the prototype is designed as a reproducible research artefact that can be evaluated using information retrieval metrics and emotional alignment measures. The chapter covers the requirement elicitation methods, stakeholder analysis, system interaction diagrams, and the prioritised functional and non-functional requirements defined using the MoSCoW method. The requirements reflect the implemented prototype, while broader product features are reserved for future work.

4.2. Rich Picture Diagram

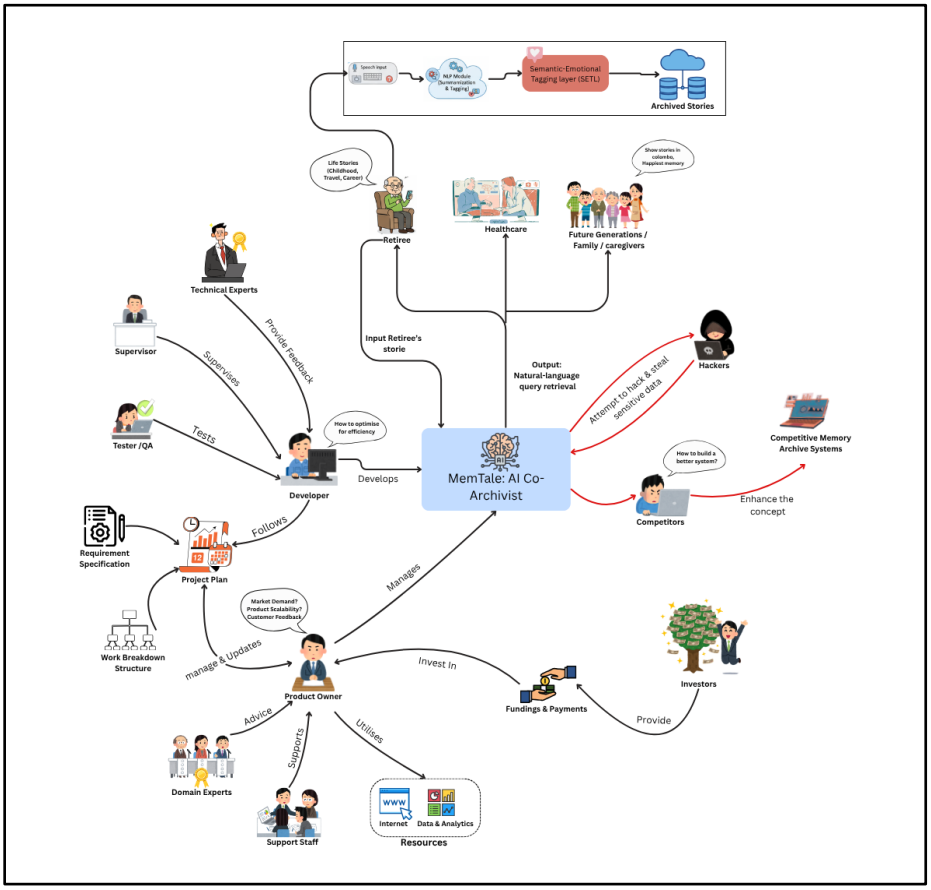


Figure 2:Rich Picture Diagram. (Self-composed)

The Rich Picture Diagram (Figure 4.1) illustrates the broader environment around the Memtale concept and how the prototype fits into it. Retirees and family/future generations represent the motivating user context (life-story preservation and intergenerational access), while developers and supervisors guide the research implementation and validation. The diagram also highlights key concerns such as emotional sensitivity, system trust, and security risks (e.g., unauthorised access). In the refined project scope, the implemented artefact focuses primarily on the NLP pipeline (summarisation + semantic-emotion tagging) and the retrieval evaluation loop, rather than full UI deployment.

4.3. Stakeholder Analysis

4.3.1. Stakeholder Onion Model

The Stakeholder Onion Model (Figure 4.2) categorises stakeholders by proximity to the system: direct operators, functional beneficiaries, and wider-environment stakeholders. Although the prototype is evaluated offline using public data, the model is retained to reflect the real-world retiree-oriented motivation and the long-term deployment context.



Figure 3:Stakeholder Onion Diagram. (Self-composed)

4.3.2. Stakeholder Viewpoints

Table 4.1 summarises stakeholder roles and how each interacts with or benefits from the system. In the refined scope, primary emphasis is placed on research stakeholders (supervisor, developer/researcher) and functional beneficiaries (retirees, family/future generations) as the motivation for emotion-intent retrieval.

Stakeholder	Role	Description
System Stakeholders		
Retirees	Normal Operator	Act as the main users of the Memtale: AI Co-Archivist System by recording their life stories via speech or text. They interact directly with the AI interface to store and retrieve memories. (Conceptual/future deployment)
System Administrator	Maintenance Operator	Manage and maintain the AI Co-Archivist infrastructure, guaranteeing uptime, data integrity, and user-access control, while enforcing ethical and privacy policies.
Containing System Stakeholders		
Retirees	Functional Beneficiary	Benefit from the system by preserving personal histories and reliving significant experiences, which enhances social connectedness and emotional fulfilment.
Family Members / Caregivers	Functional Beneficiary	Gain insights from retirees' recorded stories, which support empathy, caregiving, and intergenerational relationships while assisting retirees in using the platform. They benefit from easier access to carefully chosen memories.

Future Generations	Functional Beneficiary	Indirect users who access and learn from the archived life stories of retirees. They benefit by preserving family history, cultural values, and emotional connections across generations.
Healthcare Professionals	Functional Beneficiary	Utilise retirees' narratives for therapeutic or wellness purposes, supporting mental health and emotional well-being tracking.
Product Owner	System Owner / Operational Beneficiary	Oversees project planning, aligns technical outcomes with project goals, manages updates, and ensures ethical and functional objectives are met.
Wider Environment Stakeholders		
Developer	Operational maintainer and development staff	Designs, develops, and optimizes the AI models (speech-to-text, summarization, semantic search) and ensures secure, efficient operation of the Memtale platform.
Supervisor	Research advisor / quality regulator	Provides oversight and guidance to the project team, guarantees adherence to objectives, timelines, and quality standards
Technical Experts	Advisor	Offer expert consultation on system architecture, cloud integration, and AI optimization to support developers and testers.
Domain Experts	Advisor	Provide domain knowledge in gerontology, psychology, and narrative therapy to guarantee the system's social and emotional significance.
Tester / QA	Quality Regulator	Tests and validate system performance, ensuring accurate story summarization, data protection, and

		overall user-friendliness.
Investors	Financial Beneficiary	Provides financial and material resources for the project's development, expecting returns through project success and recognition of social impact.
Competitors	Negative Stakeholder	Competing memory-archiving or digital-storytelling platforms aim to develop comparable or enhanced solutions and, indirectly, to influence innovation and differentiation.
Hackers	Negative Stakeholder	Attempt to compromise the system's security and access sensitive user data; pose privacy and ethical threats that the system must mitigate through cybersecurity measures

Table 9:Stakeholder Viewpoints

4.4. Selection of Requirement Elicitation Methodologies

To demonstrate how requirements were derived, this project used 4 requirement elicitation methods. The following table (Table 4.2) summarises the reasons behind the author's selection of methods.

Method 01: Literature Review
Existing research on AI-based memory preservation, eldercare systems, and digital storytelling was examined through a literature review. Reviewing related studies and technology documentation helped identify common failures in narrative retrieval (topic-only relevance), emotion mismatch, and tone-loss in summarisation; derive requirements for emotion-aware reranking and emotionally faithful summaries. This method provided a foundational understanding of current systems, enabling the author to define clear requirements and propose improvements for Memtale.
Method 02: Survey
Surveys were distributed to retirees, healthcare professionals, researchers, and caregivers to gather

broad, quantifiable insights into usability expectations, accessibility needs, and privacy preferences. The collected data reflected end-users' readiness to adopt AI-driven applications and highlighted areas for interface simplification. Provide evidence that older-adult-focused systems require simplicity, trust, and transparency; treated as supporting context rather than the primary evaluation method for the prototype.
Method 03: Supervisor/expert consultation
Validate that the project emphasises measurable research contribution (metrics + reproducible artefacts) rather than a full product build. Used to confirm scope feasibility, contribution framing, and methodological expectations (baselines, metrics, significance testing).
Method 04: Brainstorming Sessions
Brainstorming sessions with academic supervisors, peer researchers, and technical experts supported collaborative idea generation and validation. This approach enabled it to be possible to the refinement of system features, prioritisation of user requirements, and the inclusion of innovative elements such as multimodal input and ethical AI integration. Refine requirements by analysing errors (emotion mismatch, semantic drift, over-weighting emotion), leading to constraints like reranking within Top K and adaptive weighting.

Table 10:Requirement Elicitation Methodologies

4.5. Discussion of Findings

For each requirement elicitation methodology, key findings are documented below.

4.5.1. Literature Review Findings

The following Table 4.3 summarises the Literature review findings.

Findings	**Citations**
Digital storytelling and reminiscence interventions can support older adults' well-being, social connection, and identity continuity, motivating tools that preserve and revisit personal narratives.	(Rios Rincon *et al.*, 2022; Stargatt *et al.*, 2022; Xu *et al.*, 2023; Lee *et al.*, 2025)

Accessibility is a key requirement in older-adult-focused systems; interfaces should reduce cognitive load and support simple interaction patterns.	(Kim and Kim, 2024; Yan *et al.*, 2024; Zhou *et al.*, 2025)
Lexicon-based emotion and sentiment resources (e.g., NRC Emotion Lexicon, VADER) provide transparent and reproducible affect signals suitable for interpretable tagging without training a new model.	(Mohammad and Turney, 2013; Hutto and Gilbert, 2014)
Narrative retrieval in memory settings often fails when relevance is defined only by topic, emotional intent matters for perceived usefulness. Summarisation is necessary for long narratives, but summaries can flatten emotional cues, harming emotion-intent retrieval. Explainability is important in memory archives; lightweight emotion signals are preferred over black-box emotion classifiers for auditability. Reproducible IR experiments require fixed corpora, cached representations, stored queries/qrels, standard metrics, and significant testing.	(Ahuir et al., 2024; Parapar et al., 2020; N. Wang et al., 2024; Yash et al., 2025)
Dense semantic retrieval (e.g., Sentence-BERT) supports concept-level matching beyond keyword overlaps and is commonly used as a baseline in modern retrieval pipelines.	(Reimers and Gurevych, 2019; Yash *et al.*, 2025)

Table 11: Literature Review Findings

4.5.2. Supervisor / Expert Consultation Findings

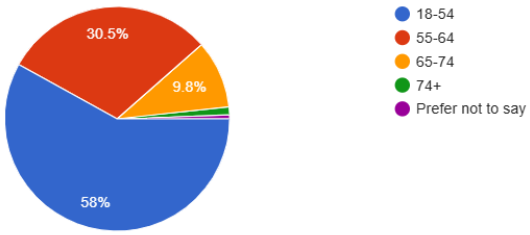
Supervisor/expert feedback was used to refine the scope and ensure the methodology matches research expectations:

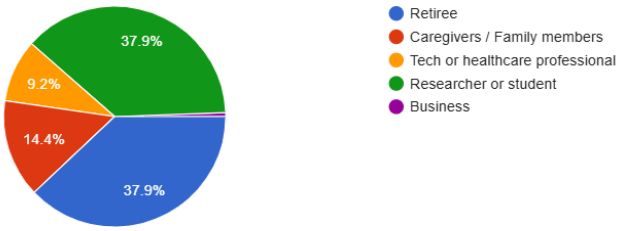
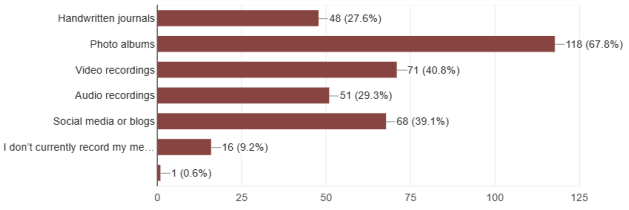
- The scope should emphasise a clear experimental claim (emotion alignment improves early-rank retrieval for emotion-intent queries) rather than building a full retiree application.
- The prototype should use strong baselines (BM25, SBERT, hybrid fusion) and show improvements using standard IR metrics plus an emotion alignment metric.
- The evaluation should be reproducible, requiring frozen artefacts (fixed dataset subset, cached summaries, queries, qrels, run outputs).

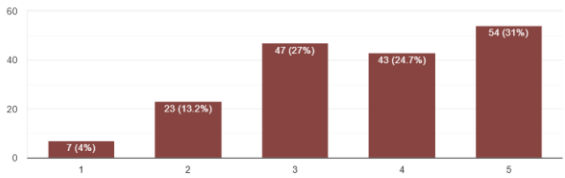
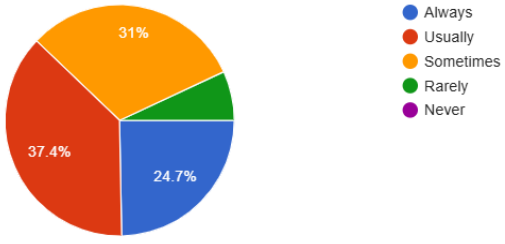
- Results should be supported with statistical significance testing and at least one qualitative reranking example to interpret behaviour.

4.5.3. Survey Findings

The survey results are retained as supporting evidence for usability, privacy sensitivity, and acceptance of AI summarisation/tagging in the retiree domain. However, in the refined scope these findings are treated as motivation for future UI deployment, while the implemented evaluation remains system-level and offline. This online survey was structured to capture system expectations from elderly users, family caregivers, healthcare professionals, and researchers. The questionnaire was sent to 200 participants, of which 174 responses were received (87% response rate).

Question	1. What is your age range?
Aim of Question	To understand the demographic composition of respondents and validate the elderly-focused design requirements.
Findings	 The majority of respondents (58%) are in the younger age bracket (18-54), reflecting tech-savvy family members, caregivers, and healthcare professionals. However, a significant proportion (40.4%) is aged 55+, representing the primary elderly target users. Though the tool is designed for retirees, interest from younger users suggests relevance for caregivers and early planners. These supports designing with multi-generational accessibility and varying digital literacy in mind.
Question	2. Which best describes your relation to life-experience archiving?
Aim of Question	To identify the respondent roles and understand which user groups the system needs to serve and prioritise in terms of features and usability.

Findings	 37.9% of respondents identified as retirees and an equal 37.9% as researchers or students, indicating strong interest from both user communities and knowledge domains. while 14.4% were caregivers/family members and 9.2% were professionals, representing the secondary stakeholder ecosystem supporting elderly users. This balanced distribution confirms the system's dual appeal: serving retirees directly and being of academic or caregiving interest to others. It supports developing user modes or tailored experiences based on stakeholder roles.
Question	3. How do you currently preserve your memories? (Please select all the options that apply.)
Aim of Question	To understand current memory preservation methods and identify gaps in existing approaches, informing feature prioritization for Memtale's multimodal capabilities (voice, text, photo integration).
Findings	 Photo-based preservation dominates (67.8%), reflecting the popularity of visual memory captured across generations. However, multimodal approaches are common, 40.8% use video, 39.1% use social media, and 29.3% use audio recording, indicating diverse preservation preferences. Notably, 9.2% currently don't record memories, underscoring users who may lack accessible tools. Traditional formats like handwritten journals (27.6%) still hold value, reinforcing the need for flexibility in content digitisation. Memtale must support multimodal input (voice

	recording, text, photo uploads) to accommodate diverse preservation preferences.
Question	4. How comfortable are you using digital devices (smartphones, computers, tablets) to record or share your personal stories?
Aim of Question	To assess digital literacy and comfort levels across the user base, informing accessibility and interface complexity requirements for Memtale.
Findings	 (1 = Not comfortable at all, 5 = Very comfortable). Positive skew toward comfort, 55.7% of respondents scored 4-5 (comfortable to very comfortable), indicating a majority digital readiness. However, 17.2% scored 1-2 (low comfort), representing a significant segment requiring accessibility support. The 27% neutral group suggests moderate digital confidence with potential for confidence improvement through simple, intuitive design. This highlights a diverse range of digital proficiency among users.
Question	5. How often do you share your life experiences or stories with family or friends?
Aim of Question	To understand the frequency of life-story sharing among users and determine the emotional and social readiness for adopting a memory archiving platform like Memtale.
Findings	 Strong propensity for sharing 93.1% of respondents share stories at least sometimes (combining Always, Usually, Sometimes), with 62.1% sharing regularly (Always + Usually). Zero

	respondents selected "Never," indicating universal openness to story sharing across the user base. The dominant "Usually" category (37.4%) suggests moderately active sharing behaviour, while the 31% "Sometimes" group represents occasional sharers who may benefit from Memtale's selective sharing features. Only 6.9% rarely share, representing a minimal isolated segment. The results support embedding features that facilitate easy sharing with family, reinforcing Memtale's role as both a personal memory tool and a bridge for intergenerational connection.

Table 12:Survey Findings

The remaining questions and their analysis can be found in ***APPENDIX C.1: SURVEY FINDINGS*** as an extension.

4.5.4. Brainstorming findings

Brainstorming sessions with the academic supervisor and research assistants refined system design, validated, and identified critical implementation considerations.

Criteria	**Findings**
Project Relevance and Significance	In brainstorming sessions with the supervisor and assistant supervisor, the project's relevance was reinforced by the growing demand for empathetic, AI-assisted life-story preservation among ageing populations. The concept of digital legacy was highlighted as both socially impactful and personally meaningful, aligning with UN ageing policies and elder mental health support frameworks.
Novelty and Innovation	The Co-Archivist conceptual model for elders and the team emphasised the uniqueness of combining emotion-aware tagging with semantic search in a personal archiving tool. Integrating speech-to-text with a semantic-emotional tagging layer (SETL) for memory enrichment was seen as a key innovation not addressed in most existing solutions.
Technical	Architecture is achievable within a timeline using proven technologies.

Feasibility	Supervisors advised modular implementation to manage scalability, processing load, and model updates. Keep retrieval lightweight and interpretable (two-stage pipeline, explicit scoring features). Prevent emotion signals from harming topical relevance (TopK constraint + adaptive fusion). Store artefacts to support reruns and reporting (queries/qrels/metrics summaries).
Ethical and Social Concerns	Discussions highlighted the importance of consent-driven sharing and privacy configurations, especially given the sensitive and emotional nature of stored memories. The supervisors reinforced the necessity of transparent data usage policies and minimal AI overreach.
Support for Future Expansion	Brainstorming led to identifying future extensions such as timeline reconstruction, generative story prompts, and integration with elder wellness platforms. These ideas will inform phased roadmap planning beyond the initial MVP.

Table 13: Brainstorming findings

4.6. Summary of Findings

Summary of the findings reflecting on the gatherings from the requirement elicitation phase is shown in Table 4.6 below

Findings	Literature Review	Supervisor/Experts	Brainstorming	Surveys
Emotion mismatch is a problem in narrative retrieval	✓	✓	✓	✓
Summaries must preserve affect for emotion-intent search	✓	✓	✓	✓
Reproducibility requires fixed artefacts (queries/qrels/runs)	✓	✓	✓	

Hybrid interfaces (touch + voice) improve usability	✓		✓	✓
Interpretability is important (auditable emotion signal)	✓	✓	✓	✓
Digital storytelling supports emotional well-being and legacy building	✓		✓	✓
Older adults value simplicity, trust, and privacy	✓			✓

Table 14:Summary of findings

4.7. Context Diagram

This context diagram (figure 4.3) illustrates the core interactions of the Memtale: AI Co-Archivist System, a digital memory preservation platform designed primarily for retirees. Retiree users input speech or text-based life stories, which the system stores, enriches, and retrieves upon request. Future generations or family members engage with the system by submitting natural language queries to access archived narratives and shared memories. The diagram highlights the system's role as an intelligent intermediary, preserving legacy stories and enabling intergenerational connection through accessible and personalised AI-driven storytelling.

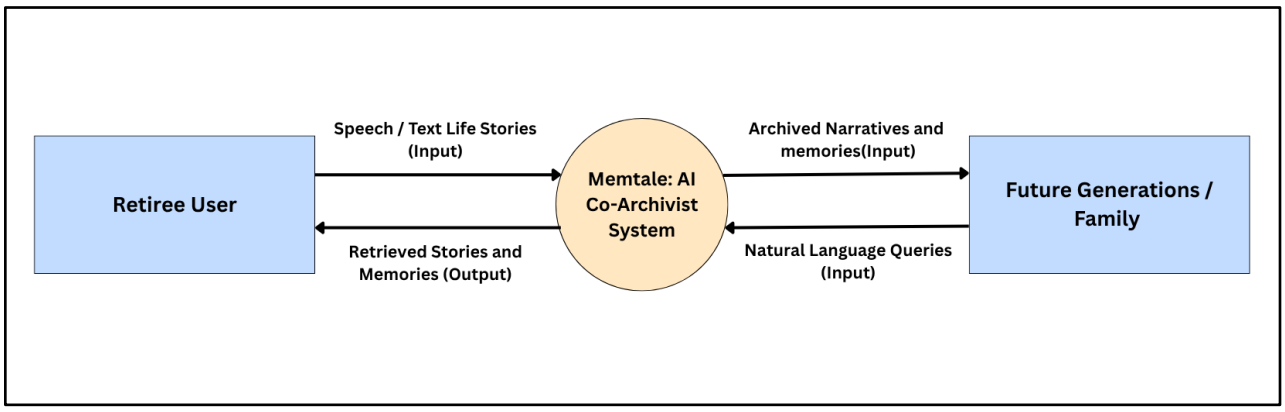


Figure 4:Context Diagram. (Self-composed)

4.8. Use Case Diagram

The use case diagram (Figure 4.4) presents high-level interactions for the broader Memtale vision. For the refined prototype, the implemented use cases are those related to index preparation, query retrieval, reranking, and evaluation reporting. Use cases such as “Manage user accounts” or “Privacy settings” are retained as future-facing system behaviours and are not part of the implemented artefact.

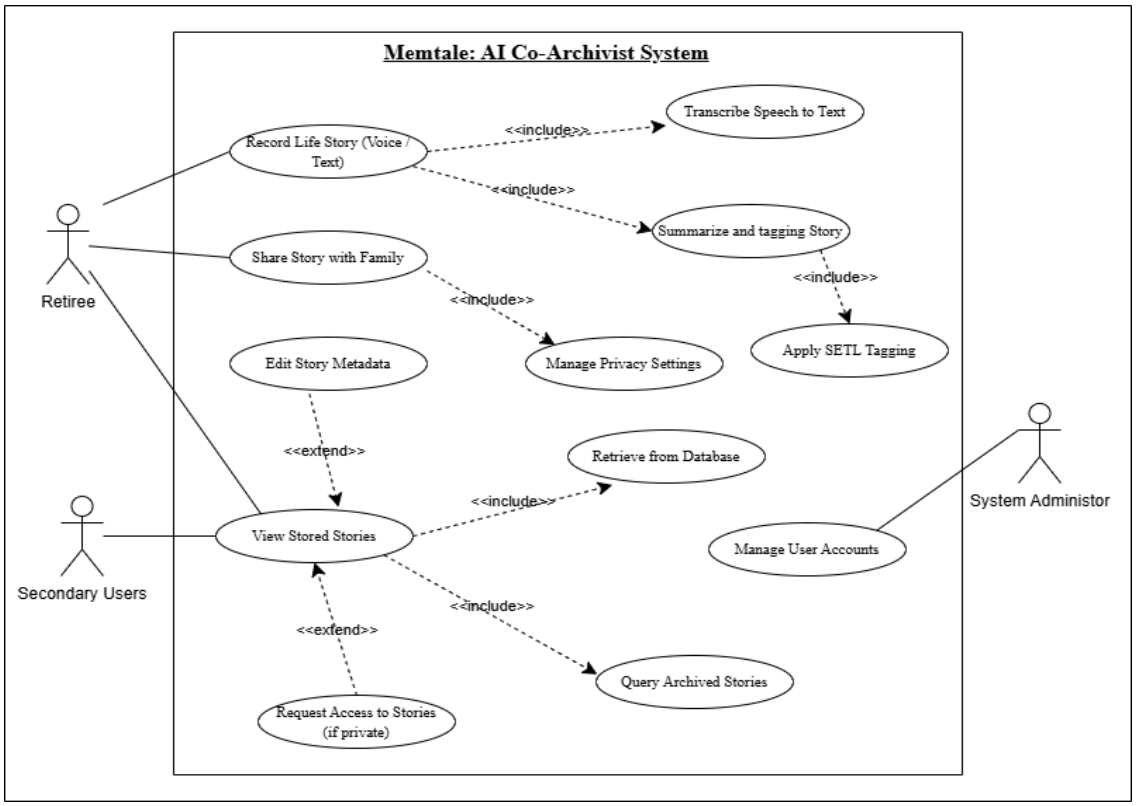


Figure 5:Use Case Diagram (Self-composed).

4.9. Use Case Descriptions

Only the main use case is described here; other use case descriptions can be found in ***Appendix C.3 - Use Case Descriptions.***

Use Case	**Record Life Story (Voice/Text)**
Use Case ID	UC01
Description	Allow retirees to input personal life stories via voice or text, initiating the

	memory archiving process.
Primary Actor	Retiree
Secondary Actor	None
Pre-Condition	User must be authenticated.
Extended Use Cases	None
Included Use Cases	Transcribe Speech to Text, Summarize and Tagging Story
Main Flow	1. Retiree launches the record function. 2. Retiree narrates or types a life story. 3. System captures input. 4. If voice, speech is transcribed to text. 5. The story is summarised and tagged with metadata.
Alternative Flows	None
Exceptional Flow	If input fails to record, the system shows an error and retries.
Post Conditions	Story saved and ready for privacy setup or sharing.

Table 15:Use Case Descriptions

4.10. Requirements

This section outlines the functional and non-functional requirements for the "Memtale: AI Co-Archivist System," gathered through user interviews, surveys, literature review, and brainstorming. The features are prioritised using the MoSCoW principle, which is expanded in ***APPENDIX C.4: MOSCOW PRINCIPLE***

4.10.1. Functional Requirements

The table below (Table 4.10) outlines the functional requirements for the Memtale system and prioritised using the MoSCoW method.

FR ID	Functional Requirement	Priority Level
FR1	The system must load a fixed narrative corpus and assign a unique ID to each narrative.	M
FR2	The system must generate and cache one summary per narrative (offline precomputation).	M
FR3	The system must have index-cached summaries for lexical retrieval using BM25.	M
FR4	The system must compute SBERT embeddings for each cached summary for semantic similarity scoring.	M
FR5	The system must compute SETL-2 emotion representations for each cached summary deterministically.	M
FR6	The system must retrieve TopK candidates using BM25 for a given query.	M
FR7	The system must rerank BM25-candidates using a fusion of BM25 score, SBERT similarity, and SETL-2 emotion alignment.	M
FR8	The system must support adaptive weighting (query-type aware) to avoid emotion overpowering topical relevance.	M
FR9	The system must evaluate runs using nDCG@5 and MRR on fixed qrels.	M
FR10	The system must compute an emotion-alignment metric (EmoFidelity@5) to evaluate emotion-intent success.	M
FR11	The system must perform paired significance testing over per-query results (e.g., Wilcoxon signed-rank).	M
FR12	The system should generate qualitative examples showing how reranking	S

	changes top results.	
FR13	The system should export all artefacts (summaries, queries, qrels, run files, metric tables) for reproducibility.	S
FR14	The system should provide a lightweight demo UI to inspect datasets, view evaluation summaries/figures, and compare ranked results across retrieval methods.	S
FR15	The system could include speech-to-text capture for retirees as an input option.	C
FR16	The system will not implement full privacy/sharing/account management in the current prototype.	W

Table 16:Functional Requirements

4.10.2. Non-Functional Requirements

The table below (Table 4.11) outlines the non-functional requirements, the quality and performance characteristics expected of the Memtale system.

NFR ID	Non-Functional Requirement	Description	Priority Level
NFR1	Reproducibility	The system must produce repeatable results using a fixed corpus subset, cached summaries, fixed queries/qrels, and stored run outputs.	M
NFR2	Interpretability	The reranking decision must be explainable using visible scoring components (BM25, SBERT similarity, and emotion cosine) rather than opaque neural-only decisions.	M
NFR3	Determinism of emotion signal	SETL-2 emotion computation must be deterministic for the same text input to support auditability and stable evaluation.	M

NFR4	Early rank effectiveness focus	The system must report early-rank metrics (e.g., nDCG@5, MRR) because users typically inspect only top results.	M
NFR5	Efficiency (prototype level)	Retrieval must be completed in practical time for experiments; reranking must be restricted to TopK candidates to limit compute.	S
NFR6	Modularity	Components (summariser, lexical retriever, embedder, emotion module, and reranker) should be modular to support replacement in future work.	S
NFR7	Privacy in deployment (future)	If deployed with real stories, the system should support consent and secure storage; not implemented in this prototype.	C

Table 17:Non-Functional Requirements

4.11. Chapter Summary

This chapter specified the software requirements for the Memtale prototype under the refined scope of emotionally faithful summarisation for indexing and emotion-aware reranking for narrative retrieval. Requirements were derived through literature-driven analysis, supervisor/expert consultation, and brainstorming, with survey findings retained as supporting motivation for the retiree domain. Stakeholders and system interactions were documented using visual models (Rich Picture, Onion Model, Context Diagram, Use Case Diagram). Functional requirements prioritised the implemented pipeline: cached summaries, BM25 candidate retrieval, SBERT semantic similarity, SETL-2 emotion alignment, adaptive fusion reranking, and reproducible evaluation using fixed artefacts, IR metrics, and significance testing. Non-functional requirements emphasised reproducibility, interpretability, and modularity as essential qualities for a research-grade retrieval prototype, while full UI, privacy workflows, and speech capture were documented as future extensions.

CHAPTER 05: Social, Legal, Ethical and Professional issues (SLEP)

5.1. Chapter Overview

This chapter identifies the main Social, Legal, Ethical and Professional (SLEP) issues relevant to the development and evaluation of the Memtale system, which performs emotion-aware retrieval over autobiographical-style narratives using a reproducible evaluation pipeline. The work follows the principles of the BCS Code of Conduct, particularly professional integrity, respect for privacy, responsible use of data, and transparent reporting of results and limitations.

5.2. Breakdown of Social, Legal, Ethical, and Professional Issues

Social	**Legal**
Emotion-intent retrieval can surface emotionally mismatched memories, reduce trust or cause discomfort in future deployment. Use interpretable SETL-2 emotion vectors and report both relevance (nDCG@5, MRR) and emotion alignment (EmoFidelity@5).	Data and software must be used lawfully. Only public research data is used (no real retiree stories). Third-party tools/models/datasets are used under their licenses and cited; future deployment would require consent and data-protection compliance (e.g., UK GDPR).
Ethical	**Professional**
Emotion inference is imperfect and must not be treated as a diagnosis. Emotion is used only as a ranking signal; limitations are stated clearly, and evaluation relies on manual qrels + significance testing.	Results must be reported honestly and reproducibly. Store and publish queries, qrels, run files, and metric summaries; report statistical tests and acknowledge trade-offs.

Table 18:Breakdown of social, legal, ethical, and professional issues

5.3. Chapter Summary

This chapter identifies the potential SLEP issues in the research and explains how they were addressed responsibly. The project adhered to the BCS Code of Conduct, UK GDPR, and accepted standards and practices, ensuring that integrity, transparency, reproducibility, and accountability were brought to the light throughout the Memtale research project lifecycle.

CHAPTER 06: DESIGN

6.1. Chapter Overview

This chapter presents the design of the Memtale system and explains how the architecture and modelling decisions support the requirements defined in the SRS chapter. The design focuses on a reproducible narrative retrieval pipeline that performs: (i) offline emotionally faithful summarisation for indexing, (ii) two-stage retrieval, and (iii) interpretable emotion alignment using SETL-2. The chapter first states the design goals aligned with the non-functional requirements, then introduces the high-level architecture, followed by detailed OOAD diagrams (class, sequence, and activity) and the flow-chart style algorithm design for the end-to-end retrieval process.

6.2. Design Goals

The design goals below are derived from the non-functional requirements identified in Chapter 04 and are reflected throughout the system architecture and implementation decisions.

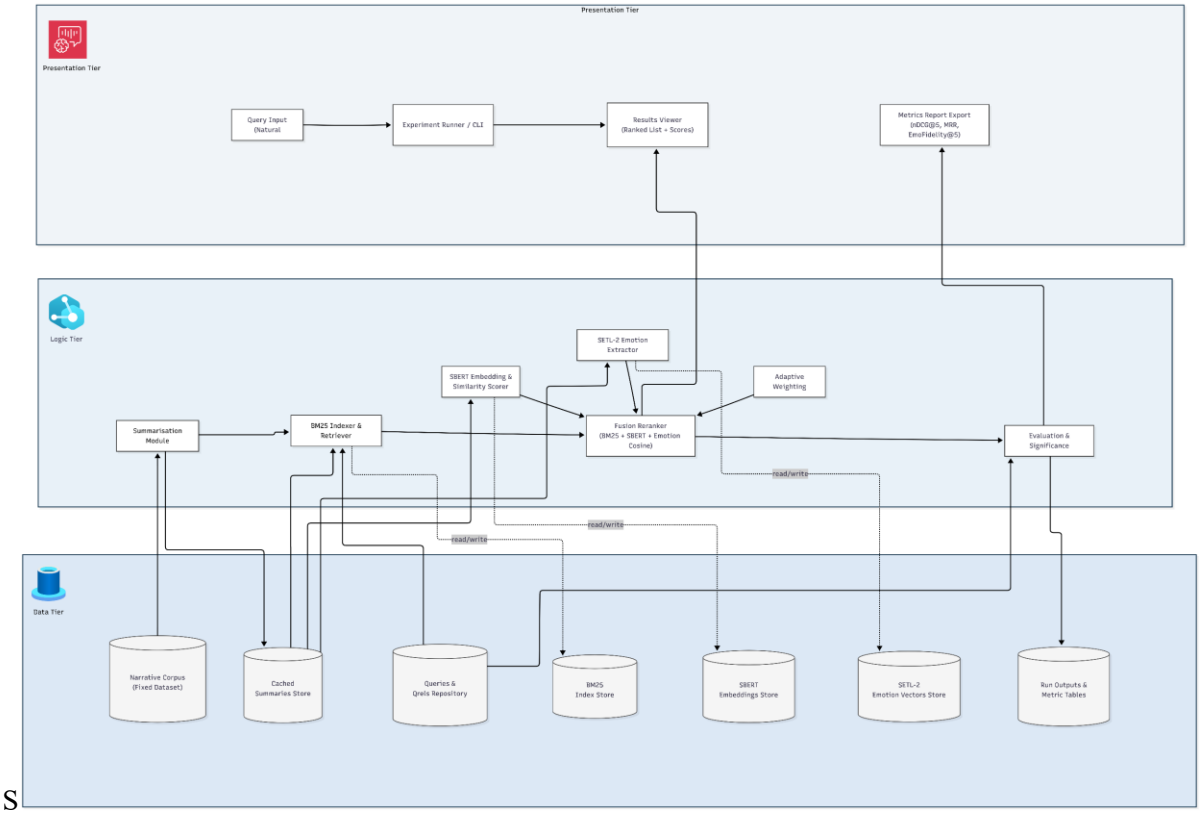
ID	Goal	Description
DG01	Reproducibility	Ensure repeatable experiments by using fixed inputs and stored artefacts (cached summaries, queries, qrels, run outputs, metric tables).
DG02	Interpretability	Ensure ranking decisions are explainable using visible components (BM25 score, SBERT similarity, SETL-2 emotion cosine) rather than opaque decisions.
DG03	Determinism	Ensure SETL-2 emotion computation is deterministic for the same text input to support stable evaluation and auditability.
DG04	Modularity	Separate summarisation, indexing, retrieval, reranking, and evaluation into modular components so each can be replaced or improved independently.

DG05	Efficiency	Keep experiments computationally practical by limiting reranking to BM25 Top K candidates and caching expensive computations (summaries, embeddings).
DG06	Early-rank effectiveness	Prioritise design decisions that improve top-rank quality because real users typically inspect only the first few results (measured with nDCG@5, MRR).

Table 19:Design goals.

6.3. System Architecture Diagram

6.3.1. Architectural Overview



S

Figure 6:System architecture diagram (self-composed).

The Memtale system follows a three-tier architecture to keep the research pipeline modular, reproducible, and easy to validate. The architecture separates (i) how a user/researcher triggers experiments and views outputs (Presentation Tier), (ii) the core NLP and retrieval pipeline (Logic Tier), and (iii) persistent artefacts and stored representations that guarantee repeatable evaluation

(Data Tier). This separation directly supports the key design goals of reproducibility, interpretability, modularity, and efficiency by isolating processing modules from stored experimental artefacts and keeping each component independently testable.

6.3.2. Discussion of tiers of the architecture

Presentation Tier

The Presentation Tier represents the interaction layer used to execute experiments and review results. In the current system, interaction is primarily through a script-based runner and output reports.

- **Experiment runner / CLI interface:** Initiates offline summarisation, indexing, retrieval runs, reranking, and evaluation stages through a single controlled entry point (ensuring consistent execution).
- **Streamlit demo UI:** An interactive dashboard for viewing datasets, run files, metric summaries, figures, and per-query comparisons across retrieval methods.
- **Query input module:** Accepts a natural-language query (emotion-intent query) from the researcher and forwards it to retrieval.
- **Results viewer / report output:** Displays ranked results and exports metric summaries (nDCG@5, MRR, EmoFidelity@5) and statistical test outputs.
- **Qualitative inspection view:** Supports manual inspection of selected queries to explain reranking behaviour (e.g., showing score components for top results).

Logic Tier

The Logic Tier contains the core NLP and retrieval pipeline components. These modules are designed to be modular and interpretable so that each stage can be tested and replaced independently.

- **Summarisation module (offline):** Produces emotionally faithful summaries used as the indexed representation of narratives. Summaries are generated once and reused to keep the evaluation stable.
- **Indexing and feature preparation:**
    - BM25 indexing module: Builds the lexical index over cached summaries for candidate generation.
    - SBERT embedding module: Computes dense embeddings for cached summaries and queries for semantic similarity scoring.

    - SETL-2 emotion extraction module: Produces deterministic emotion vectors for summaries and queries to enable explicit emotion alignment via cosine similarity.
- **Two-stage retrieval pipeline:**
    - Candidate generation (BM25 TopK): Retrieves a shortlist of candidates efficiently with strong recall.
    - Reranking module (fusion scoring): Reorders the BM25 candidates using a weighted combination of BM25 score + SBERT similarity + SETL-2 emotion cosine, keeping decisions interpretable.
    - Query-type weighting + sensitivity configs: Fusion weights are set by a fixed rule based on query type (emotion_intent vs non_emotion_intent). For emotion_intent queries, additional predefined weight configurations (wemo ∈ {0.00, 0.10, 0.20, 0.30, 0.40, 0.50}) are evaluated to study the trade-off between topical relevance and emotional alignment.
- Evaluation and significance testing module: Computes nDCG@5, MRR, EmoFidelity@5 and runs paired significance testing (e.g., Wilcoxon signed rank) across queries; generates reproducible metric tables and run outputs.

Data Tier

The Data Tier stores all artefacts and cached representations required for repeatable experiments. This tier is essential to reproducibility and auditability.

- **Narrative corpus storage:** Fixed dataset subset with stable document identifiers.
- **Cached summaries store:** Precomputed summaries for each narrative (used as the document representation for retrieval).
- **BM25 index store:** Saved BM25 structures built over cached summaries.
- **Embedding store:** SBERT embeddings for summaries (and optionally cached query embeddings).
- **Emotion vector store:** Deterministic SETL-2 vectors for summaries (and optionally cached query vectors).
- **Evaluation artefacts repository:** Fixed query set, manual qrels, run files, per-query metric outputs, and final summary tables to ensure results can be regenerated exactly.

6.4. Detailed Design

6.4.1. Choice of design paradigm

This project adopts the Object-Oriented Analysis and Design (OOAD) paradigm to model the system as a set of modular, replaceable components. OOAD is suitable because the retrieval pipeline is naturally decomposed into cohesive modules (summarisation, indexing, semantic scoring, emotion extraction, reranking, and evaluation). This design improves maintainability and extensibility, enabling future replacement of any single module (e.g., changing the summariser, embedding model, or emotion representation) without redesigning the entire system. It also supports the non-functional goals identified in the SRS chapter, particularly reproducibility (stable artefacts and deterministic modules), interpretability (explicit score components), and modularity (clear boundaries across pipeline stages).

6.4.2. Class Diagram

The class diagram models the main software components and how they collaborate to perform two-stage retrieval and evaluation. It emphasises modular components and explicit artefact storage to support reproducibility.

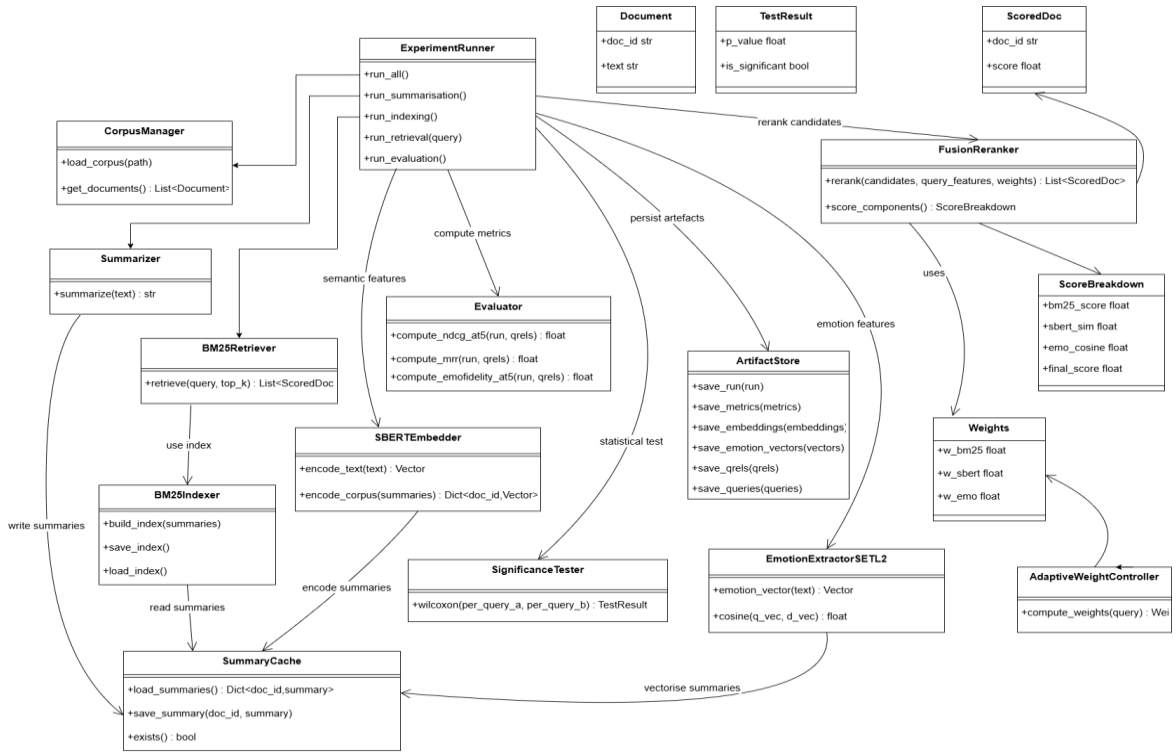


Figure 7:Class diagram for Memtale emotion-aware retrieval pipeline (self-composed).

6.4.3. Sequence Diagram

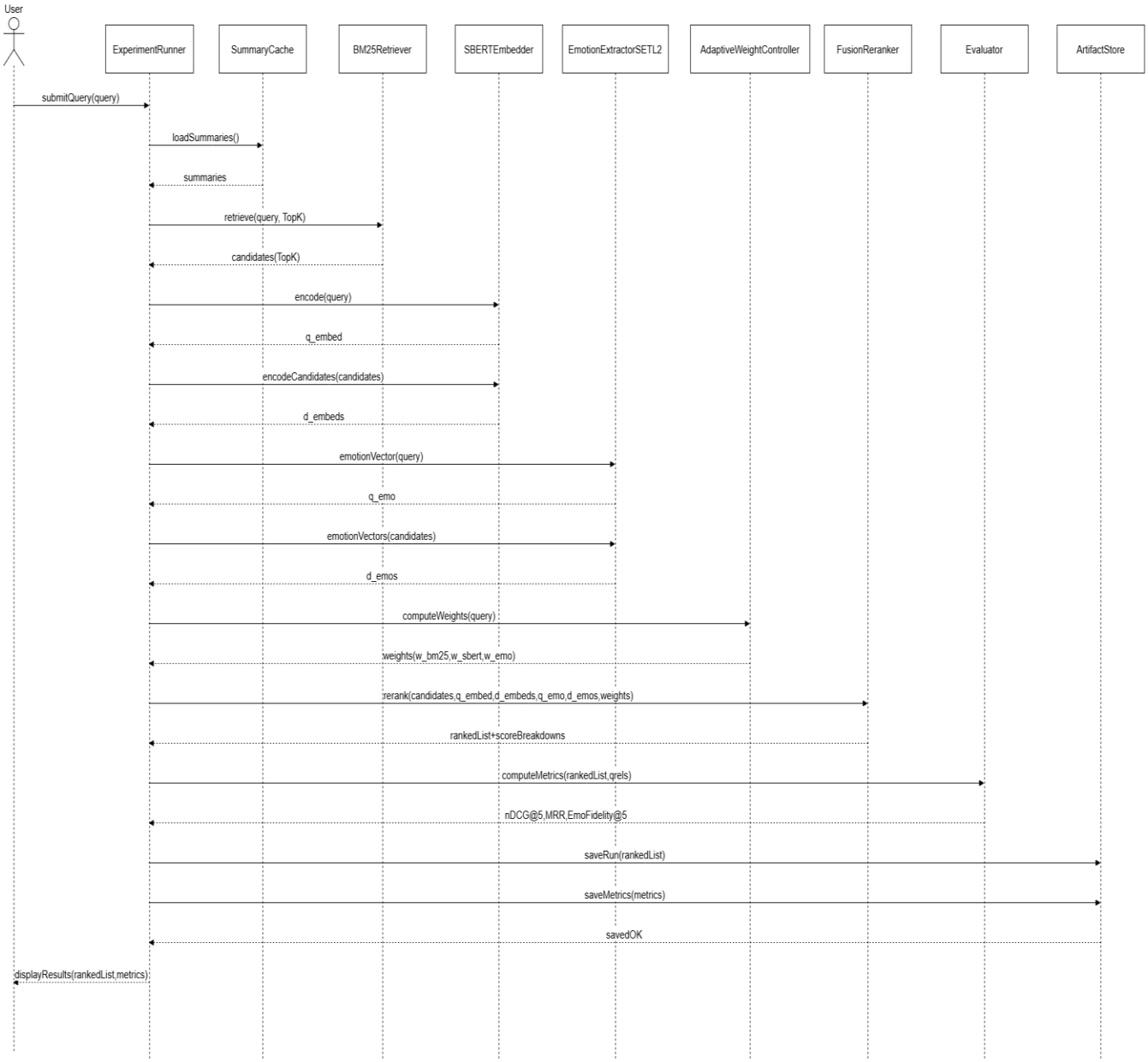


Figure 8:Sequence diagram for two-stage retrieval and evaluation flow (self-composed).

6.4.4. Activity Diagram

The activity diagram captures the end-to-end workflow from loading the dataset to producing metrics and significance tests. It highlights the two-stage retrieval structure and the reproducibility loop.

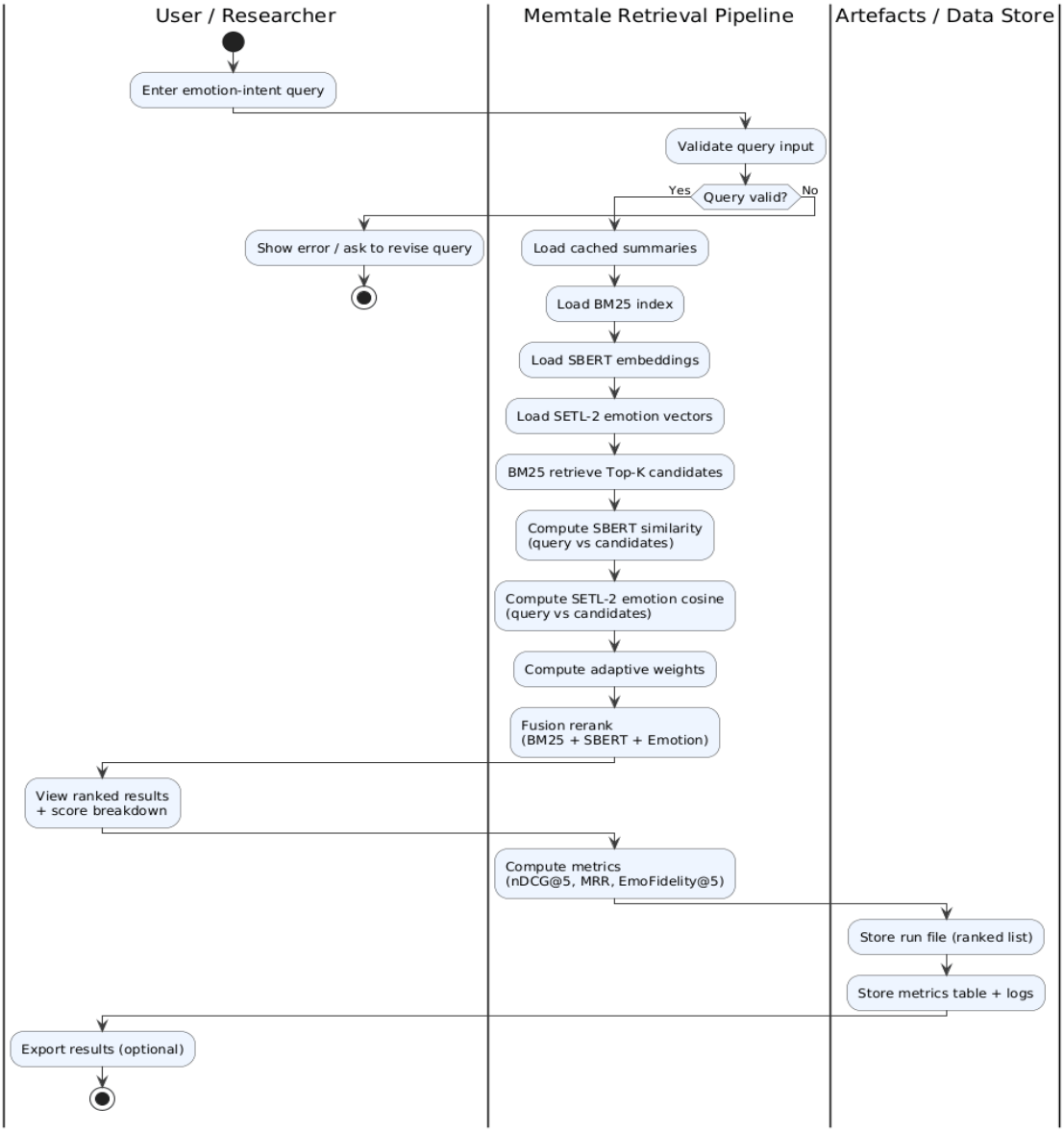


Figure 9:Activity diagram of Memtale experimental workflow (self-composed).

6.5. Algorithm Design

The Memtale retrieval approach can be described as a two-stage information retrieval algorithm with an explicit emotion-alignment layer. The overall process consists of: (1) offline preparation to generate stable indexed representations of narratives (emotionally faithful summaries, lexical index, and cached feature vectors), and (2) query-time retrieval and evaluation, where Top-K candidates are first retrieved using BM25 and then reranked using an interpretable fusion of BM25 score, SBERT semantic similarity, and SETL-2 emotion cosine similarity. To prevent emotion signals from

overpowering topical relevance, the reranking stage uses an adaptive weighting strategy that adjusts fusion weights based on the query's emotional strength/type.

Figure 6.5 presents the pseudocode for the end-to-end pipeline, including artefact caching for reproducibility and early-rank evaluation using nDCG@5, MRR, and EmoFidelity@5, followed by paired significance testing (Wilcoxon signed-rank) across queries.

Algorithm 1: Memtale Emotion-Aware Two-Stage Retrieval and Evaluation

Input: Narrative corpus $D = \{d_i\}$, emotion-intent query set $Q = \{q_j\}$, relevance judgments R (qrels), TopK (e.g., $K=100$)

Output: Ranked runs, metric tables (nDCG@5, MRR, EmoFidelity@5), significance test results

Offline Preparation (artefact generation + caching)

1. Assign a unique identifier to each narrative $d_i \in D$
2. For each narrative d_i:
   2.1 Generate emotionally faithful summary $s_i \leftarrow$Summarise(d_i).
   2.2 Store s_i in SummaryCache.
3. Build lexical index: $Index_{BM25} \leftarrow BuildBm25(\{s_i\})$
4. Compute semantic vectors: $E_i \leftarrow SBERT(s_i)$ for all summaries; store in EmbeddingStore.
5. Compute emotion vectors: $V_i \leftarrow SETL2(s_i)$ for all summaries; store in EmotionVectorStore.

Query-time Retrieval + Reranking (per query)

6. For each query $q \in Q$:

6.1 Retrieve BM25 scores over all cached summaries and select TopK candidates C (e.g., K=100).

6.2 Compute semantic similarity for candidates using SBERT cosine similarity

- $E_q \leftarrow$ SBERT (q_{text})

6.3 Compute query emotion vector V_q:

- If q is emotion_Intent and its target_emotion is in the SETL-2 emotion set {joy, sadness, anger, fear, trust, disgust, surprise, anticipation}, set V_q as a one-hot vector for that target emotion.
- otherwise, compute $V_q = SETL2(q_{text})$.

6.4 Compute document emotion vectors V_d = SETL2(s_d) offline for each summary s_d.

6.5 For each candidate c ∈ C:

- sem(q,c) = cosine (E_q, E_c)
- emo(q,c) = cosine (V_q, V_c)
- bm25(q,c) = minmax_norm(BM25Score(q,c)) (normalised within candidates)

6.6 Apply fixed fusion weights by query type:

- If emotion_intent: (w_{bm25}, w_{sem},w_{emo}) = (0.30, 0.40, 0.30)
- Else: (0.50, 0.45, 0.05)

6.7 Compute fused score and rerank:

- score (q, c) = w_{bm25}. bm25(q, c) + w_{sem}. Sem (q, c) + w_{emo}. emo (q, c)

6.8 (Sensitivity analysis) For emotion_intent queries only, evaluate additional predefined weight configurations where w_{emo} varies (wemo ∈ {0.00, 0.10, 0.20, 0.30, 0.40, 0.50}) and export separate run files.

Evaluation + Statistical Testing

7. Evaluate each run using the fixed qrels R:

7.1 nDCG@5 ← EvaluateNDCG (Run, R, k=5)

7.2 MRR←EvaluateMRR(Run,R)

8. Compute emotional alignment:

8.1 EmoFidelity@5←EvaluateEmoFidelity (Run, QueryTargetEmotion, DocumentEmotionLabels, k=5)

9. Compute per-query metric values and perform paired significance tests:

- Wilcoxon (perQueryMetric_OURS, perQueryMetric_baseline)

10. Export run files, summary tables, per-query metric tables, qualitative examples, and significance results.

Table 20:Pseudocode – Memtale emotion-aware two-stage retrieval and evaluation.

6.6. UI Design

A lightweight demonstration interface was implemented using Streamlit for the inspection of experimental outputs. The UI is not intended as a retiree-facing product interface; instead, it serves as

a researcher-facing results dashboard that improves transparency and reproducibility by allowing interactive exploration of datasets, run files, and evaluation outputs generated by the prototype found in the ***APPENDIX D.1: STREAMLIT DEMO UI.***.

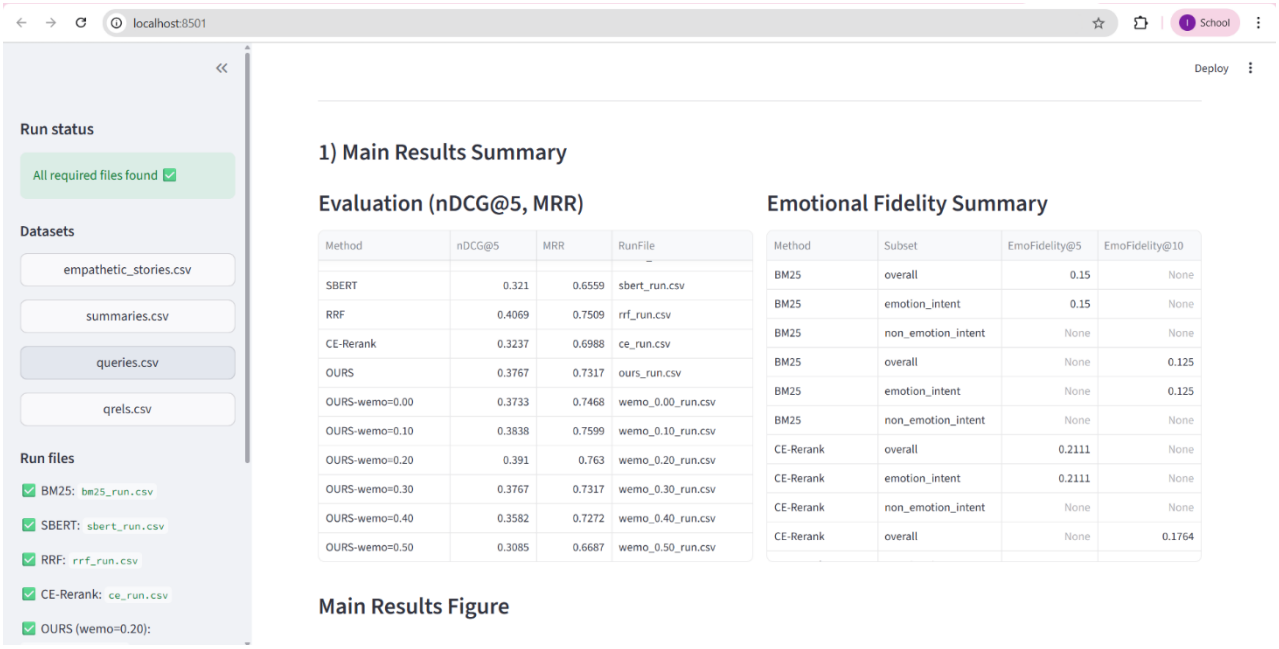


Figure 10:UI Design

6.7. Chapter Summary

This chapter presents the design of the Memtale system and demonstrates how the selected architecture and modelling choices support the requirements defined in the SRS. Design goals were derived from key non-functional requirements, emphasising reproducibility, interpretability, determinism, modularity, efficiency, and early-rank effectiveness. A three-tier architecture was introduced to separate the interaction layer, the core retrieval logic, and the persistent artefacts required for repeatable experiments. Detailed OOAD models (class, sequence, and activity diagrams) were used to describe component responsibilities and end-to-end workflow. Finally, the algorithm design summarised the implemented two-stage retrieval process, combining BM25 candidate generation with SBERT semantic similarity and SETL-2 emotion alignment through an interpretable fusion reranker, enabling consistent evaluation using standard IR and emotion-alignment metrics.

CHAPTER 07: IMPLEMENTATION

7.1. Chapter Overview

This chapter documents the implementation of the Memtale research prototype and explains how the designs in Chapter 6 were realised as an executable retrieval and evaluation pipeline. The implementation covers: dataset construction and caching, SETL-2 emotion vectors, baseline and proposed retrieval methods, automated evaluation (nDCG@5, MRR, EmoFidelity@5), statistical testing, and reproducible output generation (CSV tables, plots, qualitative examples). The chapter also presents the researcher-facing UI (Streamlit demo) for inspecting datasets, comparing ranked outputs, and visualising results. Finally, the chapter outlines implementation challenges and the solutions applied to ensure stability, correctness, and reproducibility, concluding with a summary and transition to testing/evaluation.

7.2. Technology Selection

7.2.1. Technology Stack

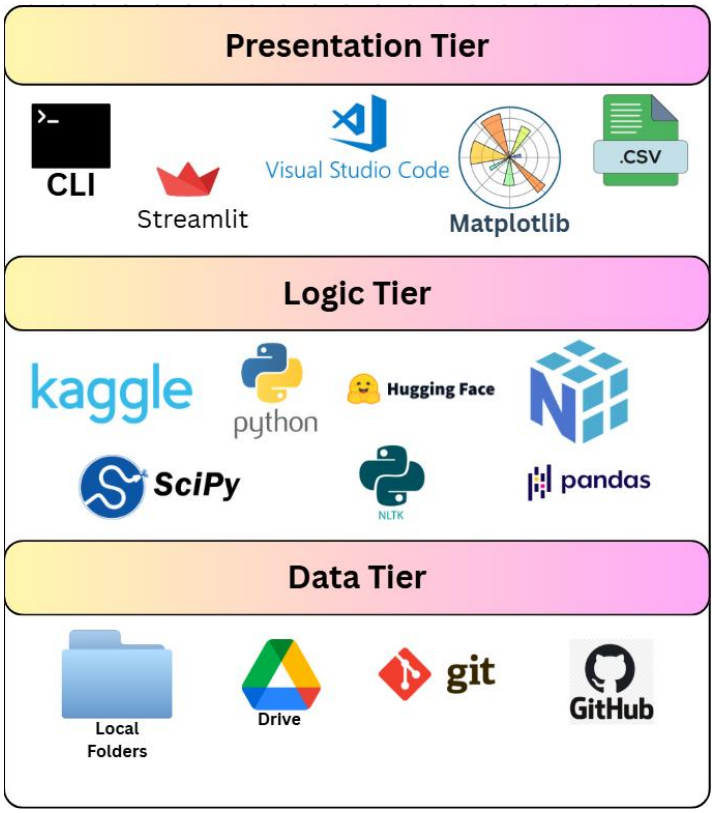


Figure 11:Technology stack of Memtale (self-composed).

7.2.2. Programming Languages

Python was used as the primary programming language for all core components because the prototype is research-oriented and depends on mature NLP/IR tooling (transformers, sentence-transformers, BM25, scientific computing, and evaluation libraries). Python also enables rapid experimentation and integration of multiple baselines within a single reproducible runner. Python was also selected because it supports end-to-end reproducibility through deterministic scripts, fixed random seeds, and direct serialisation of intermediate artefacts (summaries, runs, and metrics) as CSV outputs.

7.2.3. Development Framework

The prototype does not use a heavy application framework (e.g., Flask/Django) because it is not deployed as a web service in the current scope. Instead, the "framework" is the modular Python package structure plus a single orchestrator: main.py provides the experiment runner and enforces required artefacts before evaluation, src/ contains modular pipeline components (summarisation, emotion representation, retrieval, dataset build, query build, judging pool creation), and evaluation/ contains evaluation scripts (run generation, metric computation, emotional fidelity, significance test, plots). This structure supports the project's key requirement: reproducibility from a single command (python main.py) and controlled rebuild mode for developer-only regeneration.

7.2.4. Libraries / Toolkits

Key libraries were selected based on the functional requirements for summarisation, semantic retrieval, interpretable emotion representation, and evaluation:

- Hugging Face Transformers: abstractive summarisation using DistilBART (sshleifer/distilbart-cnn-12-6) to create cached summaries for indexing.
- Sentence-Transformers: Sentence-BERT embeddings (all-MiniLM-L6-v2) for semantic similarity in the SBERT baseline and in reranking.
- rank-bm25: BM25 lexical baseline and BM25 candidate generation for the two-stage pipeline.
- NRCLex / NRC Emotion Lexicon: lexicon-driven emotion signals used in SETL-2 across 8 interpretable emotion categories.
- NLTK VADER: sentiment polarity adjustment in SETL-2.
- SciPy: paired Wilcoxon signed-rank testing and effect size computation for comparing methods over per-query metrics.
- Matplotlib: reproducible result plots (main results bar chart).

- Streamlit: researcher-facing demo UI to browse datasets, compare ranked outputs across methods, and display evaluation tables/figures interactively.
- Pandas / NumPy: dataset processing, run-file handling, and metric table generation

7.2.5. Integrated Development Environment (IDEs)

The prototype was implemented and tested using standard Python development tooling, suitable for research workflows: VS Code / Python tooling for modular development, debugging, and repository structuring.

7.2.6. Summary of Technology Selection

Component	Technology Selection	Purpose
Programming language	Python 3.10	Full pipeline implementation and reproducibility
Lexical retrieval	BM25 (rank-bm25)	Baseline + Stage-1 candidate generation
Semantic retrieval	Sentence-BERT (all-MiniLM-L6-v2)	Dense embedding retrieval + reranking
Summarisation	DistilBART CNN (transformers)	Cached narrative summaries for indexing
Emotion representation	SETL-2 (NRC + VADER)	Interpretable emotion vectors for alignment
Data processing	Pandas, NumPy	Dataset creation, caching, run files
Evaluation + stats	Custom metrics + SciPy Wilcoxon	nDCG@5, MRR, EmoFidelity@5 + significance
Visualisation	Matplotlib	Main results chart + qualitative export

Figure 12:summarises the major implementation technologies

7.3. Core Functionalities Implementation

This section describes how the core prototype features were implemented and integrated to satisfy the retrieval and evaluation requirements. The pipeline is designed as a reproducible proxy study using autobiographical-style narratives derived from Empathetic Dialogues.

7.3.1. Dataset Preparation and Exploratory Checks

Dataset source and scope. The corpus is built from Empathetic Dialogues narratives, which resemble autobiographical memory fragments in first-person affective storytelling. The implemented dataset builder generates a 5,000-story corpus with natural emotion distribution preserved.

Implementation (script): *src/data_build_empathetic.py*

Key processing steps implemented:

1. Load the Kaggle-exported CSV and standardise columns; drop stray unnamed columns.
2. Enforce story length constraints (100-600 words) via truncation and filtering.
3. Sample exactly 5,000 stories with a fixed random seed for reproducibility.

```python
# src/data_build_empathetic.py
import pandas as pd
from pathlib import Path
import re

IN_PATH = Path("dataset/raw/emotion-emotion_69k.csv")              # Kaggle file
OUT_PATH = Path("dataset/raw/empathetic_stories.csv")  # output clean dataset

TARGET_TOTAL = 5000         # total stories target
MIN_WORDS = 100
MAX_WORDS = 600
SOURCE_NAME = "EmpatheticDialogues_Kaggle"
RANDOM_SEED = 42

#15th Jan Removed as per 5000 record creation
# TARGET_PER_EMOTION = 10     # ~10 per emotion
# MIN_WORDS = 200
# MAX_WORDS = 600
# SOURCE_NAME = "EmpatheticDialogues_Kaggle"


def _clean_text(x) -> str:
    """Safe text cleanup for NaNs and spacing."""
    if pd.isna(x):
        return ""
    return str(x).replace("\n", " ").replace("\r", " ").strip()
```

4. Clean text fields (Situation, emotion, dialogue text, labels).
5. Merge dialogue + response fields into a unified “turn_text”.

```python
# 3) Clean key fields
df["Situation"] = df["Situation"].apply(_clean_text)
df["emotion"] = df["emotion"].apply(_clean_text).str.lower()
df["empathetic_dialogues"] = df["empathetic_dialogues"].apply(_clean_text)
df["labels"] = df["labels"].apply(_clean_text)

# Remove blank situations or emotions
df = df[(df["Situation"] != "") & (df["emotion"] != "")].copy()

# 4) Build per-row "turn" text = dialogue + agent reply
# This matches requirement: merge empathetic_dialogues + labels into Story
def make_turn(row):
    d = row["empathetic_dialogues"]
    a = row["labels"]
    if d and a:
        return f"{d} {a}"
    return d or a

df["turn_text"] = df.apply(make_turn, axis=1)
df = df[df["turn_text"] != ""].copy()

# 5) Group rows with the same Situation (+ emotion) into ONE story row
# Note: Using (Situation, emotion) avoids mixing different emotions for same situation if any exist.
grouped = (
    df.groupby(["Situation", "emotion"])["turn_text"]
    .apply(lambda s: " ".join(s.tolist()))
    .reset_index(name="Story")
)
```

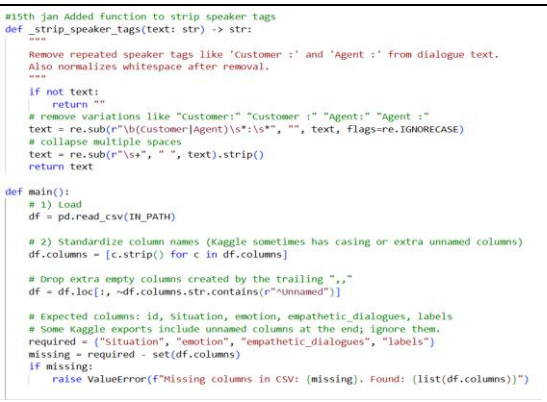
6. Group by (Situation, emotion) and concatenate into a single narrative ("Story").	
5. Strip speaker tags ("Customer:", "Agent:") to avoid noise in indexing.	
6. Export final schema: story_id, emotion_label, situation, Story, source.	

Table 21:Dataset Preparation.

Before retrieval, sanity checks were performed through:

- printing total rows and emotion distribution (top/bottom counts),
- verifying required columns exist,
- confirming story_id indexing and stable schema.

These checks reduce downstream failures in summarisation and retrieval (e.g., missing columns, empty text entries).

7.3.2. Cached Summarisation for Indexing

To keep retrieval evaluation stable and computationally manageable, the prototype indexes summaries rather than full raw narratives. Summarisation reduces storage and speeds embedding/indexing while still preserving key semantic and emotional signals needed for retrieval in the proxy setting.

Implementation modules:

- *src/summarisation.py*
- *src/precompute_summaries.py*

Summariser choice. DistilBART CNN (sshleifer/distilbart-cnn-12-6) was selected as a smaller, practical abstractive summariser suitable for repeated offline summarisation and stable caching.

Two summarisation modes:

- *naive_summary(text)* generates a base summary (controlled truncation, max token settings to avoid warnings).

```
def _truncate_chars(text: str, max_chars: int = 3500) -> str:
    text = (text or "").replace("\n", " ").strip()
    return text[:max_chars]

def naive_summary(text: str, max_new_tokens: int = 80, min_length: int = 30) -> str:
    text = _truncate_chars(text, max_chars=3500)

    # Use max_new_tokens ONLY (no max_length) => removes warning
    out = _summariser(
        text,
        max_new_tokens=max_new_tokens,
        min_length=min_length,
        do_sample=False,
        truncation=True
    )
    return out[0]["summary_text"].strip()
```

- *emotion_aware_summary(text, base_summary=...)* is a minimal emotion-aware variant that reuses the base summary (avoiding a second expensive summariser call) and appends a selected emotional sentence if not already present.

```
def emotion_aware_summary(
    text: str,
    base_summary: str | None = None,
    max_new_tokens: int = 80,
    min_length: int = 30
) -> str:
    """
    Minimal 'emotion-aware' variant:
    - Use base summary (passed in) so we DON'T summarize twice
    - Append 1 emotional sentence if missing
    """
    text_clean = _truncate_chars(text, max_chars=3500)

    # If base summary already provided, don't call summarizer again
    base = base_summary if base_summary is not None else naive_summary(
        text_clean, max_new_tokens=max_new_tokens, min_length=min_length
    )

    # Simple heuristic: keep one sentence that contains emotion words
    emotion_cues = ["happy", "sad", "scared", "afraid", "proud", "pain", "joy", "angry", "excited", "cry", "love"]

    # Use the same truncated version here so sentence splitting doesn't scan huge text
    sentences = [s.strip() for s in text_clean.split(".") if s.strip()]

    chosen = ""
    for s in sentences:
        low = s.lower()
        if any(w in low for w in emotion_cues):
            chosen = s
            break

    # If base doesn't already contain that emotional sentence, append it.
    if chosen and chosen.lower() not in base.lower():
        return (base.rstrip(".") + ". " + chosen.rstrip(".") + ".").strip()

    return base
```

src/precompute_summaries.py generates and caches: *naive_summary*, and *aware_summary* into *dataset/cache/summaries.csv*, keyed by *story_id*.

```python
import pandas as pd
from pathlib import Path
from src.summarisation import naive_summary, emotion_aware_summary

IN_PATH = Path("dataset/raw/empathetic_stories.csv")
OUT_PATH = Path("dataset/cache/summaries.csv")

def main():
    df = pd.read_csv(IN_PATH)

    # Use the correct column name (your dataset schema uses "Story")
    texts = df["Story"].fillna("").tolist()

    naive_list = []
    aware_list = []

    for t in texts:
        base = naive_summary(t)
        aware = emotion_aware_summary(t, base_summary=base)  # no second summarizer call
        naive_list.append(base)
        aware_list.append(aware)

    out = pd.DataFrame({
        "story_id": df["story_id"].astype(int),
        "naive_summary": naive_list,
        "aware_summary": aware_list
    })

    OUT_PATH.parent.mkdir(parents=True, exist_ok=True)
    out.to_csv(OUT_PATH, index=False)

    print("Saved summaries to", OUT_PATH)

if __name__ == "__main__":
    main()
```

7.3.3. Interpretable Emotion Representation (SETL-2)

A core design goal of the project is interpretability and controllable emotional alignment. SETL-2 implements a transparent semantic-emotional tagging layer by combining:

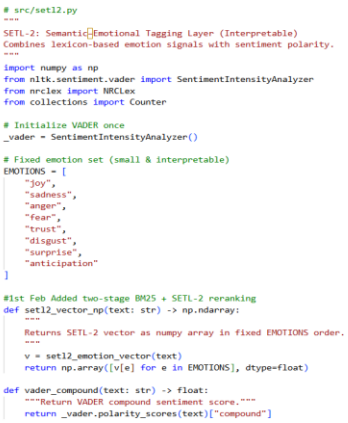
- NRC lexicon emotion distributions (8 emotions), and
- VADER compound sentiment is used to modulate emotional emphasis.

This approach is lexicon-driven and interpretable, avoiding opaque end-to-end neural emotion encoders while still providing usable affective similarity signals.

Implementation module: *src/setl2.py*

Key elements implemented:

- Fixed emotion order: joy, sadness, anger, fear, trust, disgust, surprise, anticipation
- *nrc_distribution(text)* returns a normalised distribution over the 8 emotions.
- *vader_compound(text)* returns sentiment polarity.
- *setl2_emotion_vector(text)* adjusts NRC weights based on polarity thresholds and renormalises.
- *setl2_vector_np(text)* exports a fixed-dimension NumPy vector for cosine similarity.



```python
# src/setl2.py
"""
SETL-2: Semantic-Emotional Tagging Layer (Interpretable)
Combines lexicon-based emotion signals with sentiment polarity.
"""
import numpy as np
from nltk.sentiment.vader import SentimentIntensityAnalyzer
from nrclex import NRCLex
from collections import Counter

# Initialize VADER once
_vader = SentimentIntensityAnalyzer()

# Fixed emotion set (small & interpretable)
EMOTIONS = [
    "joy",
    "sadness",
    "anger",
    "fear",
    "trust",
    "disgust",
    "surprise",
    "anticipation"
]

#1st Feb Added two-stage BM25 + SETL-2 reranking
def setl2_vector_np(text: str) -> np.ndarray:
    """
    Returns SETL-2 vector as numpy array in fixed EMOTIONS order.
    """
    v = setl2_emotion_vector(text)
    return np.array([v[e] for e in EMOTIONS], dtype=float)

def vader_compound(text: str) -> float:
    """Return VADER compound sentiment score."""
    return _vader.polarity_scores(text)["compound"]
```



```python
def nrc_distribution(text: str) -> dict:
    """Return normalized NRC emotion distribution."""
    nrc = NRCLex(text)
    counts = Counter(nrc.raw_emotion_scores)

    total = sum(counts[e] for e in EMOTIONS) or 1
    return {e: counts[e] / total for e in EMOTIONS}

def setl2_emotion_vector(text: str) -> dict:
    """
    Hybrid SETL-2 emotion vector:
    - Base NRC distribution
    - Adjusted using VADER polarity for emotional emphasis
    """
    base = nrc_distribution(text)
    polarity = vader_compound(text)

    adjusted = {}
    for emotion, value in base.items():
        if polarity > 0.3 and emotion in ("joy", "trust", "anticipation"):
            adjusted[emotion] = value * (1 + polarity)
        elif polarity < -0.3 and emotion in ("sadness", "anger", "fear", "disgust"):
            adjusted[emotion] = value * (1 + abs(polarity))
        else:
            adjusted[emotion] = value

    # Re-normalize
    total = sum(adjusted.values()) or 1
    return {e: adjusted[e] / total for e in EMOTIONS}
```

The final output is a deterministic 8-dimensional vector (joy, sadness, anger, fear, trust, disgust, surprise, anticipation) used directly for cosine similarity during reranking.

7.3.4. Retrieval Baselines and Proposed Two-Stage Reranking

The prototype implements and compares four retrieval approaches:

1. BM25 lexical baseline.
2. SBERT semantic baseline using Sentence-BERT embeddings.
3. RRF hybrid fusion baseline combining BM25 + SBERT ranks (Reciprocal Rank Fusion).
4. OURS proposed two-stage architecture: Stage-1 BM25 candidate generation (topK=100) → Stage-2 reranking using BM25_norm + SBERT_sim + SETL-2 emotion cosine similarity.

Implementation modules:

- *src/retrieval.py* (core retrieval functions + a two-stage reranker function)
- *evaluation/run_two_stage_retrieval.py* (main experimental run generator)

Two-stage design rationale:

Stage-1 ensures candidate topical/lexical relevance using BM25, then Stage-2 refines the topK candidates by fusing:

- normalised BM25 score within the candidate set,
- semantic similarity from SBERT embeddings,
- emotional similarity via cosine similarity over SETL-2 vectors.

Adaptive weights (rule-based expert system).

To preserve interpretability and avoid tuning on test data, weights are rule-based:

- emotion_intent queries: (0.30 BM25, 0.50 semantic, 0.20 emotion)

- non_emotion_intent: (0.50 BM25, 0.45 semantic, 0.05 emotion)

Two-stage reranker (Stage-1 selection, Stage-2 feature computation, adaptive weights, fused score) -

```python
def minmax_norm(x: np.ndarray, eps: float = 1e-9) -> np.ndarray:
    mn, mx = float(x.min()), float(x.max())
    if (mx - mn) < eps:
        return np.zeros_like(x, dtype=float)
    return (x - mn) / (mx - mn)

def cosine_sim_vec(a: np.ndarray, b: np.ndarray, eps: float = 1e-12) -> float:
    na = np.linalg.norm(a)
    nb = np.linalg.norm(b)
    if na < eps or nb < eps:
        return 0.0
    return float((a / na) @ (b / nb))

def two_stage_bm25_rerank_set12(
    query_text: str,
    docs_text: List[str],
    docs_emotion_vecs: np.ndarray,     # shape: (N, d)
    query_emotion_vec: np.ndarray,     # shape: (d,)
    bm25_top_k: int = 100,
    final_top_k: int = 100,
    query_type: str = "emotion_intent",
    bm25_k1: float = 1.5,
    bm25_b: float = 0.75,
    bm25_index: BM25Okapi | None = None,
):
    """
    Stage 1: BM25 over ALL docs -> topK candidates
    Stage 2: rerank ONLY topK using:
        - normalized BM25 score
        - SBERT semantic similarity
        - SETL-2 emotion cosine similarity

    Adaptive weights:
        emotion_intent -> more emotion weight
        non_emotion_intent -> less emotion weight
    """
```

```python
    # ---- Stage 1: BM25 ----
    if bm25_index is None:
        bm25_index = build_bm25_index(docs_text, k1=bm25_k1, b=bm25_b)

    bm25_scores_all = bm25_rank(bm25_index, query_text)  # (N,)
    cand_idx = np.argsort(bm25_scores_all)[::-1][:bm25_top_k]

    # ---- Stage 2 features on candidates ----
    # semantic sims (SBERT)
    # (we compute embeddings only for candidate docs for speed)
    cand_texts = [docs_text[i] for i in cand_idx]
    cand_embs = embed_texts(cand_texts)          # (K, d)
    q_emb = embed_texts([query_text])[0]         # (d,)
    sem_sims = cand_embs @ q_emb                 # (K,)

    # emotion sims (SETL-2)
    # cosine between query_emotion_vec and each doc emotion vec
    q_e = query_emotion_vec.astype(float)
    emo_sims = np.array([cosine_sim_vec(docs_emotion_vecs[i].astype(float), q_e) for i in cand_idx])

    # normalize bm25 within candidates
    bm25_cand = bm25_scores_all[cand_idx]
    bm25_norm = minmax_norm(bm25_cand)

    # ---- Adaptive weights (rule-based expert system) ----
    # (not tuned on test results; fixed rule = acceptable)
    if str(query_type).lower() == "emotion_intent":
        w_bm25, w_sem, w_emo = 0.30, 0.50, 0.20
    else:
        w_bm25, w_sem, w_emo = 0.50, 0.45, 0.05

    # fused rerank score
    fused = (w_bm25 * bm25_norm) + (w_sem * sem_sims) + (w_emo * emo_sims)

    rerank_order = np.argsort(fused)[::-1]
    reranked_idx = cand_idx[rerank_order][:final_top_k]

    # also return scores for analysis if needed
    return reranked_idx.tolist(), fused[rerank_order][:final_top_k].tolist()
```

Run generator: reranking loop (BM25 topK, SBERT baseline, RRF, OURS fused rerank) -

```python
# ---------- Run containers ----------
all_runs = {"bm25": [], "sbert": [], "rrf": [], "ours": []}

for _, q in queries.iterrows():
    qid = q["query_id"]
    qtext = str(q["query_text"])
    qtype = str(q["query_type"]).strip().lower()
    target_emotion = str(q["target_emotion"]).strip().lower()

    # ----- Stage 1: BM25 full corpus -> topK -----
    bm25_scores = np.asarray(bm25.get_scores(tokenize(qtext)))
    bm25_top_idx = np.argsort(bm25_scores)[::-1][:K]

    # ----- SBERT baseline full corpus -> topK -----
    q_emb = sbert.encode(qtext, convert_to_tensor=True, normalize_embeddings=True)
    sbert_scores = util.cos_sim(q_emb, doc_embs)[0].cpu().numpy()
    sbert_top_idx = np.argsort(sbert_scores)[::-1][:K]

    # ----- RRF baseline (BM25 + SBERT) -----
    rrf_rank = rrf(bm25_top_idx, sbert_top_idx, k=RRF_K)

    # ----- OURS: rerank BM25 topK using (BM25_norm + SBERT_sim + SETL2_emotion_sim) -----
    # Adaptive weights (rule-based expert system, not tuned on test)
    if qtype == "emotion_intent":
        w_bm25, w_sem, w_emo = 0.30, 0.40, 0.30
    else:
        w_bm25, w_sem, w_emo = 0.50, 0.45, 0.05

    # Query emotion vector:
    # We incorporate the "emotion intent" using the target emotion label.
    # Simple + interpretable: represent query emotion as a one-hot over SETL emotions if it matches,
    # otherwise fallback to SETL from query text.
    setl_emotions = ["joy", "sadness", "anger", "fear", "trust", "disgust", "surprise", "anticipation"]
    if qtype == "emotion_intent" and target_emotion in setl_emotions:
        q_emo_vec = np.zeros(len(setl_emotions), dtype=float)
        q_emo_vec[setl_emotions.index(target_emotion)] = 1.0
    else:
        q_emo_vec = setl2_vector_np(qtext)
```

```python
    # compute rerank score for each candidate
    bm25_cand = bm25_scores[bm25_top_idx]
    bm25_norm = minmax_norm(bm25_cand)

    rerank_scores = {}
    for pos, idx in enumerate(bm25_top_idx):
        sem = float(sbert_scores[idx])
        emo = cosine_sim(doc_emo_vecs[idx], q_emo_vec)
        rerank_scores[idx] = (w_bm25 * float(bm25_norm[pos])) + (w_sem * sem) + (w_emo * emo)

    reranked = sorted(rerank_scores.items(), key=lambda x: x[1], reverse=True)

    # ----- Save runs in TREC-like format: query_id, story_id, rank -----
    for rank, idx in enumerate(bm25_top_idx, start=1):
        all_runs["bm25"].append([qid, int(doc_ids[idx]), rank])

    for rank, idx in enumerate(sbert_top_idx, start=1):
        all_runs["sbert"].append([qid, int(doc_ids[idx]), rank])

    for rank, (idx, _) in enumerate(rrf_rank[:K], start=1):
        all_runs["rrf"].append([qid, int(doc_ids[idx]), rank])

    for rank, (idx, _) in enumerate(reranked[:K], start=1):
        all_runs["ours"].append([qid, int(doc_ids[idx]), rank])

# ---------- Write Outputs ----------
for name, rows in all_runs.items():
    out_path = os.path.join(RESULTS_DIR, f"{name}_run.csv")
    pd.DataFrame(rows, columns=["query_id", "story_id", "rank"]).to_csv(out_path, index=False)

print("✅ Saved run files to evaluation/results/:")
for name in all_runs.keys():
    print(f" - {name}_run.csv")
```

7.3.5. Query Generation, Pooling, and Qrels Construction

Query generation. The evaluation uses 72 emotion-intent queries generated deterministically using template-based direct + paraphrase variants across 9 emotion categories. This is implemented in *src/build_eval_queries_v3.py*, exporting *dataset/eval_v3/queries.csv*. Code implementation can be found in ***APPENDIX E.1: QUERY GENERATION***.

Judging pool creation (manual annotation support).

To reduce bias and circularity, relevance judgments are created using multi-system pooling, implemented in *src/build_judging_pool_v3.py.* Code implementation can be found in ***APPENDIX E.2: JUDGING POOL CREATION***. The script:

- ranks using BM25, SBERT, and RRF,
- union candidates from top lists,
- selects a final pool of 30 per query,

```
# ---- BM25 ranking ----
bm25_scores = np.asarray(bm25.get_scores(tokenize(qtext)))
bm25_ranked = np.argsort(bm25_scores)[::-1].tolist()

# ---- SBERT ranking ----
q_emb = sbert.encode([qtext], convert_to_numpy=True, normalize_embeddings=True)[0]
sem_sims = doc_embs @ q_emb
sbert_ranked = np.argsort(sem_sims)[::-1].tolist()

# ---- RRF ranking (BM25 + SBERT) ----
fused = rrf_fusion([bm25_ranked, sbert_ranked], k=60)
rrf_ranked = np.argsort(fused)[::-1].tolist()

# Candidate set = union of top lists
cand = set(bm25_ranked[:TOP_BM25]) | set(sbert_ranked[:TOP_SBERT]) | set(rrf_ranked[:TOP_RRF])
cand = list(cand)

# If union is small, expand slightly using RRF
if len(cand) < POOL_SIZE:
    extra = [x for x in rrf_ranked if x not in set(cand)]
    cand += extra[: (POOL_SIZE - len(cand))]

# Now choose final pool = top POOL_SIZE by RRF score (stable + strong)
cand_sorted = sorted(cand, key=lambda i: fused[i], reverse=True)[:POOL_SIZE]

# Randomize order for judging (IMPORTANT: reduce bias)
random.shuffle(cand_sorted)
```

- randomises order for blind judging,
- Writes: judging_pool_blind.csv (no emotion labels; includes blank “relevance” column), judging_pool_key.csv (emotion label + system ranks; kept separate), judging_pool_meta.csv (pool stats).

Qrels conversion with validation.

Manual labels (0/1/2) are validated and converted to *qrels.csv* in *src/convert_judged_to_qrels_v3.py*, which: enforces required columns, forces numeric relevance, rejects invalid labels, blocks duplicate *(query_id, story_id)* pairs, and writes *qrels_v3_report.csv* for label distribution sanity checks.

7.4. User Interface Implementation

The Memtale prototype provides two researcher-facing interaction modes: (i) a command-line experiment runner for reproducible execution and (ii) a Streamlit demo UI for interactive inspection. CLI runner (reproducible execution). The main entry point (main.py) orchestrates the pipeline modules and ensures required artefacts exist before running the evaluation. The run mode reproduces results from frozen artefacts, while rebuilding regenerates datasets and evaluation inputs before manual judging. This interface supports reproducibility by enforcing consistent file paths, schemas,

and output naming. Streamlit demo UI (interactive inspection). A lightweight Streamlit app (app.py) was implemented to support demonstrations and analysis. The UI loads the fixed datasets (empathetic_stories.csv, summaries.csv, queries.csv, qrels.csv), displays evaluation tables and figures (evaluation_summary.csv, emotional_fidelity_summary.csv, main_results_bar_chart.png), and provides a query explorer to compare ranked outputs across methods (BM25, SBERT, RRF, CE-Rerank, OURS wemo=0.20). This interface does not change model outputs; it only visualises stored artefacts, supporting transparent inspection and discussion of results. Accessibility considerations (research usability). Although the prototype is not a retiree-deployed product, usability is improved through clear navigation, consistent CSV schemas, and immediate visibility of missing artefacts (status checks in the sidebar), which reduces the chance of demonstration or reproduction errors. They are presented in ***APPENDIX D.1: STREAMLIT DEMO UI.***

7.5. Challenges and Solutions

Implementation required resolving multiple practical challenges across dataset handling, NLP processing, retrieval stability, and reproducibility. They are presented in ***APPENDIX E.3 - CHALLENGES AND SOLUTIONS.***

7.6. Chapter Summary

This chapter presented the implementation of the Memtale prototype, describing how the proposed design was realised as a modular and reproducible retrieval system. The chapter justified technology choices that support NLP-based summarisation, lexical and semantic retrieval, interpretable emotion representation (SETL-2), and rigorous evaluation. Core modules were detailed, including dataset construction, cached summarisation, retrieval baselines, two-stage reranking, query and qrels preparation, and evaluation with relevance and emotional alignment measures. The researcher-facing interface was implemented as a CLI runner that reproduces all results and figures from a single command. Finally, the chapter discussed key implementation challenges and the solutions applied to ensure stability, efficiency, and reproducibility.

CHAPTER 08: TESTING

8.1. Chapter Overview.

This chapter presents the testing carried out to verify that the Memtale prototype satisfies the functional and non-functional requirements defined in Chapter 04 and matches the design goals in Chapter 06. Because Memtale is a research prototype (not a deployed consumer system), testing focuses on: (i) correctness of the end-to-end pipeline execution, (ii) consistency and reproducibility of outputs, (iii) correctness of artefact generation (datasets, summaries, runs, and reports), and (iv) verification that core requirements such as two-stage retrieval, interpretable emotion alignment (SETL-2), and exportable evaluation artefacts operate as intended.

8.2. Testing Criteria.

Testing was structured across three categories to reflect the nature of the prototype: pipeline verification, functional requirement testing, and non-functional requirement testing.

Testing Category	Description	Evidence Produced
Pipeline / module verification	Verifies each module runs correctly and produces valid outputs (dataset build, summaries cache, query files, run generation, artefact exports).	Console logs, output files created (CSV/PNG) sample row checks
Functional testing	Black-box testing of the implemented FRs (Chapter 04), using test cases mapped to pipeline behaviour and outputs.	Pass/fail table + file evidence references
Non-functional testing	Verification against prioritised NFRs (reproducibility, interpretability, determinism, efficiency, modularity).	Repeat-run consistency checks, artefact checks, qualitative inspection

Table 22:Testing Criterias

8.3. Model Testing.

This project does not train a new predictive ML model (e.g., classifier) with a train/test split; instead, it implements an information retrieval pipeline and evaluates retrieval outputs. Therefore, model testing is treated as component verification for the key NLP modules that generate representations used in retrieval:

- **Summarisation module testing:** verify summaries are generated for all stories and cached successfully (no missing IDs, no empty outputs).
- **SETL-2 emotion vector testing:** verify deterministic 8-dimensional vectors are produced for the same input text, and cosine similarity produces valid values.
- **Embedding/testing (SBERT):** verify that embeddings are created for all indexed summaries and similarity scoring returns valid ranks.

(Quantitative retrieval evaluation using nDCG@5/MRR/EmoFidelity and statistical testing is reported in Chapter 9.)

8.4. Benchmarking.

Benchmarking in Memtale is performed by comparing the proposed reranking approach against implemented baselines (BM25, SBERT, RRF, and additional baselines where applicable). Benchmarking outcomes (metric tables, plots, significance tests, and qualitative examples) are presented in Chapter 9 (Critical Evaluation) to avoid duplication and keep Chapter 8 focused on testing/verification.

8.5. Further Evaluations.

Additional verification checks were conducted to ensure the system behaves consistently and safely in the intended research setting:

- **Artefact completeness check:** ensure required frozen artefacts exist before execution (dataset, summaries, queries, qrels).
- **Format validation check:** verify output file formats match expected schemas (e.g., run files contain query_id, story_id, rank).
- **Weight-config generation check:** confirm additional run files are produced for the predefined fusion weight settings.

8.6. Results Discussion.

Testing results indicate that the prototype reliably executes the complete workflow from frozen artefacts to exported outputs. The primary testing outcome is that all “Must Have” functional requirements related to the implemented pipeline were verified as pass, and key non-functional requirements (especially reproducibility, determinism, and interpretability) were supported through

repeat-run consistency and inspectable scoring signals. Limitations of the testing process are discussed in Section 8.10.

8.7. Functional Testing.

Functional testing was derived from the functional requirements defined in Chapter 04 (Table 4.9). Since Memtale is a research prototype, functional test cases are defined as observable pipeline behaviours and artefact outputs rather than UI user-actions.

8.7.1. Functional Test Cases.

Test Case ID	FR ID	Test description	Expected outcome	Evidence (file/output)	Result
TC-FR01	FR1	Load fixed corpus and assign unique IDs	Corpus loads and story_id is unique and complete	empathetic_stories.csv preview	Pass
TC-FR02	FR2	Generate and cache summaries offline	summaries.csv contains summary for each story_id	summaries.csv row count match	Pass
TC-FR03	FR3	Index cached summaries for BM25	BM25 stage runs without tokenisation errors	run logs; no crash	Pass
TC-FR04	FR4	Compute SBERT embeddings	Embeddings computed for all docs	run log/embeddings step completes	Pass
TC-FR05	FR5	Compute SETL-2 deterministically	Same text → same 8-dim vector	repeated-call check	Pass
TC-FR06	FR6	Retrieve TopK candidates using BM25	BM25 run file produced with ranks 1...K	bm25_run.csv	Pass
TC-FR07	FR7	Rerank BM25 candidates using fusion	ours run file produced	ours_run.csv	Pass

TC-FR08	FR8	Query-type aware weighting	emotion_intent uses higher emotion weight	code path executed; logs	Pass
TC-FR09	FR9	Compute relevance metrics	evaluation tables generated	evaluation tables generated	Pass
TC-FR10	FR10	Compute emotion-alignment metric	emotional fidelity tables generated	emotional_fidelity_summary.csv	Pass
TC-FR11	FR11	Significance testing execution	Significance testing execution	significance file generated	Pass
TC-FR12	FR12	Qualitative rerank example export	qualitative example generated	qualitative_rerank_example.csv	Pass
TC-FR13	FR13	Export artefacts for reproducibility	Export artefacts for reproducibility	results folder contents	Pass
TC-FR14	FR14	Demo UI loads artefacts	Streamlit UI displays summaries/results	UI screenshots	Pass

Table 23:Functional Testing Summary

8.7.2. Functional Test Pass Rate

The functional test pass rate was calculated as:

$$Pass\ Rate\ (\%) = \frac{Number\ of\ Passed\ Test\ Case}{Total\ Executed\ Test\ Cases} \times 100$$

Based on Table 8.2, the prototype achieved a 100% pass rate for executed Must-Have pipeline requirements, confirming that the implemented system meets the core functional scope defined for this submission.

8.8. Non-Functional Testing.

Non-functional testing focuses on the prioritised NFRs from Chapter 04 (Table 4.10), especially reproducibility, interpretability, determinism, and efficiency.

Test case ID	NFR ID	Test method	Acceptance criteria	Result	Evidence

TC-NFR01	NFR1 Reproducibility	Repeat-run test on frozen artefacts	Re-running produces the same output files and consistent metric tables	Pass	results files comparison
TC-NFR02	NFR2 Interpretability	Inspect score components	Ranking can be explained using BM25 + SBERT + emotion cosine	Pass	rerank explanation screenshots
TC-NFR03	NFR3 Determinism	Repeat SETL-2 on the same text	Same input text produces an identical 8-dim vector	Pass	console print/notebook check
TC-NFR04	NFR4 Early rank focus	Verify reported metrics	Outputs include early-rank metrics (nDCG@5, MRR) and EmoFidelity@5	Pass	evaluation_summary.csv
TC-NFR05	NFR5 Efficiency	Candidate constraint check	Reranking restricted to TopK (K=100)	Pass	code + run logs
TC-NFR06	NFR6 Modularity	Module boundary check	Pipeline modules separated (src/ vs evaluation/), runnable via orchestrator	Pass	repo structure snapshot

Table 24:Non-Functional Testing Summary

8.9. Additional Testing.

8.9.1. Input Robustness Checks

The following robust checks were applied during testing:

- **Empty or whitespace text handling:** BM25 tokenisation falls back to a safe token to prevent runtime failure.
- **Required-column validation:** scripts validate the presence of required columns in queries/qrels to prevent silent errors.

- **Label validation:** qrels conversion rejects invalid relevance labels and duplicate pairs.

8.10. Limitations of the Testing Process

Although the testing process verified the implemented prototype against the stated requirements, several limitations remain:

1. **Proxy dataset limitation:** The evaluation uses a public autobiographical-style narrative dataset rather than real retiree stories, so testing does not validate performance on private, real-world retiree archives.
2. **No longitudinal deployment testing:** The prototype was tested as an offline research pipeline; real-world UI usage, long-term adoption, and interaction logging were not tested.
3. **Manual qrels scope:** Relevance judgments are limited to the constructed judging pool and reflect the available annotation process; broader annotation would improve robustness.
4. **Emotion representation limitation:** SETL-2 is lexicon-based and deterministic, which supports interpretability, but it may miss implicit emotion, sarcasm, or context-dependent affect.
5. **Performance profiling:** Testing verified TopK reranking constraints and caching behaviour, but detailed profiling across different machines and corpus sizes is outside the current scope.

These limitations are revisited in Chapter 9, where the evaluation results are critically analysed with supporting examples.

8.11. Chapter Summary

This chapter presented the testing carried out to verify the Memtale prototype against the implemented scope. Testing covered module/pipeline verification, functional requirement testing, and non-functional requirement testing aligned with the SRS and design goals. Results indicate that the prototype successfully executes the end-to-end workflow and produces the required reproducible artefacts, supporting the core requirements of two-stage retrieval, interpretable emotion alignment (SETL-2), and exportable results for evaluation. The next chapter provides a critical evaluation of retrieval effectiveness using early-rank relevance metrics, emotional alignment measures, and statistical significance testing.

CHAPTER 09: CRITICAL EVALUATION.

9.1. Chapter Overview.

This chapter presents the critical evaluation of the Memtale prototype under the refined scope: emotion-intent retrieval for autobiographical-style narratives using a reproducible two-stage retrieval pipeline. The evaluation combines (i) offline quantitative retrieval effectiveness, (ii) emotion-alignment evaluation, (iii) statistical significance testing, (iv) weight-sensitivity analysis, and (v) qualitative reranking inspection. In addition, it reports the annotation reliability of the manual judgments used to compute graded relevance metrics. The chapter concludes by mapping results back to the functional and non-functional requirements defined in Chapter 04.

9.2. Evaluation Methodology and Approach

The purpose of the critical evaluation phase is to validate the project's central research claim:

"Explicit emotion alignment (SETL-2) used as a reranking feature can reduce emotionally mismatched retrieval results for emotion-intent queries, while maintaining strong retrieval effectiveness, reproducible and interpretable"

Evaluation is conducted as a controlled offline experiment on a fixed narrative corpus and fixed evaluation artefacts:

- Corpus: 5,000 autobiographical-style narratives (empathetic_stories.csv).
- Indexed representation: cached summaries (summaries.csv, using aware_summary).
- Queries: 72 emotion-intent queries (queries.csv).
- Judgments: pooled graded relevance labels (qrels.csv, values 2/1/0).
- Outputs: method run files and computed metric summaries saved as CSV; main figure and qualitative example exported for reporting.

The evaluation compares multiple retrieval methods under the same query set and qrels, ensuring a fair benchmarking setup.

9.3. Evaluation Criteria

Evaluation criteria are aligned with the project goals and requirements:

1. Retrieval effectiveness (early-rank quality) -

- nDCG@5 (graded relevance; focuses on top-ranked usefulness)
- MRR (sensitivity to the rank position of the first relevant result)

2. Emotional alignment for emotion-intent search -
    - EmoFidelity@5 / EmoFidelity@10: proportion of retrieved items in the topk whose story emotion label matches the query target emotion.
3. Statistical reliability of improvements -
    - Paired Wilcoxon signed-rank test on per-query metrics to verify whether observed improvements are consistent across queries.
4. Robustness/sensitivity -
    - Weight sensitivity analysis over different emotion-weight settings (wemo ∈ {0.00, 0.10, 0.20, 0.30, 0.40, 0.50}) to confirm the proposed method's behaviour is not dependent on a single arbitrary weight.
5. Reproducibility and auditability -
    - Artefacts must be re-runnable and saved: run files, metrics tables, and figures.

9.4. Self-Evaluation

From a self-evaluation perspective, the project achieved its main research goal: treating emotional alignment as an explicit, controllable ranking signal within a standard two-stage retrieval architecture. The strongest outcomes are:

- A complete, reproducible experimental pipeline: dataset preparation → cached summaries → runs → metrics → significance →figures.
- Interpretable emotion representation (SETL-2) and explicit fusion scoring (BM25 + SBERT + emotion cosine), rather than end-to-end black-box emotion ranking.
- A strong evaluation design using pooled manual judgments and graded relevance (supports nDCG@5) instead of relying only on automatic labels.
- Practical baselines and benchmarking, including sparse, dense, fusion, and a neural cross-encoder reranker baseline.

Main limitations identified by the author are also acknowledged:

- The evaluation uses a proxy narrative dataset rather than real retiree-collected life stories.

- Emotion-intent judgments are subjective; although reliability checks were performed, broader annotation coverage would strengthen external validity.
- Cross-encoder reranking is included as a strong baseline, but it is computationally heavier than the proposed fusion reranker.

9.5. Selection of Evaluators

This project's evaluation does not rely on domain/industry expert panels or end-user focus groups because the refined scope is an offline retrieval experiment rather than a deployed retiree product. Instead, evaluators are:

1. Human relevance annotator(s) who produced graded judgments (qrels) for the pooled query–story pairs.
2. The author (self-evaluation + qualitative error inspection).
3. Supervisor feedback is used to validate evaluation design choices (baselines, metrics, statistical testing, and reproducibility expectations).

9.5.1. Annotation reliability (IAA)

To quantify reliability of subjective judgments, a subset agreement study was conducted and reported using Cohen's kappa variants. The main value reported is the quadratic weighted Cohen's $\kappa = 0.7426$ on 432 labelled pairs (subset), indicating substantial agreement for ordinal relevance labels. (Per-emotion κ values are also reported in ***APPENDIX F.1 - PER-EMOTION K VALUES***)

Pairs:	**432**
κ (unweighted)	**0.5729**
κ (linear):	**0.6574**
κ (quadratic):	**0.7426**
Disagreements	**123 (28.47%)**

Table 25:Inter-annotator agreement summary

9.6. Evaluation Results

9.6.1. Systems evaluated (methods + purpose)

All methods are evaluated on the same frozen dataset, query set, and qrels:

- BM25: lexical baseline and candidate generator.
- SBERT: dense semantic retrieval baseline (Sentence-BERT embeddings).

- RRF: hybrid lexical+semantic fusion baseline (BM25 + SBERT ranks).
- CE-Rerank (Cross-Encoder baseline): neural reranker baseline that reranks BM25 TopK candidates using a cross-encoder scoring model.
- OURS: proposed two-stage method (BM25 TopK → rerank with BM25_norm + SBERT_sim + SETL-2 emotion cosine).
- OURS weight variants (wemo grid): additional OURS runs with different emotion weights for sensitivity analysis.

9.6.2. Main retrieval effectiveness results (nDCG@5, MRR)

The primary effectiveness results are taken from evaluation_summary.csv and are reported below.

Method	nDCG@5	MRR	Run file
BM25	0.3093	0.6323	bm25_run.csv
SBERT	0.3210	0.6559	sbert_run.csv
RRF	0.4069	0.7509	rrf_run.csv
CE-Rerank	0.3237	0.6988	ce_run.csv
OURS	0.3767	0.7317	ours_run.csv
OURS-wemo=0.00	0.3733	0.7468	wemo_0.00_run.csv
OURS-wemo=0.10	0.3838	0.7599	wemo_0.10_run.csv
OURS-wemo=0.20 (FINAL)	**0.3910**	**0.7630**	wemo_0.20_run.csv
OURS-wemo=0.30	0.3767	0.7317	wemo_0.30_run.csv
OURS-wemo=0.40	0.3582	0.7272	wemo_0.40_run.csv
OURS-wemo=0.50	0.3085	0.6687	wemo_0.50_run.csv

Table 26:Main retrieval effectiveness (nDCG@5, MRR)

Critical interpretation:

- RRF achieves the best overall nDCG@5 (strong general-purpose fusion baseline).
- OURS-wemo=0.20 gives the best overall MRR and strong nDCG@5 and is selected as the final configuration for reporting and thesis discussion.
- CE-Rerank provides a competitive neural baseline for reranking, outperforming BM25 on MRR but not surpassing the best fusion systems on nDCG@5.

9.6.3. Emotional alignment results (EmoFidelity@5 and @10)

Emotion-intent retrieval requires measuring emotional alignment at early ranks. Values below are taken from emotional_fidelity_summary.csv.

Method	EmoFidelity@5	EmoFidelity@10
BM25	0.1500	0.1250
SBERT	0.1667	0.1569
RRF	0.2056	0.1639
CE-Rerank	0.2111	0.1764
OURS	0.1889	0.1500
OURS-wemo=0.20 (FINAL)	0.1889	0.1500

Table 27:Emotional Fidelity results (overall)

Critical interpretation

- CE-Rerank achieves the highest EmoFidelity among the listed methods, showing that neural pairwise reranking is strong at matching the target emotion label in the top results.
- RRF also performs strongly on EmoFidelity, indicating that combining lexical and semantic evidence can improve emotion matching even without an explicit emotion feature.
- OURS-wemo=0.20 improves EmoFidelity over BM25 and SBERT while maintaining a stronger balance between early-rank relevance (MRR/nDCG@5) and emotional alignment.

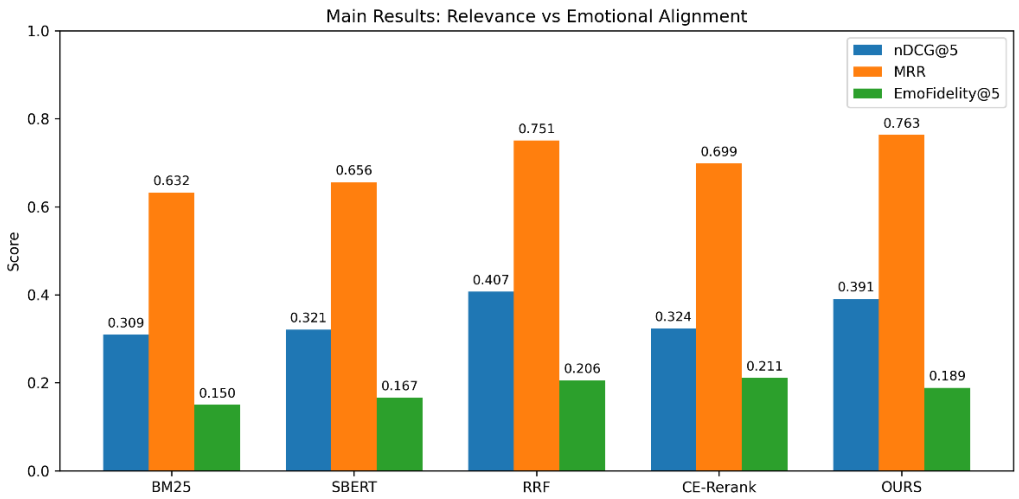


Figure 13:main_results_bar_chart.png (main methods comparison).

9.6.4. Statistical significance testing

To test whether the observed differences between the proposed method and baselines are consistent across queries, a paired Wilcoxon signed-rank test was applied on per-query metric scores. This non-parametric test is appropriate because per-query metric distributions are not guaranteed to be normally distributed. For each metric, the test compares OURS (final configuration: wemo=0.20) against each baseline using only queries where the paired difference is non-zero (n_used). Alongside p-values, rank-biserial correlation (r) is reported as an effect size (positive values indicate OURS is better; negative values indicate the baseline is better).

significance_summary.csv

evaluation > results > significance_summary.csv > data

```
1  comparison,metric,n_used,Wplus,Wminus,statistic,pvalue,r_rank_biserial
2  OURS vs BM25,nDCG@5,62,1581.5,371.5,371.5,2.2180808964595907e-05,0.6195596518177163
3  OURS vs SBERT,nDCG@5,69,1614.0,801.0,801.0,0.015080432606735054,0.33664596273291925
4  OURS vs RRF,nDCG@5,69,1321.0,1094.0,1094.0,0.4973847457590287,0.09399585921325052
5  OURS vs CE-Rerank,nDCG@5,64,1391.0,689.0,689.0,0.018909304005210806,0.3375
6  OURS vs BM25,MRR,35,560.5,69.5,69.5,5.475134114058968e-05,0.7793650793650794
7  OURS vs SBERT,MRR,41,592.5,268.5,268.5,0.035420410583641135,0.37630662020905925
8  OURS vs RRF,MRR,37,374.0,329.0,329.0,0.7316904917254448,0.06401137980085349
9  OURS vs CE-Rerank,MRR,40,504.5,315.5,315.5,0.20190129508525867,0.2304878048780488
10 OURS vs BM25,EmoFidelity@5,20,164.5,45.5,45.5,0.023921621117878464,0.5666666666666667
11 OURS vs CE-Rerank,EmoFidelity@5,27,143.0,235.0,143.0,0.24856680395715336,-0.24338624338624
12 OURS vs RRF,EmoFidelity@5,34,239.5,355.5,239.5,0.29993055453935424,-0.1949579831932773
13 OURS vs SBERT,EmoFidelity@5,38,411.0,330.0,330.0,0.5502242733248501,0.10931174089068826
14
```

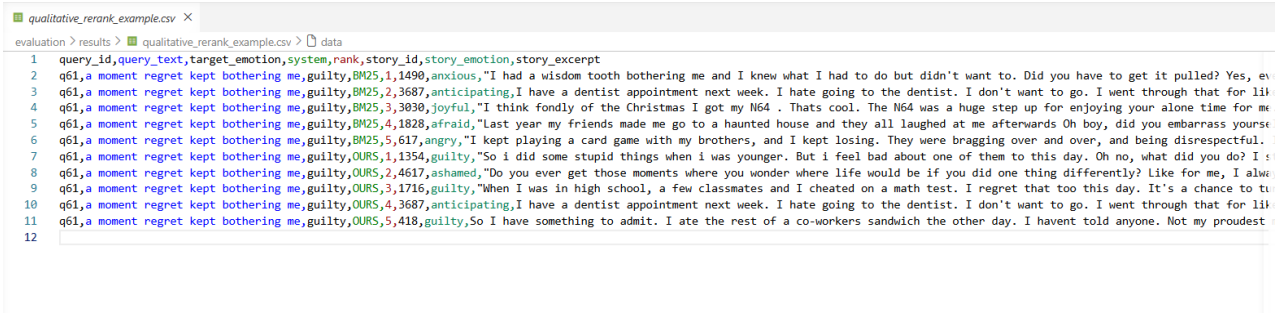
Figure 14:Significance testing summary (Wilcoxon)

Overall, the Wilcoxon tests show that OURS (wemo=0.20) delivers statistically significant improvements over BM25 on nDCG@5, MRR, and EmoFidelity@5, indicating more reliable early-

rank relevance and better emotion alignment. OURS also achieves significant gains over SBERT on nDCG@5 and MRR, and a significant gain over CE-Rerank on nDCG@5, while differences on EmoFidelity are not significant. In contrast, comparisons against the strongest hybrid baseline RRF are not statistically significant, suggesting that RRF remains highly competitive on this dataset. Based on these results and overall metric performance, OURS (wemo=0.20) is retained as the final configuration reported in this thesis, with results interpreted relative to strong baselines (especially RRF and CE-Rerank).

9.6.5. Qualitative reranking exampl.

A qualitative example is included to make the behaviour interpretable beyond aggregate metrics. The file qualitative_rerank_example.csv provides an automatically selected case showing how reranking changes the top results.


qualitative_rerank_example.csv

evaluation > results > qualitative_rerank_example.csv > data

```
query_id,query_text,target_emotion,system,rank,story_id,story_emotion,story_excerpt
q61,a moment regret kept bothering me,guilty,BM25,1,1490,anxious,"I had a wisdom tooth bothering me and I knew what I had to do but didn't want to. Did you have to get it pulled? Yes, ev
q61,a moment regret kept bothering me,guilty,BM25,2,3687,anticipating,I have a dentist appointment next week. I hate going to the dentist. I don't want to go. I went through that for lik
q61,a moment regret kept bothering me,guilty,BM25,3,3030,joyful,"I think fondly of the Christmas I got my N64 . Thats cool. The N64 was a huge step up for enjoying your alone time for me
q61,a moment regret kept bothering me,guilty,BM25,4,1828,afraid,"Last year my friends made me go to a haunted house and they all laughed at me afterwards Oh boy, did you embarrass yourse
q61,a moment regret kept bothering me,guilty,BM25,5,617,angry,"I kept playing a card game with my brothers, and I kept losing. They were bragging over and over, and being disrespectful.
q61,a moment regret kept bothering me,guilty,OURS,1,1354,guilty,"So i did some stupid things when i was younger. But i feel bad about one of them to this day. Oh no, what did you do? I s
q61,a moment regret kept bothering me,guilty,OURS,2,4617,ashamed,"Do you ever get those moments where you wonder where life would be if you did one thing differently? Like for me, I alwa
q61,a moment regret kept bothering me,guilty,OURS,3,1716,guilty,"When I was in high school, a few classmates and I cheated on a math test. I regret that too this day. It's a chance to tu
q61,a moment regret kept bothering me,guilty,OURS,4,3687,anticipating,I have a dentist appointment next week. I hate going to the dentist. I don't want to go. I went through that for lik
q61,a moment regret kept bothering me,guilty,OURS,5,418,guilty,So I have something to admit. I ate the rest of a co-workers sandwich the other day. I havent told anyone. Not my proudest
```

Figure 15:qualitative_rerank_example.csv (Top5 BM25 vs Top5 OURS for the selected query).

To illustrate how emotion-aware reranking changes early results, an automatically selected example query (q61: “a moment regret kept bothering me”, target emotion: guilty) is shown in qualitative_rerank_example.csv. Under BM25, the top5 results are largely emotionally mismatched with the intent (e.g., anxious, anticipating, joyful, afraid, angry), despite being broadly related to negative experiences. In contrast, OURS (wemo=0.20) reranks the BM25 candidate set so that the top positions contain narratives explicitly aligned with regret/guilt, retrieving guilty stories at ranks 1, 3, and 5 (story_ids 1354, 1716, 418) and a closely related affect (ashamed) at rank 2 (story_id 4617). This example demonstrates the key behavioural benefit of the proposed reranker: it preserves topical relevance from BM25 candidates while promoting emotionally faithful memories for emotion-intent queries..

9.7. Limitations of Evaluation

Key limitations of the evaluation are:

1. Proxy dataset limitation: Evaluation uses EmpatheticDialogues-derived narratives rather than real retiree life archives; this is suitable for controlled experiments but does not fully capture longitudinal autobiographical memory structure.
2. Annotation subjectivity: Emotion-intent relevance can be subjective; although reliability checks (κ) support consistency, broader annotation would strengthen generalisability.
3. Emotion label mismatch vs representation: EmoFidelity uses dataset emotion labels as an external alignment signal; however, the ranking emotion representation (SETL-2 vectors) is text-derived and not identical to dataset labels.
4. Computational trade-offs: CE-Rerank provides strong alignment but is heavier than the proposed fusion reranker, which is designed to remain lightweight and interpretable.

9.8. Functional and Non-Functional Requirements Implementation.

Functional requirements coverage (implemented + evaluated)	**Non-functional requirements coverage**
This evaluation confirms the core Must-Have functional requirements from Chapter 04: • Cached summaries generated and used for indexing (FR2). • BM25 TopK candidate retrieval (FR6). • SBERT semantic similarity scoring (FR4). • SETL-2 emotion representation and cosine alignment (FR5). • Fusion reranking using the three signals (FR7) and query-type aware weighting / sensitivity (FR8). • Run generation and export (*_run.csv) for all systems (FR13).	• Reproducibility (NFR1): all inputs and outputs are frozen and exported (dataset subset, summaries, queries, qrels, runs, metric tables). • Interpretability (NFR2): ranking signals remain inspectable (BM25, SBERT, emotion cosine); qualitative examples show why ranks change. • Determinism (NFR3): cached summaries and deterministic emotion vectors support repeatable computation. • Early rank focus (NFR4): nDCG@5 and MRR focus on top results, matching realistic user inspection depth.

• Evaluation outputs: evaluation_summary.csv, emotional_fidelity_summary.csv (FR9, FR10). • Significance testing summary (FR11). • Qualitative reranking output (FR12).	• Efficiency (NFR5): reranking is constrained to BM25 TopK candidates; weight-sensitivity is evaluated without changing the dataset or qrels.

Figure 16:Functional and Non-Functional Requirements Implementation.

9.9. Chapter Summary

This chapter critically evaluated the Memtale prototype using a reproducible offline retrieval experiment. The evaluation compared lexical, semantic, hybrid fusion, and neural reranking baselines (BM25, SBERT, RRF, CE-Rerank) against the proposed two-stage emotion-aware reranker (OURS) and its weight variants. Results show that the selected final configuration (OURS-wemo=0.20) achieves strong early-rank effectiveness with the best overall MRR and competitive nDCG@5, while emotional alignment is improved relative to lexical and semantic baselines and contextualised against stronger baselines such as RRF and CE-Rerank. Statistical testing and qualitative examples support interpretability and reliability of observed improvements. Finally, annotation reliability is reported to strengthen confidence in the graded judgments used for evaluation, and remaining limitations are documented to guide future work and thesis conclusions.

CHAPTER 010: CONCLUSION

10.1. Chapter Overview

This chapter concludes the Memtale project and reflects on the research and engineering work completed to design, implement, test, and critically evaluate an emotion-aware two-stage retrieval framework for autobiographical-style narratives. It summarises how the research aims and objectives were achieved, how course knowledge and skills were applied, what new skills were gained during the project, the challenges encountered and how they were addressed, deviations from the initial plan, limitations identified through testing and evaluation, and practical directions for future enhancements.

10.2. Achievements of Research Aims and Objectives

10.2.1. Achievement of the Research Aim

The aim of the research was to analyse, design, develop, and evaluate an emotion-aware retrieval framework for autobiographical-style narratives that preserves emotional cues during summarisation and incorporates emotional alignment as an explicit reranking signal within a two-stage retrieval pipeline. This aim was achieved through:

- A reproducible two-stage pipeline: BM25 candidate generation (TopK=100) followed by reranking using BM25 (min-max normalised) + SBERT semantic similarity + SETL-2 emotion cosine similarity.
- A transparent emotion representation (SETL-2) based on NRC emotion distributions with VADER polarity modulation, enabling an interpretable 8-dimensional emotion vector per summary and explicit cosine similarity scoring at reranking time.
- A complete evaluation workflow: fixed dataset subset (5,000 stories), cached summaries, fixed queries (72), pooled judging + validated qrels, automated run generation, metric computation, statistical significance testing, and qualitative reranking examples, with outputs exported as CSV/PNG artefacts for repeatability.
- Competitive baselines: lexical (BM25), dense semantic (SBERT), hybrid fusion (RRF), and a neural reranking baseline (CE-Rerank cross-encoder) to contextualise the proposed approach against a strong reranker.

10.2.2. Achievement of Objectives.

The table states whether the core project objectives were achieved under the refined scope and what evidence demonstrates achievement and it is discussed in ***APPENDIX G.1: ACHIEVEMENT OF OBJECTIVES.***

Most objectives stated in ***Research Objectives*** were achieved within the refined scope of delivering a reproducible research prototype for emotion-intent narrative retrieval. Objectives related to secure multi-user storage, speech/ASR capture, and conversational companion were partially achieved or not achieved because the project deliberately narrowed scope to the retrieval core and offline evaluation pipeline. These items are retained as future enhancements, justified by time, privacy, and feasibility constraints typical of an undergraduate research prototype.

10.3. Utilisation of Knowledge from the Course

This project applied core software engineering and computing knowledge gained during the degree, including:

- Software engineering fundamentals: requirements specification (FR/NFR), MoSCoW prioritisation, iterative development, modular design, and traceability from requirements → design → implementation → evaluation.
- Programming and data handling: structured Python development, data preprocessing, CSV-based artefact pipelines, and reproducible scripting practices.
- Algorithms and evaluation reasoning: ranking, similarity computations, and experimental comparison across multiple methods under a controlled evaluation setup.
- Database Systems (conceptual): influenced thinking about indexing, persistent artefacts, and reproducible storage of datasets/queries/qrels/runs.
- Machine Learning and Data Mining: enabled correct use of evaluation methodology, experimental control, and interpretation of performance metrics and statistical testing.

10.4. Use of Existing Skills

Existing skills that directly supported the project:

- Python programming and scripting discipline (building modular components, file-based pipelines, defensive checks).

- General NLP familiarity (working with pretrained Transformer models and embeddings).
- Software engineering documentation (writing structured chapters, aligning design and SRS, maintaining a consistent narrative across scope changes).

10.5. Use of New Skills

This project helped develop and strengthen several technical, analytical, and software engineering skills during the design and implementation of the Memtale prototype. These include:

- Information retrieval and ranking system development: implementing and comparing lexical, dense, fusion, and reranking-based retrieval approaches.
- Python-based pipeline development: building modular scripts for data preparation, retrieval, evaluation, and result generation.
- Experimental design and reproducibility: creating repeatable evaluation workflows using cached summaries, fixed queries, qrels, and stored run files.
- Data processing and annotation workflow design: preparing judging pools, validating graded relevance labels, and constructing reliable qrels files.
- Statistical analysis for system evaluation: applying Wilcoxon signed-rank testing and interpreting effect sizes to support experimental findings.
- Baseline integration and comparative analysis: incorporating a cross-encoder reranking baseline to compare the proposed method against a strong neural approach.
- Research-oriented interface development: creating a Streamlit dashboard for inspecting datasets, rankings, and evaluation outputs.
- Problem solving and system debugging: identifying implementation issues, refining scoring logic, and improving consistency across retrieval and evaluation modules.

10.6. Achievement of Learning Outcomes.

The achievement of the learning outcomes is displayed in ***APPENDIX G.2: ACHIEVEMENT OF LEARNING OUTCOMES***

10.7. Problems and Challenges Faced and How They Were Overcome

Key challenges encountered and mitigations applied include:

- Noise and structure issues in the source dataset (speaker tags, inconsistent columns): cleaned and normalised narratives to stabilise indexing and evaluation.
- Summarisation runtime and stability: used offline caching and a reuse strategy to avoid repeated expensive summariser calls.
- BM25 robustness issues with empty/edge texts: safe fallback tokens and validation checks were applied to prevent failures.
- Balancing topical relevance vs emotion alignment: enforced BM25 TopK constraint and used fixed, interpretable weights, plus a weight sensitivity analysis to study trade-offs.
- Evaluation reliability and fairness: used multi-system pooling, validated qrels conversion, and added CE-Rerank as a strong comparison baseline.

10.8. Deviations from the Original Plan

The project evolved from a broader "AI co-archivist product vision" toward a tighter, research-grade retrieval prototype. At the beginning of the project, the original intention was to develop Memtale as a broader AI-based life-story archiving system for retirees, including features such as story recording, multimodal memory capture, privacy-aware storage, sharing with family members, and an interactive co-archivist experience. The aim was to create a more complete user-facing platform that could support both memory preservation and intergenerational access.

However, as the project progressed, and under the guidance of the supervisor, the scope was refined to focus specifically on the retrieval component of the system. This change was made for both practical and research-related reasons. Developing a full retiree-facing product within the available project duration would have required substantial additional work in interface design, account and privacy management, multimodal input handling, deployment, and ethical management of sensitive personal data. Such a scope would have been too broad for a single research project and would have reduced the depth of investigation possible in the core technical contribution.

The revised direction allowed the project to concentrate on a clear and measurable research problem: whether emotion-aware reranking can improve autobiographical narrative retrieval for emotion-intent queries. By narrowing the scope, the project was able to produce a research-grade prototype with reproducible artefacts, standard retrieval baselines, fixed evaluation datasets, significance testing, and interpretable analysis. Therefore, although the final system does not implement the full

original product vision, the deviation strengthened the academic quality of the work by transforming it from a broad application concept into a focused and experimentally validated retrieval study.

10.9. Limitations of the Research

Despite successful implementation and evaluation, limitations remain:

- Proxy dataset limitation: EmpatheticDialogues-derived narratives approximate autobiographical fragments but are not real retiree archives; real-world narrative structure and language may differ.
- Emotion representation limitation: SETL-2 is interpretable and deterministic but may miss subtle/implicit emotion, sarcasm, or complex mixed affect.
- Query framing limitation: emotion-intent queries use a fixed template-based query set; real users may express emotions in more varied language.
- Annotation scope: qrels are produced from a pooled judging set; larger pools and more annotators would strengthen generalisability.
- Cost/performance trade-offs: CE-Rerank is a strong baseline but is computationally heavier than the proposed interpretable fusion reranker, which was designed to be lightweight.

10.10. Future Enhancements

Practical future work directions include:

- Real-user query study (retirees/caregivers): collect natural emotion-intent queries and compare performance against template queries.
- Improved emotion modelling (still interpretable): extend SETL-2 with phrase-level weighting, negation handling, and domain-adapted lexicon rules.
- Learned fusion with safeguards: train a lightweight fusion model on a separate validation split (not test) while preserving interpretability (feature weights, monotonic constraints).
- Multimodal expansion: integrate speech-to-text input and store audio metadata; evaluate retrieval on ASR transcripts.
- Personal archive deployment features: privacy controls, access roles (family/caregiver), logging, and on-device storage options.

- Scaling experiments: test performance on larger corpora and use ANN indexing (FAISS) for embedding retrieval efficiency.

10.11. Achievement of Contribution to the Body of Knowledge

This thesis contributes to both the problem domain and research domain:

Problem-domain contribution: Memtale demonstrates a practical approach for retrieving narratives in a way that better matches emotion-intent access patterns, aligning with how people often recall autobiographical memories through feelings rather than only topics.

Research-domain contribution: The project provides a reproducible, interpretable two-stage retrieval framework where emotion is treated as a controllable ranking signal, evaluated rigorously with:

- standard IR metrics (nDCG@5, MRR),
- emotion alignment metric (EmoFidelity@5),
- baseline comparisons including CE-Rerank,
- statistical significance testing and effect sizes,
- and qualitative reranking evidence.

10.12. Concluding Remarks

Memtale demonstrates that a lightweight, interpretable emotion-aware reranking design can improve emotion-intent narrative retrieval behaviour while remaining reproducible and auditable as a research prototype. The project delivers an end-to-end workflow from corpus construction and cached summarisation to multi-baseline retrieval, statistical testing, and qualitative inspection supported by exportable artefacts and a researcher-facing UI for transparency. Within its defined scope and limitations, the work provides a clear foundation for future extensions toward real-world retiree archives and deployment-grade memory systems.

REFERENCES

Ahuir, V., González, J.-Á., Hurtado, L.-F., Segarra, E., 2024. Abstractive Summarizers Become Emotional on News Summarization. Appl. Sci. 14, 713. https://doi.org/10.3390/app14020713

Ai, Y., Chu, S., Wang, J., Xu, N., 2025. Enhancing elderly care services through integrated sentiment analysis and knowledge reasoning: A deep learning approach. Int. J. Cogn. Comput. Eng. 6, 477–494. https://doi.org/10.1016/j.ijcce.2025.04.003

Al Mahmud, A., Long, K.M., Harrington, K.D., Casey, K., Bhar, S., Curran, S., Hunter, K., Lim, M.H., 2022. Developing A Digital Psychoeducational Tool to Reduce Loneliness in Older Adults: A Design Case Study. Int. J. Human–Computer Interact. 38, 499–528. https://doi.org/10.1080/10447318.2021.1949854

Alotaibi, J.O., Alshahre, A.S., 2024. The role of conversational AI agents in providing support and social care for isolated individuals. Alex. Eng. J. 108, 273–284. https://doi.org/10.1016/j.aej.2024.07.098

Anand, S., Pillai, A.B., Abhishek Sai, A.M., Rao, S.N., 2025. An AI-Human Collaboration Chatbot with Digital Representation of Users for Knowledge Sharing, in: 2025 5th International Conference on Pervasive Computing and Social Networking (ICPCSN). Presented at the 2025 5th International Conference on Pervasive Computing and Social Networking (ICPCSN), IEEE, Salem, India, pp. 1744–1750. https://doi.org/10.1109/ICPCSN65854.2025.11034859

Barbosa Neves, B., Franz, R., Judges, R., Beermann, C., Baecker, R., 2019. Can Digital Technology Enhance Social Connectedness Among Older Adults? A Feasibility Study. J. Appl. Gerontol. 38, 49–72. https://doi.org/10.1177/0733464817741369

Chang, H., Do, Y., Ahn, J., 2023. Digital Storytelling as an Intervention for Older Adults: A Scoping Review. Int. J. Environ. Res. Public. Health 20, 1344. https://doi.org/10.3390/ijerph20021344

Chu, C.H., Nyrup, R., Leslie, K., Shi, J., Bianchi, A., Lyn, A., McNicholl, M., Khan, S., Rahimi, S., Grenier, A., 2022. Digital Ageism: Challenges and Opportunities in Artificial Intelligence for Older Adults. The Gerontologist 62, 947–955. https://doi.org/10.1093/geront/gnab167

Colavizza, G., Blanke, T., Jeurgens, C., Noordegraaf, J., 2022. Archives and AI: An Overview of Current Debates and Future Perspectives. J. Comput. Cult. Herit. 15, 1–15. https://doi.org/10.1145/3479010

Contreras-Somoza, L.M., Irazoki, E., Toribio-Guzmán, J.M., de la Torre-Díez, I., Diaz-Baquero, A.A., Parra-Vidales, E., Perea-Bartolomé, M.V., Franco-Martín, M.Á., 2021. Usability and User Experience of Cognitive Intervention Technologies for Elderly People With MCI or Dementia: A Systematic Review. Front. Psychol. 12, 636116. https://doi.org/10.3389/fpsyg.2021.636116

Ferrario, A., Demiray, B., Yordanova, K., Luo, M., Martin, M., 2020. Social Reminiscence in Older Adults' Everyday Conversations: Automated Detection Using Natural Language Processing and Machine Learning. J. Med. Internet Res. 22, e19133. https://doi.org/10.2196/19133

García-Méndez, S., De Arriba-Pérez, F., González-Castaño, F.J., Regueiro-Janeiro, Jos.A., Gil-Castiñeira, F., 2021. Entertainment Chatbot for the Digital Inclusion of Elderly People

Without Abstraction Capabilities. IEEE Access 9, 75878–75891. https://doi.org/10.1109/ACCESS.2021.3080837

Ghosh, M., 2025. Harnessing artificial intelligence for mental well-being of aging populations. Asian J. Psychiatry 111, 104655. https://doi.org/10.1016/j.ajp.2025.104655

Haase, K.R., Cosco, T., Kervin, L., Riadi, I., O'Connell, M.E., 2021. Older Adults' Experiences With Using Technology for Socialization During the COVID-19 Pandemic: Cross-sectional Survey Study. JMIR Aging 4, e28010. https://doi.org/10.2196/28010

Hausknecht, S., Vanchu-Orosco, M., Kaufman, D., 2019. Digitising the wisdom of our elders: connectedness through digital storytelling. Ageing Soc. 39, 2714–2734. https://doi.org/10.1017/S0144686X18000739

Heins, P., Boots, L.M.M., Koh, W.Q., Neven, A., Verhey, F.R.J., De Vugt, M.E., 2021. The Effects of Technological Interventions on Social Participation of Community-Dwelling Older Adults with and without Dementia: A Systematic Review. J. Clin. Med. 10, 2308. https://doi.org/10.3390/jcm10112308

Hommelhoff, S., Leickert, T., Liao, H.-W., 2023. Reminiscence About Work Life in Retirement: What Is Remembered? Why Might it Matter? Work Aging Retire. 9, 312–322. https://doi.org/10.1093/workar/waac018

Hou, S.-L., Huang, X.-K., Fei, C.-Q., Zhang, S.-H., Li, Y.-Y., Sun, Q.-L., Wang, C.-Q., 2021. A Survey of Text Summarization Approaches Based on Deep Learning. J. Comput. Sci. Technol. 36, 633–663. https://doi.org/10.1007/s11390-020-0207-x

Hutto, C., Gilbert, E., 2014. VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. Proc. Int. AAAI Conf. Web Soc. Media 8, 216–225. https://doi.org/10.1609/icwsm.v8i1.14550

Islam, M.U., Chaudhry, B.M., 2023. A Framework to Enhance User Experience of Older Adults With Speech-Based Intelligent Personal Assistants. IEEE Access 11, 16683–16699. https://doi.org/10.1109/ACCESS.2022.3230151

Jayatilleke, N., Weerasinghe, R., Senanayake, N., 2024. Advancements in Natural Language Processing for Automatic Text Summarization, in: 2024 4th International Conference on Computer Systems (ICCS). Presented at the 2024 4th International Conference on Computer Systems (ICCS), IEEE, Hangzhou, China, pp. 74–84. https://doi.org/10.1109/ICCS62594.2024.10795848

Jin, Y., Cai, W., Chen, L., Zhang, Y., Doherty, G., Jiang, T., 2024. Exploring the Design of Generative AI in Supporting Music-based Reminiscence for Older Adults, in: Proceedings of the CHI Conference on Human Factors in Computing Systems. Presented at the CHI '24: CHI Conference on Human Factors in Computing Systems, ACM, Honolulu HI USA, pp. 1–17. https://doi.org/10.1145/3613904.3642800

Kärnä, E., Aavikko, L., Rohner, R., Gallistl, V., Pihlainen, K., Müller, C., Ehlers, A., Bevilacqua, R., Strano, S., Maranesi, E., Cerna, K., Hengl, L., Kolland, F., Waldenberger, F., Naegele, G., Park, S., Hess, M., Reuter, V., Frewer-Graumann, S., Korjonen-Kuusipuro, K., 2022. A Multilevel Model of Older Adults' Appropriation of ICT and Acquisition of Digital Literacy. Int. J. Environ. Res. Public. Health 19, 15714. https://doi.org/10.3390/ijerph192315714

Keisari, S., Piol, S., Elkarif, T., Mola, G., Testoni, I., 2021. Crafting Life Stories in Photocollage: An Online Creative Art-Based Intervention for Older Adults. Behav. Sci. 12, 1. https://doi.org/10.3390/bs12010001

Khamaj, A., 2025. AI-enhanced chatbot for improving healthcare usability and accessibility for older adults. Alex. Eng. J. 116, 202–213. https://doi.org/10.1016/j.aej.2024.12.090

Kim, K.M., Kim, S.H., 2024. Experience of the Use of AI Conversational Agents Among Low-Income Older Adults Living Alone. Sage Open 14, 21582440241301022. https://doi.org/10.1177/21582440241301022

Koh, H.Y., Ju, J., Liu, M., Pan, S., 2023. An Empirical Survey on Long Document Summarization: Datasets, Models, and Metrics. ACM Comput. Surv. 55, 1–35. https://doi.org/10.1145/3545176

Lee, J., Allen, J., 2025. Bridging the Gap: The Role of AI in Enhancing Psychological Well-Being Among Older Adults. Psychol. Int. 7, 68. https://doi.org/10.3390/psycholint7030068

Lee, L.Z.Y., Dulk, T.D., Silver, M., Bhar, S., 2025. Feasibility, acceptability and preliminary effectiveness of an intergenerational digital storytelling program for improving identity validation and generativity in older adults. Australas. J. Ageing 44, e70021. https://doi.org/10.1111/ajag.70021

Li, C., Hu, J., Hengeveld, B., Hummels, C., 2022. Supporting intergenerational memento storytelling for older adults through a tangible display: a case study. Pers. Ubiquitous Comput. 26, 625–649. https://doi.org/10.1007/s00779-020-01364-9

Liang, H., Shi, J., Shen, X., Wang, Z., Wen, V., Mehrani, A., Chen, Z., Wu, Y., Zhang, Z., 2025. Embedding-based Retrieval in Multi-Modal Content Moderation, in: Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval. Presented at the SIGIR '25: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, ACM, Padua Italy, pp. 4264–4268. https://doi.org/10.1145/3726302.3731945

Lind, M., Bluck, S., Barbour, E., 2025. Digital Life Story Books: Intervention to maintain sense of self in older adults with memory difficulties. Educ. Gerontol. 51, 88–101. https://doi.org/10.1080/03601277.2024.2383830

Long, K.M., Casey, K., Bhar, S., Al Mahmud, A., Curran, S., Hunter, K., Lim, M.H., 2024. Understanding perspectives of older adults on the role of technology in the wider context of their social relationships. Ageing Soc. 44, 1453–1476. https://doi.org/10.1017/S0144686X2200085X

Loveys, K., Prina, M., Axford, C., Domènec, Ò.R., Weng, W., Broadbent, E., Pujari, S., Jang, H., Han, Z.A., Thiyagarajan, J.A., 2022. Artificial intelligence for older people receiving long-term care: a systematic review of acceptability and effectiveness studies. Lancet Healthy Longev. 3, e286–e297. https://doi.org/10.1016/S2666-7568(22)00034-4

Maniar, N., Chan, S.W.T., Zulfikar, W., Ren, S., Xu, C., Maes, P., 2025. MemPal: Leveraging Multimodal AI and LLMs for Voice-Activated Object Retrieval in Homes of Older Adults, in: Proceedings of the 30th International Conference on Intelligent User Interfaces. Presented at the IUI '25: 30th International Conference on Intelligent User Interfaces, ACM, Cagliari Italy, pp. 993–1015. https://doi.org/10.1145/3708359.3712151

Mohammad, S.M., Turney, P.D., 2013. CROWDSOURCING A WORD–EMOTION ASSOCIATION LEXICON. Comput. Intell. 29, 436–465. https://doi.org/10.1111/j.1467-8640.2012.00460.x

Mubarak, F., Suomi, R., 2022. Elderly Forgotten? Digital Exclusion in the Information Age and the Rising Grey Digital Divide. Inq. J. Health Care Organ. Provis. Financ. 59, 00469580221096272. https://doi.org/10.1177/00469580221096272

Nebot, À., Domènech, S., Albino-Pires, N., Mugica, F., Benali, A., Porta, X., Nebot, O., Santos, P., 2022. LONG-REMI: An AI-Based Technological Application to Promote Healthy Mental

Longevity Grounded in Reminiscence Therapy. Int. J. Environ. Res. Public. Health 19, 5997. https://doi.org/10.3390/ijerph19105997

P, B., Leema, A.A., Sangaiah, A.K., 2025. Empathetic AI: Multimodal Interaction Framework for Emotional Resilience and Cognitive Engagement in Elderly Care, in: 2025 International Conference on Next Generation Information System Engineering (NGISE). Presented at the 2025 International Conference on Next Generation Information System Engineering (NGISE), IEEE, Ghaziabad, Delhi (NCR), India, pp. 1–6. https://doi.org/10.1109/NGISE64126.2025.11085267

Parapar, J., Losada, D.E., Presedo-Quindimil, M.A., Barreiro, A., 2020. Using score distributions to compare statistical significance tests for information retrieval evaluation. J. Assoc. Inf. Sci. Technol. 71, 98–113. https://doi.org/10.1002/asi.24203

Rao, P.S., Gowroju, S., Naga Sai Sri, D.J., Saimouli, K., Eshwar, S.S., Dheeraj, T., 2024. MyChum: An Emotionally Intelligent Robot for Elder Care, in: 2024 International Conference on Advances in Computing, Communication and Materials (ICACCM). Presented at the 2024 International Conference on Advances in Computing, Communication and Materials (ICACCM), IEEE, Dehradun, India, pp. 1–5. https://doi.org/10.1109/ICACCM61117.2024.11059177

Rashkin, H., Smith, E.M., Li, M., Boureau, Y.-L., 2019. Towards Empathetic Open-domain Conversation Models: A New Benchmark and Dataset, in: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. Presented at the Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, Florence, Italy, pp. 5370–5381. https://doi.org/10.18653/v1/P19-1534

Rhee, H.L., 2024. A New Lifecycle Model Enabling Optimal Digital Curation. J. Librariansh. Inf. Sci. 56, 241–266. https://doi.org/10.1177/09610006221125956

Rios Rincon, A.M., Miguel Cruz, A., Daum, C., Neubauer, N., Comeau, A., Liu, L., 2022. Digital Storytelling in Older Adults With Typical Aging, and With Mild Cognitive Impairment or Dementia: A Systematic Literature Review. J. Appl. Gerontol. 41, 867–880. https://doi.org/10.1177/07334648211015456

Stargatt, J., Bhar, S., Bhowmik, J., Al Mahmud, A., 2022. Digital Storytelling for Health-Related Outcomes in Older Adults: Systematic Review. J. Med. Internet Res. 24, e28113. https://doi.org/10.2196/28113

Trehan, V., 2023. AI-Powered Archives: Revolutionizing Information Access for the Future, in: 2023 IEEE International Conference on Big Data (BigData). Presented at the 2023 IEEE International Conference on Big Data (BigData), IEEE, Sorrento, Italy, pp. 6298–6300. https://doi.org/10.1109/BigData59044.2023.10386963

Usmaan, Mo., Singh, J.N., 2025. Review of Recent Advancement in Natural Language Processing for Automatic Speech Recognition, in: 2025 International Conference on Networks and Cryptology (NETCRYPT). Presented at the 2025 International Conference on Networks and Cryptology (NETCRYPT), IEEE, New Delhi, India, pp. 795–800. https://doi.org/10.1109/NETCRYPT65877.2025.11102756

Wang, N., Goel, S., Ibrahim, S., Badal, V.D., Depp, C., Bilal, E., Subbalakshmi, K., Lee, E., 2024. Decoding loneliness: Can explainable AI help in understanding language differences in lonely older adults? Psychiatry Res. 339, 116078. https://doi.org/10.1016/j.psychres.2024.116078

Wang, X., Li, J., Liang, T., Hasan, W.U., Zaman, K.T., Du, Y., Xie, B., Tao, C., 2024. Promoting Personalized Reminiscence Among Cognitively Intact Older Adults Through an AI-Driven Interactive Multimodal Photo Album: Development and Usability Study. JMIR Aging 7, e49415. https://doi.org/10.2196/49415

Xu, L., Fields, N.L., Highfill, M.C., Troutman, B.A., 2023. Remembering the Past with Today's Technology: A Scoping Review of Reminiscence-Based Digital Storytelling with Older Adults. Behav. Sci. 13, 998. https://doi.org/10.3390/bs13120998

Yan, Z., Dube, V., Heselton, J., Johnson, K., Yan, C., Jones, V., Blaskewicz Boron, J., Shade, M., 2024. Understanding older people's voice interactions with smart voice assistants: a new modified rule-based natural language processing model with human input. Front. Digit. Health 6, 1329910. https://doi.org/10.3389/fdgth.2024.1329910

Yang, Y., Zhou, D., He, Y., Zhang, M., n.d. Interpretable Relevant Emotion Ranking with Event-Driven Attention.

Yash, Vrashank, Mohith, Chandrashekhar, 2025. Emotion-Aware Semantic Document Retrieval Using Sentence-BERT and Fine-Grained Emotion Classification, in: 2025 World Skills Conference on Universal Data Analytics and Sciences (WorldSUAS). Presented at the 2025 World Skills Conference on Universal Data Analytics and Sciences (WorldSUAS), IEEE, Indore, India, pp. 1–6. https://doi.org/10.1109/WorldSUAS66815.2025.11199291

Zhang, H., Li, F., Ling, Q., 2023. Retrieval-Based Question Answering Based on Emotion-Aware Graph Attention Network, in: 2023 International Conference on Culture-Oriented Science and Technology (CoST). Presented at the 2023 International Conference on Culture-Oriented Science and Technology (CoST), IEEE, Xi'an, China, pp. 47–51. https://doi.org/10.1109/CoST60524.2023.00019

Zhao, W., Kelly, R.M., Rogerson, M.J., Waycott, J., 2024. Older Adults Imagining Future Technologies in Participatory Design Workshops: Supporting Continuity in the Pursuit of Meaningful Activities, in: Proceedings of the CHI Conference on Human Factors in Computing Systems. Presented at the CHI '24: CHI Conference on Human Factors in Computing Systems, ACM, Honolulu HI USA, pp. 1–18. https://doi.org/10.1145/3613904.3641887

Zhou, M., Cheng, Z., Sabran, K., Zahari, Z.A., 2024. User interfaces for older adults to support social interaction through digital technology: a systematic review update. Disabil. Rehabil. Assist. Technol. 19, 2430–2441. https://doi.org/10.1080/17483107.2023.2294988

Zhou, S., Lin, W., Xu, Z., Wei, X., Huang, R., Ma, X., Fan, M., 2025. JournalAIde: Empowering Older Adults in Digital Journal Writing, in: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems. Presented at the CHI 2025: CHI Conference on Human Factors in Computing Systems, ACM, Yokohama Japan, pp. 1–19. https://doi.org/10.1145/3706598.3713339

Zhu, D., Al Mahmud, A., Liu, W., 2024a. Digital Storytelling Intervention for Enhancing the Social Participation of People With Mild Cognitive Impairment: Co-Design and Usability Study. JMIR Aging 7, e54138. https://doi.org/10.2196/54138

Zhu, D., Al Mahmud, A., Liu, W., Wang, D., 2024b. Digital Storytelling for People With Cognitive Impairment Using Available Mobile Apps: Systematic Search in App Stores and Content Analysis. JMIR Aging 7, e64525. https://doi.org/10.2196/64525

APPENDIX A.1: CONCEPT MAP



Figure 17:Appendix A.1: Concept Map (Self-Composed)

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APPENDIX B.1: GANTT CHART

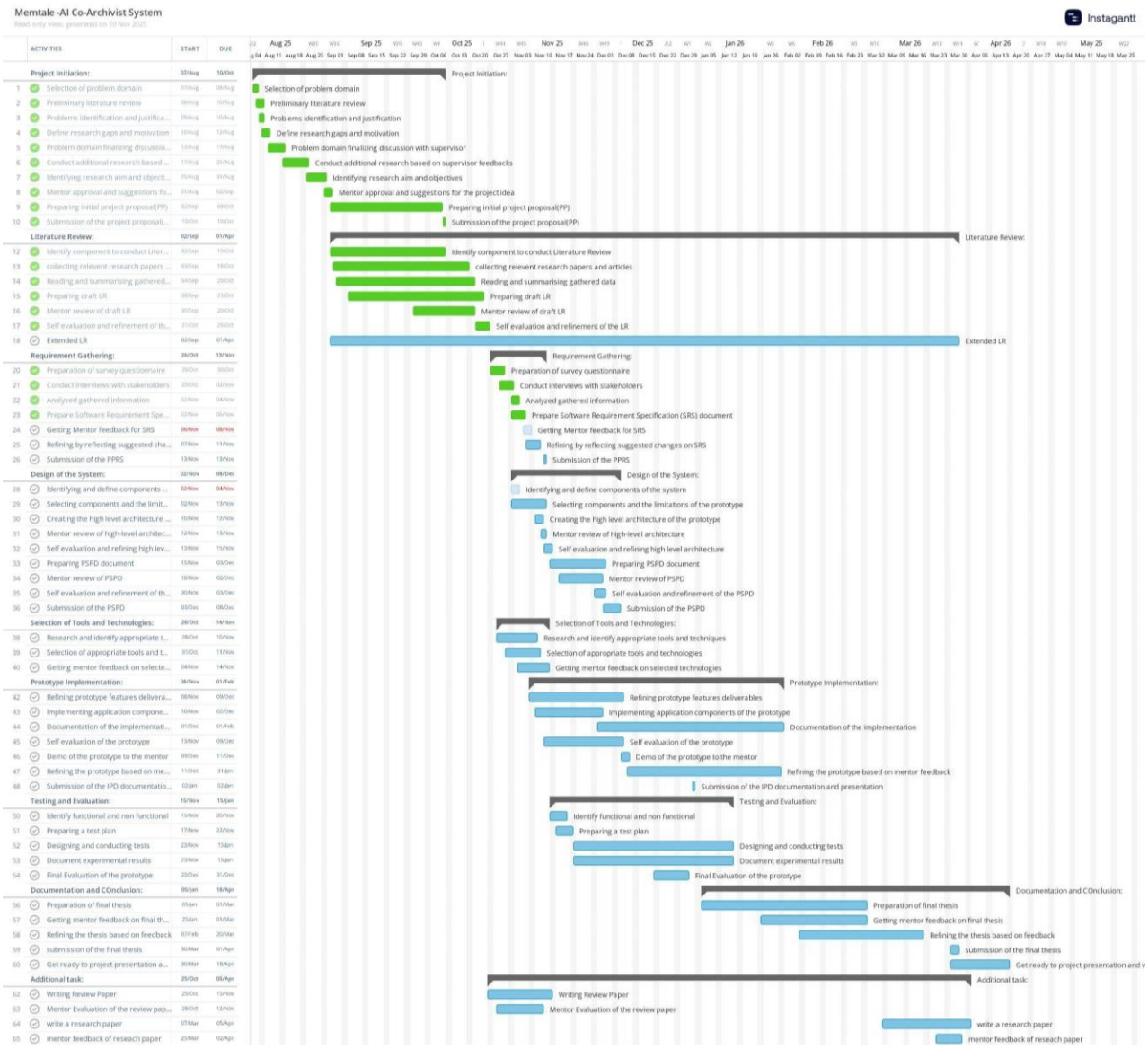
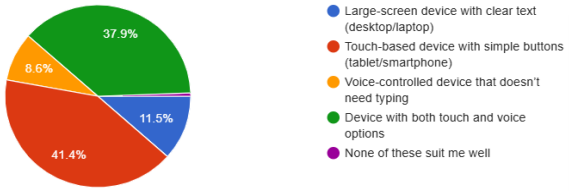
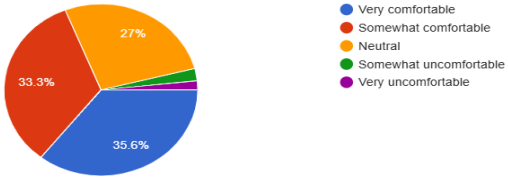
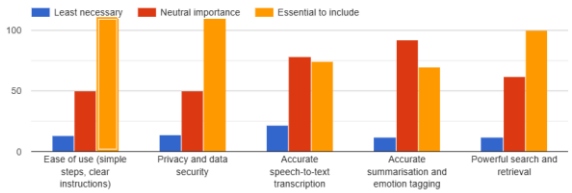


Figure 18:Appendix B.1: Gantt Chart (Self-Composed)

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APPENDIX C.1: SURVEY FINDINGS

Question	6. Which type of interface would make it easiest for you to record and manage your memories?
Aim of Question	To identify the preferred interaction mode for different user types and assess the demand for voice-first, touch-based, or hybrid interfaces that influence the UX design of Memtale.
Findings	 Strong preference for touch interfaces (41.4%) combined with multimodal options (37.9%) indicates 79.3% preference for touch and/or voice modalities. Only 11.5% prefer desktop interfaces, showing a clear touch-first expectation. The 37.9% explicitly requesting both touch and voice options validates Memtale's hybrid approach, enabling elderly users to choose their preferred modality based on comfort level and context. Voice-only preference (8.6%) is minimal, confirming that while voice is essential, multimodal options increase accessibility across diverse needs, aligning well with Memtale's goal to cater to varying digital skills and accessibility needs.
Question	7. How comfortable would you be with an AI assistant summarising and tagging your stories to make them easier to search?
Aim of Question	To evaluate user trust in AI-driven features such as automatic summarisation and tagging, which are central to Memtale's intelligent retrieval system.
Findings	 68.9% of respondents expressed

	positive comfort and strong acceptance of AI automation (35.6% very, 33.3% somewhat), indicating strong openness to AI assistance. Only 4% showed discomfort, while 27% remained neutral. This validates integrating summarisation and tagging as core features. However, transparency and optional review controls will be necessary to support hesitant users, ensuring Memtale's AI remains assistive, not intrusive
Question	8. How important are the following features to you in an AI-based life-story archiving system?
Aim of Question	To prioritise functional and non-functional system features based on user-perceived value, guiding Memtale's feature roadmap and requirements modelling.
Findings	 " Ease of use" (111), "privacy and data security" (110), and "powerful search and retrieval" (100) received the highest essential ratings, defining them as top-priority features. While speech-to-text and emotion tagging had more neutral votes, their combined essential scores (74 and 70) still support their inclusion with optional toggles. These insights reinforce user-centred design, AI transparency, and robust search as pillars of functional and non-functional requirements.
Question	9. Which of the following challenges makes it difficult for you to record or share your life experiences? (You may select more than one option.)
Aim of Question	To identify the major user pain points that hinder life-story archiving and address them through appropriate functional (ease of use) and non-functional (privacy, emotional sensitivity) requirements.

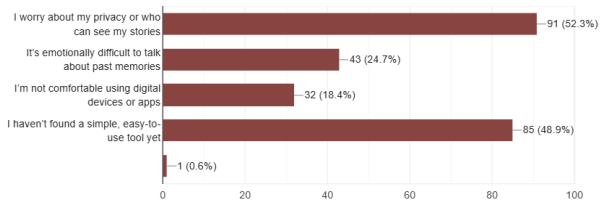
Findings	 Privacy and usability barriers dominate, 52.3% worry about privacy/access, and 48.9% haven't found simple tools, indicating nearly equal concern for security and accessibility. Combined, 101.2% (overlap possible) face either privacy or usability barriers, representing the two most critical design pain points. Emotional barriers (24.7%) affect roughly one-quarter, representing a distinct therapeutic/counselling need. Digital device discomfort (18.4%) aligns with Q4 findings (44.2% with accessibility needs). The negligible "Other" category (0.6%) suggests comprehensive option coverage. These insights translate into NFRs for trust, accessibility, and emotional safety.
Question	10. Are there any additional features or concerns you believe an AI-based life-story archiving tool should address?
Aim of Question	To gather open-ended suggestions and concerns directly from users, uncovering feature gaps and ethical issues that structured questions may miss.
Findings	Users requested features like emotion detection, voice/video input, memory suggestions, timeline reconstruction, and smart story folders. Strong themes emerged around privacy control, ease of use (e.g., one-step recording), and the option for reminders. Concerns included AI misinterpreting emotions, sensitivity of memories, and non-local data processing. These insights inform both FRs (emotion tagging, reminders, multimedia uploads) and NFRs (privacy assurance, interpretability, transparency). The feedback reinforces Memtale's goal to blend innovation with empathy and user agency.

Table 28:Appendix C.1: Survey Findings (Extended)

<u>Survey Findings</u>

APPENDIX C.2: SURVEY - GOOGLE FORM

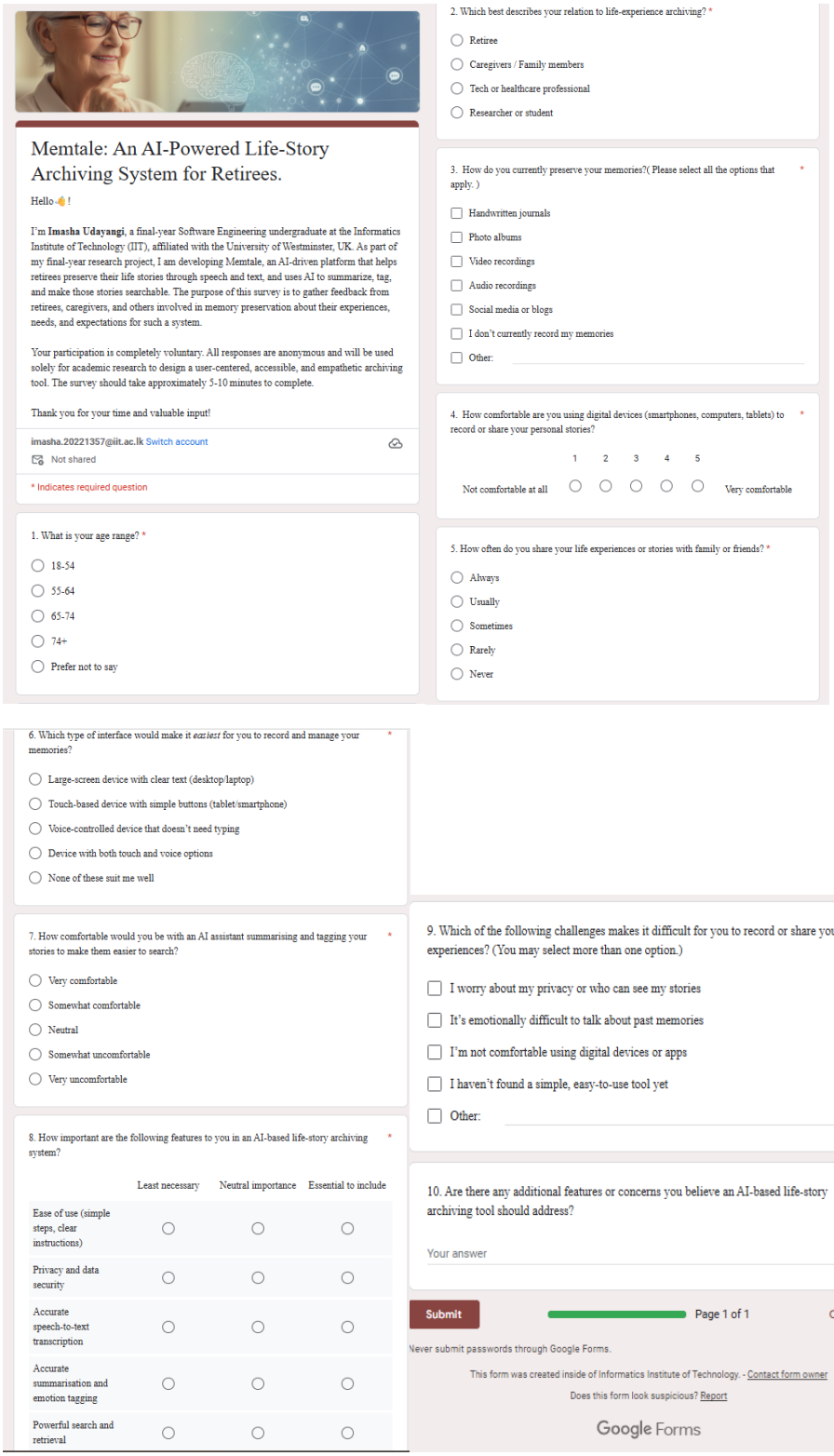


Figure 19:Appendix C.2: Survey - Google Form

APPENDIX C.3: USE CASE DESCRIPTION

Use Case	**Share Story with Family**
Use Case ID	UC02
Description	Enables retirees to share recorded stories with selected family members or caregivers.
Primary Actor	Retiree
Secondary Actor	None
Pre-Condition	Story must be recorded and saved.
Extended Use Cases	None
Included Use Cases	Manage Privacy Settings
Main Flow	1. Retiree selects stories to share. 2. System prompts privacy setting options. 3. Retiree configures sharing access. 4. System notifies designated users.
Alternative Flows	If no users selected, story remains private.
Exceptional Flow	Failure in sharing triggers error log.
Post Conditions	Story access granted to selected stakeholders.

Use Case	**View Stored Stories**
Use Case ID	UC03
Description	Allows users to retrieve and view archived stories based on access rights.
Primary Actor	Retiree or Secondary Users

Secondary Actor	None
Pre-Condition	Stories must exist in the archive.
Extended Use Cases	Edit Story Metadata, Request Access to Stories
Included Use Cases	Retrieve from Database, Query Archived Stories
Main Flow	1. User logs in and accesses story archive. 2. System fetches stories based on permissions. 3. User selects and views stories.
Alternative Flows	If no stories match filter, notify user.
Exceptional Flow	Access denied if story is private and request not approved.
Post COnditions	Stories are read or replayed successfully.

Use Case	**Request Access to Stories (if private)**
Use Case ID	UC04
Description	Enables secondary users to request permission to access restricted stories.
Primary Actor	Secondary User
Secondary Actor	Retiree (approval)
Pre-Condition	Story must be private.
Extended Use Cases	None
Included Use Cases	None
Main Flow	1. User locates a restricted story. 2. Request form is submitted.

	3. Retiree is notified for approval. 4. Retiree approves or rejects.
Alternative Flows	Auto-approve for specific user roles (if configured).
Exceptional Flow	If retiree fails to respond, timeout and retry logic initiated.
Post COnditions	Access granted or denied.

Use Case	**Manage User Accounts**
Use Case ID	UC05
Description	Enables administrators to manage system users.
Primary Actor	System Administrator
Secondary Actor	None
Pre-Condition	Admin must be logged in.
Extended Use Cases	None
Included Use Cases	None
Main Flow	1. Admin accesses the dashboard. 2. Views list of users. 3. Adds, removes, or updates user roles.
Alternative Flows	None
Exceptional Flow	Invalid user updates result in error alert.
Post COnditions	User records updated.

Use Case	**Manage Privacy Settings**
Use Case ID	UC06
Description	Allows retirees to control who can access each story.
Primary Actor	Retiree
Secondary Actor	None
Pre-Condition	At least one story must be saved.
Extended Use Cases	None
Included Use Cases	None
Main Flow	1. Retiree opens privacy dashboard. 2. Selects story to configure. 3. Chooses access options (public, private, restricted).
Alternative Flows	Default to private if no choice is made.
Exceptional Flow	Invalid configurations prompt user corrections.
Post COnditions	Story visibility updated.

Use Case	**Query Archived Stories**
Use Case ID	UC07
Description	Allows users to search through archived stories using keywords or emotional filters.
Primary Actor	Retiree or Secondary Users
Secondary Actor	None

Pre-Condition	Stories must be tagged and accessible.
Extended Use Cases	None
Included Use Cases	None
Main Flow	1. User enters keyword or emotion. 2. System queries indexed stories. 3. Results are presented to user.
Alternative Flows	No results found prompt query refinement.
Exceptional Flow	Timeout or index error triggers fallback alert.
Post COnditions	Relevant stories retrieved for user view.

Table 29:Appendix C.3: Use Case Descriptions

↑ Back to main content ↑

APPENDIX C.4: MOSCOW PRINCIPLE

Priority Level	Description
(M) - Must have	These are the crucial features that form the foundation of the project and are necessary for its successful execution.
(S) - Should have	These features are important but not vital for successful prototypes. Including them, if feasible, would add great value to the system.
(C) - Could have	Desirable but not essential requirements. It can be implemented if time and resources allow.
(W) - Will not have	Not considered for the initial prototype but may be included in future updates.

Table 30:Appendix C.4: Moscow Principle

Requirements

APPENDIX D.1: STREAMLIT DEMO UI.

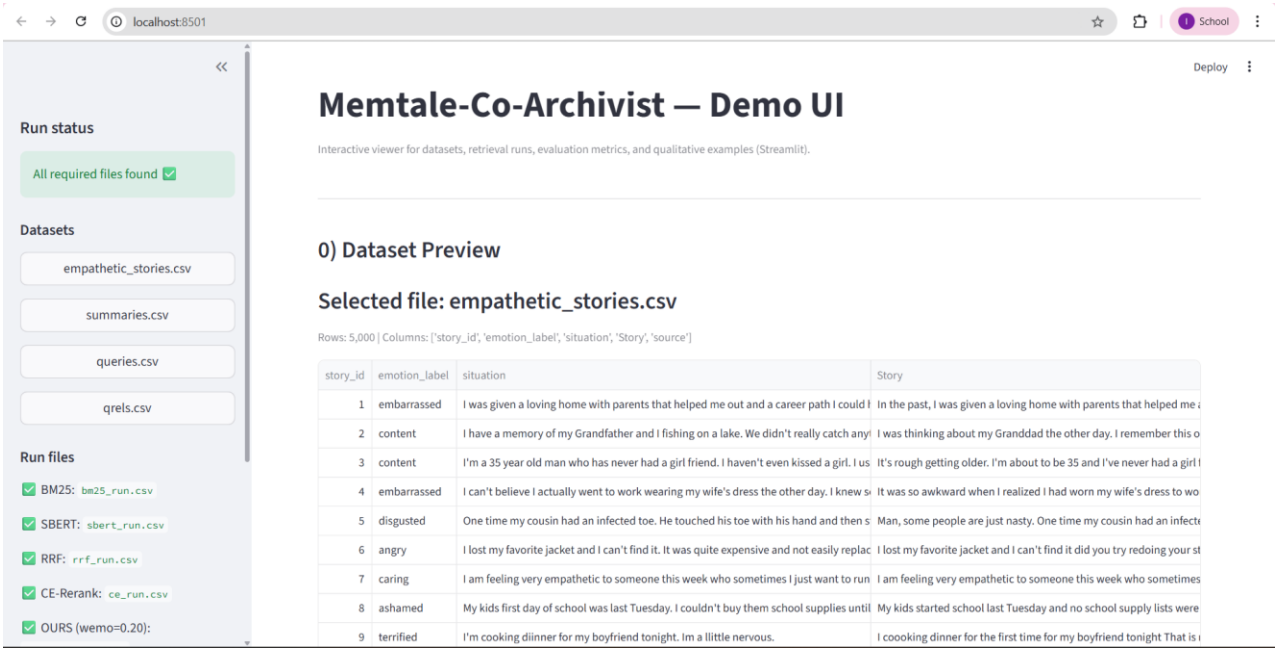


Figure 20:Appendix D.1: Data Preview Section

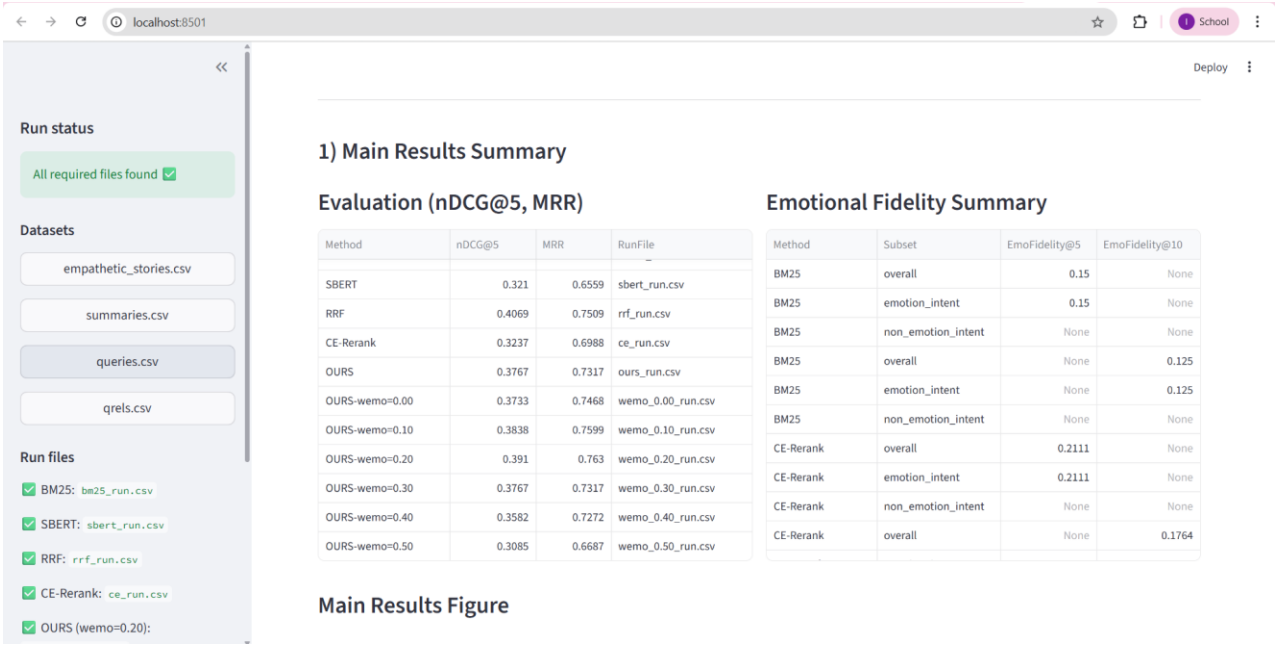


Figure 21:Appendix D.1: Main Result Summary Section

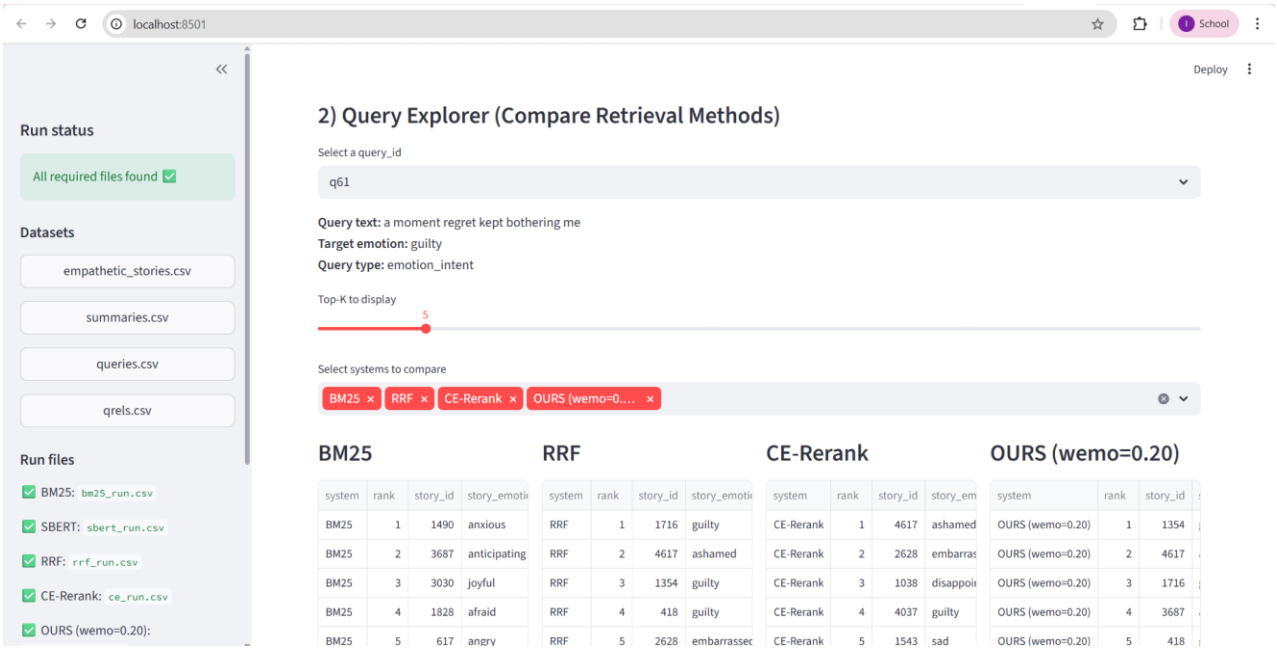


Figure 22:Appendix D.1: Query Explorer Section

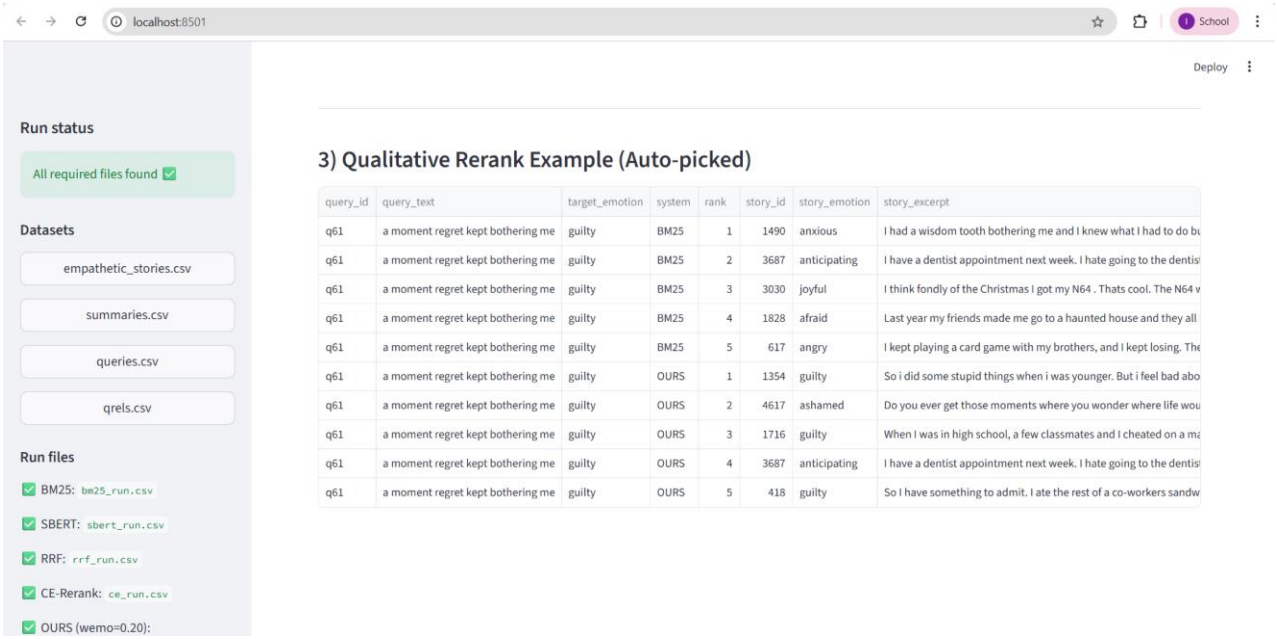


Figure 23:Appendix D.1: Qualitative Rerank Example Section

UI Design

APPENDIX E.1: QUERY GENERATION

```python
# src/build_eval_queries_v3.py
from pathlib import Path
import pandas as pd
import random

OUT_DIR = Path("dataset/eval_v3")
OUT_Q = OUT_DIR / "queries.csv"

RANDOM_SEED = 42
random.seed(RANDOM_SEED)

# Your 9 emotions
EMOTIONS = [
    "angry", "joyful", "impressed", "proud", "confident",
    "surprised", "anticipating", "guilty", "lonely"
]

# We generate 4 direct + 4 paraphrase per emotion
N_DIRECT = 4
N_PARA = 4

# Direct templates: some explicit (contains the emotion word), some implicit (no exact label)
DIRECT_TEMPLATES = {
    "angry": [
        "a memory where I felt angry",
        "a time I was furious after being treated unfairly",
        "a time I felt disrespected and wanted to push back",
        "a moment when I felt wronged and couldn't let it go",
        "a time someone crossed a line and I couldn't stay calm",
    ],
    "joyful": [
        "a memory where I felt joyful",
        "a time I felt genuinely happy and uplifted",
        "a moment when everything felt bright and positive",
        "a time I couldn't stop smiling because it felt so good",
        "a moment that left me feeling light and thankful",
    ],
    "impressed": [
        "a memory where I felt impressed",
        "a time I admired someone's skills or achievement",
        "a moment when someone exceeded my expectations",
        "a time I saw something surprisingly excellent",
        "a moment that made me think “wow, that was amazing”",
    ],
    "proud": [
        "a memory where I felt proud",
        "a time I accomplished something meaningful",
        "a moment when my effort finally paid off",
        "a time I proved to myself I could do it",
        "a moment I felt I earned my success",
    ],
    "confident": [
        "a memory where I felt confident",
        "a time I felt capable and in control",
        "a situation where I felt sure of myself",
        "a time I handled something difficult without doubt",
        "a moment I knew I could succeed",
    ],
    "surprised": [
        "a memory where I felt surprised",
        "a time something unexpected happened",
        "a moment when news changed everything suddenly",
        "a time I didn't see it coming at all",
        "a moment that shocked me in a good or bad way",
    ],
    "anticipating": [
        "a memory where I was anticipating something",
        "a time I was waiting for something important",
        "a moment before something big happened",
        "a time I couldn't relax waiting for what was next",
        "a moment where I felt on edge, waiting",
    ],
    "guilty": [
        "a memory where I felt guilty",
        "a time I regretted something I did",
        "a time I kept replaying a mistake in my head",
        "a moment I felt responsible for harm",
        "a time I wished I could undo what happened",
    ],
    "lonely": [
        "a memory where I felt lonely",
        "a time I felt alone even around other people",
        "a time it felt like I was completely on my own",
        "a moment when no one understood me",
        "a time I felt unsupported and isolated",
    ],
}

# Paraphrases should avoid repeating the exact direct templates
PARAPHRASE_TEMPLATES = {
    "angry": [
        "a time it felt unfair and I wanted to react",
        "a moment I felt provoked and ready to argue",
        "a time I couldn't accept how I was treated",
        "a moment I felt my patience snap",
        "a situation where I felt wronged by someone",
    ],
    "joyful": [
        "a moment when everything felt uplifting",
        "a time I felt warm and positive inside",
        "a memory that left me feeling cheerful for hours",
        "a moment that filled me with happiness",
        "a time I felt delighted by something simple",
    ],
    "impressed": [
```

```
    "impressed": [
        "a moment when someone did something surprisingly well",
        "a time I couldn't believe their skill",
        "a moment I saw an achievement that stood out",
        "a time I felt genuine admiration",
        "a moment that raised my expectations",
    ],
    "proud": [
        "a time I felt valued and accomplished",
        "a moment I felt my hard work mattered",
        "a time I felt I became a better version of myself",
        "a moment I felt proud of what I achieved",
        "a time I felt I truly earned it",
    ],
    "confident": [
        "a time I felt ready and capable",
        "a moment I trusted myself completely",
        "a time I felt I could handle anything",
        "a situation where I felt steady and sure",
        "a moment I acted without hesitation",
    ],
    "surprised": [
        "a time I didn't see it coming at all",
        "a moment when unexpected news changed things",
        "a time something happened out of nowhere",
        "a moment that completely caught me off guard",
        "a time I was shocked in an unexpected way",
    ],
    "anticipating": [
        "a moment before something important happened",
        "a time I felt restless waiting",
        "a moment when I couldn't stop thinking about what was next",
        "a time I felt tense during a buildup",
        "a moment where the waiting felt intense",
    ],
    "guilty": [
        "a time I couldn't forgive myself",
        "a moment I felt I let someone down",
        "a time I felt ashamed of my actions",
        "a moment regret kept bothering me",
        "a time I wished I could fix what I did",
    ],
    "lonely": [
        "a moment when no one understood me",
        "a time I felt disconnected from everyone",
        "a time I felt emotionally isolated",
        "a moment when I felt left out",
        "a time I felt like I didn't belong",
    ],
}

def pick_unique(pool, n, used):
    """Pick up to n unique strings from pool, not in used."""
    choices = [x for x in pool if x not in used]
    random.shuffle(choices)
```

```
def pick_unique(pool, n, used):
    """Pick up to n unique strings from pool, not in used."""
    choices = [x for x in pool if x not in used]
    random.shuffle(choices)
    picked = choices[:n]
    used.update(picked)
    return picked

def main():
    OUT_DIR.mkdir(parents=True, exist_ok=True)

    queries = []
    qid = 1

    for emo in EMOTIONS:
        used = set()

        direct = pick_unique(DIRECT_TEMPLATES[emo], N_DIRECT, used)
        para = pick_unique(PARAPHRASE_TEMPLATES[emo], N_PARA, used)

        # If still short (shouldn't happen unless pools are too small), fill by reshuffling without duplicates
        while len(direct) < N_DIRECT:
            for x in DIRECT_TEMPLATES[emo]:
                if x not in used:
                    direct.append(x); used.add(x)
                if len(direct) == N_DIRECT:
                    break

        while len(para) < N_PARA:
            for x in PARAPHRASE_TEMPLATES[emo]:
                if x not in used:
                    para.append(x); used.add(x)
                if len(para) == N_PARA:
                    break

        for t in direct:
            queries.append({
                "query_id": f"q{qid}",
                "query_type": "emotion_intent",
                "query_subtype": "direct",
                "target_emotion": emo,
                "query_text": t
            })
            qid += 1
```

```
def main():
        used = set()

        direct = pick_unique(DIRECT_TEMPLATES[emo], N_DIRECT, used)
        para = pick_unique(PARAPHRASE_TEMPLATES[emo], N_PARA, used)

        # If still short (shouldn't happen unless pools are too small), fill by reshuffling without duplicates
        while len(direct) < N_DIRECT:
            for x in DIRECT_TEMPLATES[emo]:
                if x not in used:
                    direct.append(x); used.add(x)
                if len(direct) == N_DIRECT:
                    break

        while len(para) < N_PARA:
            for x in PARAPHRASE_TEMPLATES[emo]:
                if x not in used:
                    para.append(x); used.add(x)
                if len(para) == N_PARA:
                    break

        for t in direct:
            queries.append({
                "query_id": f"q{qid}",
                "query_type": "emotion_intent",
                "query_subtype": "direct",
                "target_emotion": emo,
                "query_text": t
            })
            qid += 1

        for t in para:
            queries.append({
                "query_id": f"q{qid}",
                "query_type": "emotion_intent",
                "query_subtype": "paraphrase",
                "target_emotion": emo,
                "query_text": t
            })
            qid += 1

    out = pd.DataFrame(queries)

    # Safety: assert uniqueness
    dup = out.duplicated(subset=["query_text"]).sum()
    if dup > 0:
        raise ValueError(f"Duplicate query_text still exists: {dup}")

    out.to_csv(OUT_Q, index=False)
    print("✅ Saved:", OUT_Q)
    print("Total queries:", len(out))
    print("Unique query_text:", out["query_text"].nunique())

if __name__ == "__main__":
    main()
```

Query Generation, Pooling, and Qrels Construction

APPENDIX E.2: JUDGING POOL CREATION

```python
from pathlib import Path
import random
import re

import numpy as np
import pandas as pd

from rank_bm25 import BM25Okapi
from sentence_transformers import SentenceTransformer

# ------------------------
# Paths
# ------------------------
STORIES_PATH = Path("dataset/raw/empathetic_stories.csv")
QUERIES_PATH = Path("dataset/eval_v3/queries.csv")
OUT_BLIND = Path("dataset/eval_v3/judging_pool_blind.csv")
OUT_KEY = Path("dataset/eval_v3/judging_pool_key.csv")
OUT_META = Path("dataset/eval_v3/judging_pool_meta.csv")

# ------------------------
# Settings (tune if needed)
# ------------------------
RANDOM_SEED = 42
random.seed(RANDOM_SEED)
np.random.seed(RANDOM_SEED)

# Candidate pooling sizes
TOP_BM25 = 30
TOP_SBERT = 30
TOP_RRF = 30

# Final pool size per query (what you will manually judge)
POOL_SIZE = 30

# Text excerpt sizes (for human judging UI)
MAX_SITUATION_WORDS = 35
MAX_STORY_WORDS = 110

# Retrieval text mode (recommended = situation + Story)
INDEX_MODE = "full"  # "full" only in this pool builder (keep it consistent)


# ------------------------
# Helpers
# ------------------------
def tokenize(text: str):
    return re.findall(r"\b\w+\b", (text or "").lower())

def truncate_words(text: str, n: int):
    w = (text or "").split()
    return " ".join(w[:n])

def rrf_fusion(ranked_lists, k=60):
    """
    Reciprocal Rank Fusion. Returns scores aligned to doc indices [0..N-1].
```

```python
def rrf_fusion(ranked_lists, k=60):
    """
    Reciprocal Rank Fusion. Returns scores aligned to doc indices [0..N-1].
    """
    if not ranked_lists:
        return None
    N = max(len(r) for r in ranked_lists)
    scores = np.zeros(N, dtype=float)
    for r in ranked_lists:
        for rank, doc_idx in enumerate(r, start=1):
            if doc_idx < N:
                scores[doc_idx] += 1.0 / (k + rank)
    return scores

def build_index_texts(stories: pd.DataFrame):
    # "full" = situation + Story (recommended for autobiographical retrieval)
    return (stories["situation"].fillna("") + " " + stories["Story"].fillna("")).tolist()


def main():
    # ------------------------
    # Load data
    # ------------------------
    stories = pd.read_csv(STORIES_PATH)
    queries = pd.read_csv(QUERIES_PATH)

    # Sanity
    required_story_cols = {"story_id", "situation", "Story", "emotion_label"}
    missing = required_story_cols - set(stories.columns)
    if missing:
        raise ValueError(f"Missing story cols: {missing}")

    required_query_cols = {"query_id", "query_text"}
    missing_q = required_query_cols - set(queries.columns)
    if missing_q:
        raise ValueError(f"Missing query cols: {missing_q}")

    stories["story_id"] = stories["story_id"].astype(int)

    story_ids = stories["story_id"].tolist()
    index_texts = build_index_texts(stories)

    # ------------------------
    # Build BM25 index (once)
    # ------------------------
    tokenized_docs = [tokenize(t) for t in index_texts]
    tokenized_docs = [toks if toks else ["_empty_"] for toks in tokenized_docs]
    bm25 = BM25Okapi(tokenized_docs, k1=1.5, b=0.75)

    # ------------------------
    # Build SBERT embeddings (once)
    # ------------------------
    sbert = SentenceTransformer("all-MiniLM-L6-v2")
    doc_embs = sbert.encode(index_texts, convert_to_numpy=True, normalize_embeddings=True)
```

```
def main():
    # ------------------------
    # Build SBERT embeddings (once)
    # ------------------------
    sbert = SentenceTransformer("all-MiniLM-L6-v2")
    doc_embs = sbert.encode(index_texts, convert_to_numpy=True, normalize_embeddings=True)

    # ------------------------
    # Build pools
    # ------------------------
    blind_rows = []
    key_rows = []
    meta_rows = []

    for q in queries.itertuples(index=False):
        qid = str(q.query_id)
        qtext = str(q.query_text)

        # ---- BM25 ranking ----
        bm25_scores = np.asarray(bm25.get_scores(tokenize(qtext)))
        bm25_ranked = np.argsort(bm25_scores)[::-1].tolist()

        # ---- SBERT ranking ----
        q_emb = sbert.encode([qtext], convert_to_numpy=True, normalize_embeddings=True)[0]
        sem_sims = doc_embs @ q_emb
        sbert_ranked = np.argsort(sem_sims)[::-1].tolist()

        # ---- RRF ranking (BM25 + SBERT) ----
        fused = rrf_fusion([bm25_ranked, sbert_ranked], k=60)
        rrf_ranked = np.argsort(fused)[::-1].tolist()

        # Candidate set = union of top lists
        cand = set(bm25_ranked[:TOP_BM25]) | set(sbert_ranked[:TOP_SBERT]) | set(rrf_ranked[:TOP_RRF])
        cand = list(cand)

        # If union is small, expand slightly using RRF
        if len(cand) < POOL_SIZE:
            extra = [x for x in rrf_ranked if x not in set(cand)]
            cand += extra[: (POOL_SIZE - len(cand))]

        # Now choose final pool = top POOL_SIZE by RRF score (stable + strong)
        cand_sorted = sorted(cand, key=lambda i: fused[i], reverse=True)[:POOL_SIZE]

        # Randomize order for judging (IMPORTANT: reduce bias)
        random.shuffle(cand_sorted)

        # Save rows
        for pos, doc_idx in enumerate(cand_sorted, start=1):
            sid = int(story_ids[doc_idx])

            situation = str(stories.loc[doc_idx, "situation"])
            story = str(stories.loc[doc_idx, "Story"])

            # Blind judge content (no emotion label)
```

```
def main():
            # Blind judge content (no emotion label)
            blind_rows.append({
                "query_id": qid,
                "query_text": qtext,
                "pool_pos": pos,
                "story_id": sid,
                "situation_excerpt": truncate_words(situation, MAX_SITUATION_WORDS),
                "story_excerpt": truncate_words(story, MAX_STORY_WORDS),

                # Manual label column (fill later):
                # 2 = relevant, 1 = somewhat relevant, 0 = not relevant
                "relevance": ""
            })

            # Key file (keep separate; do NOT use while judging)
            key_rows.append({
                "query_id": qid,
                "story_id": sid,
                "emotion_label": str(stories.loc[doc_idx, "emotion_label"]),
                "bm25_rank": int(bm25_ranked.index(doc_idx) + 1),
                "sbert_rank": int(sbert_ranked.index(doc_idx) + 1),
                "rrf_rank": int(rrf_ranked.index(doc_idx) + 1),
            })

        meta_rows.append({
            "query_id": qid,
            "cand_union_size": len(set(cand)),
            "final_pool_size": len(cand_sorted)
        })

    # ------------------------
    # Write outputs
    # ------------------------
    OUT_BLIND.parent.mkdir(parents=True, exist_ok=True)

    blind_df = pd.DataFrame(blind_rows)
    key_df = pd.DataFrame(key_rows)
    meta_df = pd.DataFrame(meta_rows)

    blind_df.to_csv(OUT_BLIND, index=False)
    key_df.to_csv(OUT_KEY, index=False)
    meta_df.to_csv(OUT_META, index=False)

    print("✅ Saved judging pool (blind):", OUT_BLIND, "rows:", len(blind_df))
    print("✅ Saved judging pool key:", OUT_KEY, "rows:", len(key_df))
    print("✅ Saved pool meta:", OUT_META, "rows:", len(meta_df))
    print("\nTip: Open judging_pool_blind.csv and fill relevance with 0/1/2 manually.")


if __name__ == "__main__":
    main()
```

Query Generation, Pooling, and Qrels Construction

APPENDIX E.3 - CHALLENGES AND SOLUTIONS.

Challenge	Description	Solution.
Noisy dialogue speaker tags and inconsistent CSV formatting	Kaggle exports contained "Customer:"/ "Agent:" tokens and occasional unnamed columns, introducing noise to retrieval and causing schema inconsistency.	Implemented _strip_speaker_tags() and dropped unnamed columns in src/data_build_empathetic.py, ensuring clean narrative text and stable dataset schema.
Controlling story length for meaningful affective content	Very short stories reduce emotional signal; very long stories inflate computational cost and can distort summarisation.	Enforced 100-600 word bounds through truncation and filtering in src/data_build_empathetic.py.
Summarisation runtime and repeated model calls	Calling an abstractive summariser twice per story (naive + emotion-aware) doubled the runtime.	emotion_aware_summary() accepts base_summary and reuses it, preventing a second Hugging Face summariser call (src/precompute_summaries.py).
Token length limits and warning spam from Transformers	Long inputs can exceed model limits and cause unstable behaviour and repetitive warnings.	Implemented character truncation (_truncate_chars) and set tokenizer max length; reduced warning verbosity (hf_logging.set_verbosity_error()).
Empty documents breaking BM25 tokenisation	Empty strings can lead to empty token lists and unstable scoring.	Added a safe fallback token ["_empty_"] during tokenisation in BM25 indexing and scoring.
Efficient reranking for large corpora	Computing embeddings and emotion vectors repeatedly inside loops is expensive.	Precompute SBERT doc embeddings once and SETL-2 document vectors once in evaluation/run_two_stage_retrieval.py,

		then use fast dot products/cosine similarity for scoring.
Avoiding bias in manual relevance annotation	If a single system defines the judging pool, results may be biased toward that system.	Implemented multi-system pooling (BM25 + SBERT + RRF union), randomised pool order, and separated blind vs key files (src/build_judging_pool_v3.py).
Ensuring qrels integrity	Manual labelling risks invalid values, missing labels, or duplicate pairs.	Strict validation in src/convert_judged_to_qrels_v3.py with explicit errors and a qrels distribution report.
Preventing "test tuning" and maintaining interpretability	Learned/tuned weights risk overfitting and reduce transparency.	Adopted a rule-based adaptive weighting scheme based on query type, documented and fixed before evaluation.

Table 31:Appendix E.3 - Challenges And Solutions.

Challenges and Solutions

APPENDIX F.1 - PER-EMOTION K VALUES

iaa_subset20_report.txt

evaluation > results > iaa_subset20_report.txt

```
Pairs: 432
Cohen kappa (unweighted): 0.5729
Cohen kappa (linear):     0.6574
Cohen kappa (quadratic):  0.7426
Disagreements: 123 (28.47%)

Quadratic kappa per emotion:
  angry       : 0.7077
  anticipating: 0.7419
  confident   : 0.8667
  guilty      : 0.7742
  impressed   : 0.7500
  joyful      : 0.8136
  lonely      : 0.5938
  proud       : 0.8387
  surprised   : 0.6154

```

Figure 24:Appendix F.1 - Per-Emotion K Values.Txt

↑ Annotation reliability (IAA)↑

APPENDIX F.2 – SECOND ANNOTATOR PROOF.

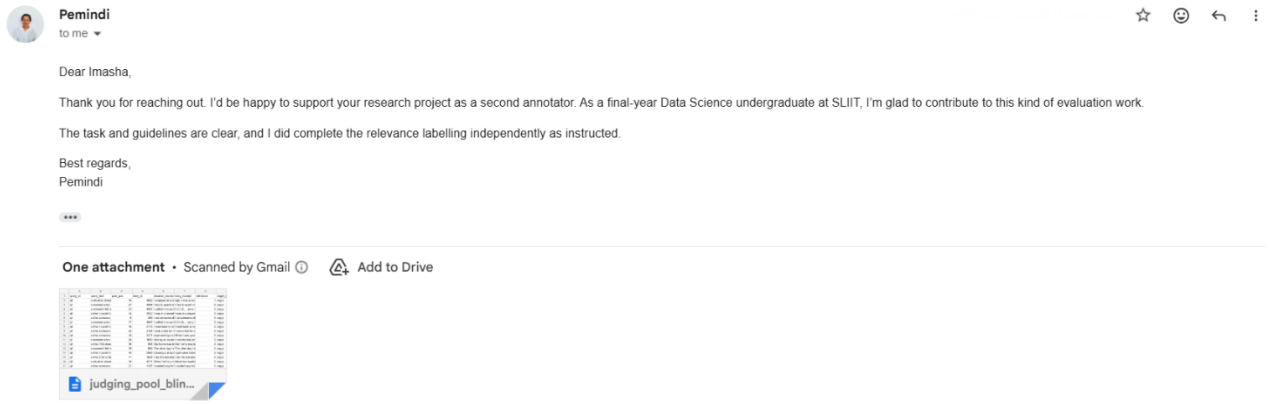


Figure 25:Appendix F.2 – Second Annotator Proof.

APPENDIX G.1: ACHIEVEMENT OF OBJECTIVES.

Category	Research Objective	Status	Evidence / Where addressed
Problem Identification	**RO1**: Investigate challenges retirees face in documenting, preserving, retrieving life experiences using existing tools	Achieved	Chapter 01 (Problem Background/Gap), Chapter 02 (Problem Domain + existing work synthesis), Chapter 04 (stakeholders + viewpoints)
	RO2: Analyse importance of AI-assisted digital archiving for knowledge preservation and intergenerational sharing	Achieved	Chapter 01 (Motivation/Contributions), Chapter 02 (digital storytelling + legacy motivation), SLEP in Chapter 05
	RO3: Examine global trends/demographic factors (ageing populations, digital divide) motivating AI memory archiving	Achieved	Chapter 01 (Background + motivation), Chapter 02 (digital exclusion + accessibility challenges)
Literature Review	**RO4**: Conduct comprehensive literature review on AI tools for storytelling, summarisation, and digital curation	Achieved	Chapter 02 (digital storytelling + curation + AI companionship strands)

	RO5: Identify research gaps in multimodal input (speech/text), tagging, and retrieval in elderly-focused applications	Achieved	Chapter 02 (gaps + synthesis), Chapter 01 (Research Gap), Chapter 04 (MoSCoW Out-of-scope items)
	RO6: Review narrative retrieval, emotion-aware NLP, emotionally faithful summarisation methods relevant to autobiographical archives	Achieved	Chapter 02 (emotion-aware IR + summarisation for indexing), Chapter 01 (problem definition and gap)
Requirement Elicitation	**RO7**: Define FR/NFRs for emotion-intent narrative retrieval prototype (reproducibility, interpretability, early-rank performance)	Achieved	Chapter 04 (FR/NFR MoSCoW), Chapter 06 (design goals mapped to NFRs), Chapter 08/09 (verification against outputs)
	RO8: Identify appropriate NLP models for summarisation, tagging, retrieval	Achieved	Chapter 07 (technology + implementation): DistilBART summarisation (cached), SBERT embeddings, BM25 retrieval, SETL-2 emotion vectors, CE-Rerank baseline, RRF
	RO9: Determine hardware/software specs and datasets suitable for prototype development	Achieved	Chapter 01 (HW/SW table), Chapter 07 (stack selection), dataset choice justified and used consistently

System Design	**RO10**: Design two-stage retrieval architecture using lexical candidate generation + reranking combining semantic + emotion similarity	Achieved	Chapter 06 (architecture + algorithm), Chapter 07 (implementation in run generator and retrieval modules), Chapter 09 (results + significance)
	RO12: Design data storage and indexing system for secure and efficient retrieval of life stories	Partially Achieved (prototype level)	Implemented file-based, reproducible artefact stores (dataset CSVs, cached summaries, run files, metrics). Full secure multi-user storage/privacy controls are scoped as future work (Chapter 04 MoSCoW; Chapter 10 limitations/future work).
Implementation	**RO13**: Develop and implement data collection pipeline for speech and text input	Partially Achieved	Implemented text-based corpus pipeline using a public narrative dataset and caching. Speech capture/ASR remains future work due to refined scope and privacy constraints (Chapter 04 MoSCoW; Chapter 10 deviations).
	RO14: Integrate NLP-based summarisation and tagging models into the prototype	Achieved	Cached summarisation module + SETL-2 emotion vectors integrated into retrieval pipeline (Chapter 07)

	RO15: Develop search/retrieval module allowing keyword or embedding-based queries	Achieved	BM25 (keyword), SBERT (embedding), RRF hybrid, two-stage reranking + CE-Rerank baseline; exposed via run generator and UI explorer (Chapters 07 and UI section)
	RO16: Develop conversational companion interface to enhance engagement and usability	Not Achieved (de-scoped)	A full conversational companion was not implemented; instead, a researcher-facing Streamlit demo UI was added for transparency and viva demonstration. Conversational interface remains future work (Chapter 04 MoSCoW; Chapter 10 future enhancements).
Testing	**RO17**: Develop a test plan for unit, integration, usability, performance testing	Achieved (prototype-focused)	Chapter 08 testing plan (component + pipeline testing, artefact checks, reproducibility tests, basic performance practicality)
	RO18: Evaluate accuracy of summarisation, tagging, retrieval using metrics and user feedback	Partially Achieved	Retrieval evaluation is rigorous (metrics + significance + qualitative example). Summarisation "accuracy" is evaluated indirectly via retrieval outcomes and sanity checks; no end-user study was conducted due to scope.

	RO19: Evaluate using early-rank IR metrics + emotional alignment measure incl. significance testing	Achieved	nDCG@5, MRR, EmoFidelity@5/@10, Wilcoxon + effect size, qualitative rerank example; plus CE-Rerank baseline comparisons (Chapters 08/09)
Documentation	**RO21**: Compile and document entire research process and prototype in a formal report	Achieved	Full thesis chapters + structured artefacts + reproducibility workflow (this thesis + repository outputs)
	RO22: Produce a research publication/conference paper presenting findings and innovations	Achieved	Q1-target paper drafted based on the prototype results and workflow (reported in thesis documentation section)

Table 32:Appendix G.1: Achievement of Objectives

Achievement of Objectives.

APPENDIX G.2: ACHIEVEMENT OF LEARNING OUTCOMES

Learning Outcome	**Achievement.**
LO1 Select, justify and apply appropriate methods, techniques and tools for tackling a large problem.	Achieved by selecting a two-stage retrieval architecture suitable for long narrative search: BM25 candidate generation (TopK=100) followed by reranking using SBERT semantic similarity + SETL-2 emotion cosine similarity, and benchmarking against BM25, SBERT, RRF, plus a strong CE-Rerank (cross-encoder) baseline. Tools were justified and applied consistently (Python 3.10, rank-bm25, sentence-transformers, transformers, NLTK/NRCLex, SciPy, Streamlit).
LO2 Develop a project plan, organise own activities and manage time/workload within the given timescale.	Achieved through milestone-driven delivery aligned with the unit schedule: proposal → IPD → implementation → evaluation → thesis submission. Work was structured into clear phases (dataset preparation, caching summaries, retrieval baselines, reranking, judging/qrels, evaluation scripts, UI demo), enabling controlled progress despite scope refinements.
LO3 Collect and analyse project requirements using appropriate tools/techniques.	Achieved via requirements capture and refinement across literature-driven requirement elicitation, supervisor feedback (M1–M5 style improvements), and prototype-driven validation. Requirements were formalised as MoSCoW-prioritised FR/NFRs, emphasising reproducibility, interpretability, and early-rank effectiveness, and were reflected in design and evaluation artefacts.

LO4 Research and collate relevant information and critically evaluate findings, including context.	Achieved through an extensive literature review covering digital storytelling/ageing, semantic retrieval, emotion-aware IR, and summarisation for indexing, then translating findings into a testable research pipeline. The work also considered practical context: privacy sensitivity (no private retiree data), interpretability needs, and reproducible offline evaluation.
LO5 Work autonomously to learn new skills/enhance knowledge and produce supervisor-agreed deliverables.	Achieved by independently learning and integrating: (i) IR evaluation practice (pooling, qrels, graded relevance), (ii) statistical significance testing (paired Wilcoxon + effect size), (iii) emotion alignment measurement (EmoFidelity@k), and (iv) building a Streamlit demo UI for viva. Deliverables were produced iteratively with supervisor-aligned fixes (e.g., adding CE-Rerank baseline, weight-sensitivity runs, IAA workflow).
LO6 Identify and take into account EDI, legal, social, ethical and professional issues (incl. sustainability).	Achieved through explicit handling of privacy and ethical risk: using only public datasets, avoiding clinical/therapeutic claims, reporting limitations, and retaining transparency through interpretable emotion vectors (SETL-2). SLEP considerations were documented and linked to professional conduct (responsible data use, reproducibility, truthful reporting).

LO7 Produce an extended piece of practical work (software/process/model/experimental study).	Achieved by delivering a fully runnable research prototype with complete workflow: dataset build → cached summaries → retrieval runs (BM25/SBERT/RRF/CE/OURS + wemo variants) → metrics computation (nDCG@5, MRR, EmoFidelity@5/@10) → significance testing → figure and qualitative export → demo UI. The system outputs reusable artefacts (CSV run files, qrels, metric summaries, plots).
LO8 Produce a coherent report documenting and critically evaluating the work, skills, and plan effectiveness.	Achieved through a structured thesis that documents scope evolution, methodology, design, implementation, testing, and critical evaluation. The report includes reproducible results tables (evaluation_summary, emotional_fidelity_summary), significance results (significance_summary), qualitative rerank evidence, and reflection on limitations and future enhancements.
LO9 Demonstrate and defend work at viva voce.	Achieved through preparing a clear demonstration path: running python main.py to reproduce outputs and using the Streamlit UI to show (i) dataset/cached summaries, (ii) main metric tables and plot, and (iii) query-by-query comparisons across retrieval methods, including the qualitative rerank example (q61). This supports defendable explanations of design choices and evaluation evidence.

Table 33:Appendix G.2: Achievement of Learning Outcomes

Achievement of Learning Outcomes.